

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]
```

```

cols = list(dict.fromkeys(target_col + features_X +
                          feats_to_balance + feats_grouping +
                          feats_reference+ feats_postproc))

df = df[cols]
#df['campaign'] = "ALL"

return df

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        #file_path = "csv/innova/switched/df_signal_decomposition_NH3_current_db2.csv"
        file_path = "csv/innova/notswitched/df_signal_decomposition_current.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv/innova/switched/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/notswitched/features_decomposition_list.json", "r") as f:
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/switched/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")

    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)
    #Removing of rabbit 4

```

```
df["column"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)
```

```
return df,dict_features
```

```
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Function print :

1.2 Metadata

Nombre de lignes df	8168
Nombre de lignes df clean	8168

Nombre de features	5
% de NaN	0.0
Taille dataset entraînement	5717
Taille dataset test	2450
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI

feats_to_balance :

feats_grouping : campaign

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

Grouping : campaign

2 Model(s) analysis

2.1 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_test < 0.95$ (under performance of the test)

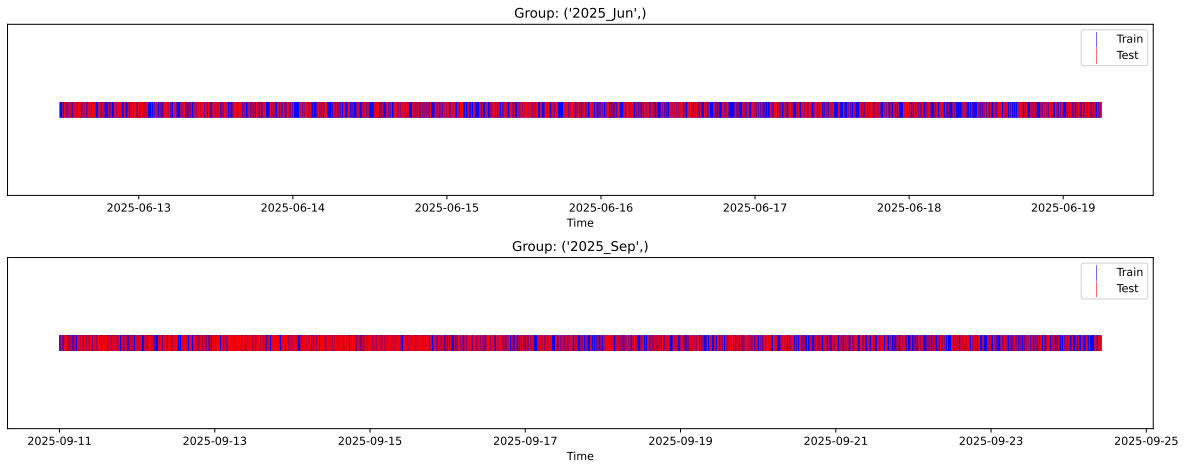
[Green] $r_test > 1.05$ (over performance of the test)

[Red] r_train not in $[0.95, 1.05]$ (gap between train/test)

2.2 Repartition of train and test

2.2.1 External Split: X_{train} vs X_{test}

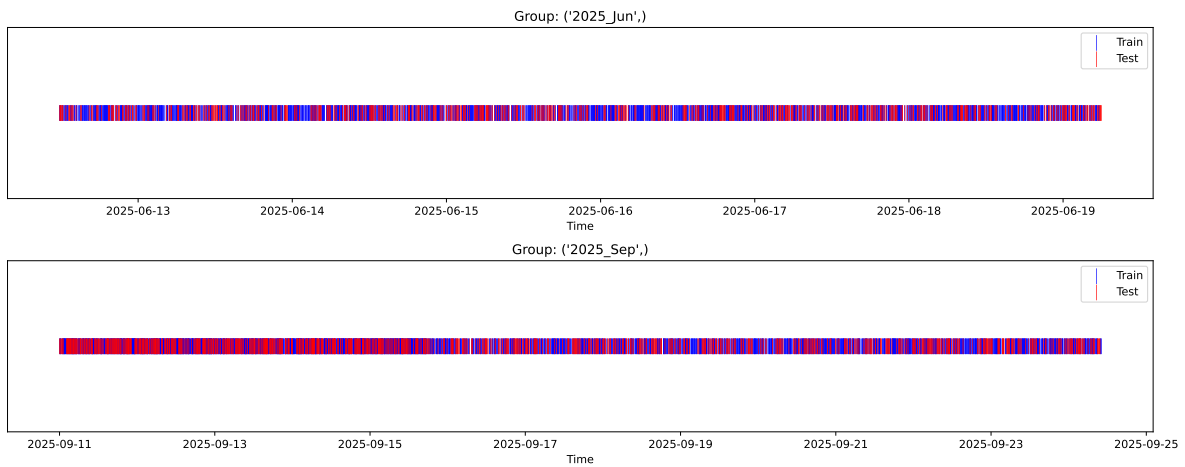
External Split: Train: 5717 samples Test: 2451 samples



2.2.2 Internal CV Folds

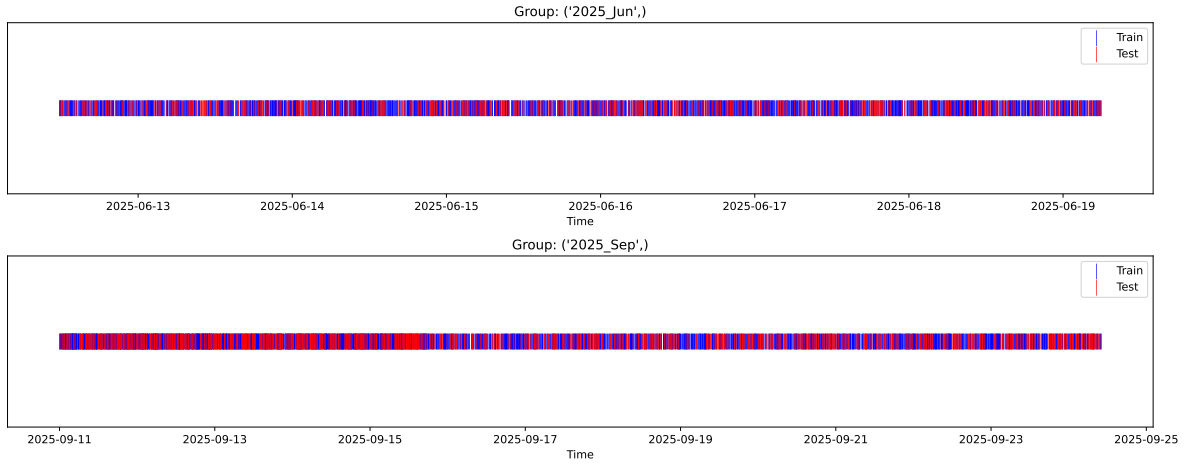
Fold 1/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



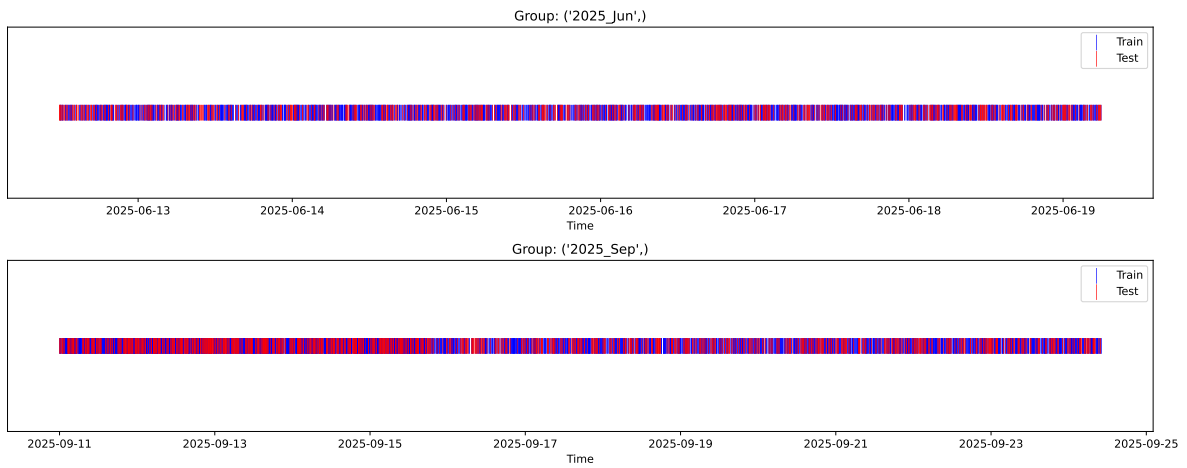
Fold 2/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



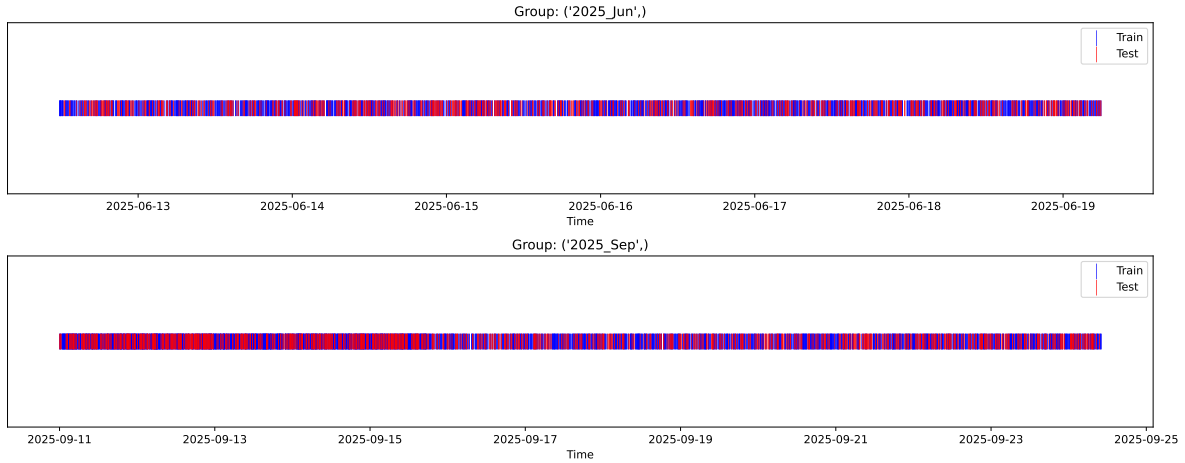
Fold 3/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



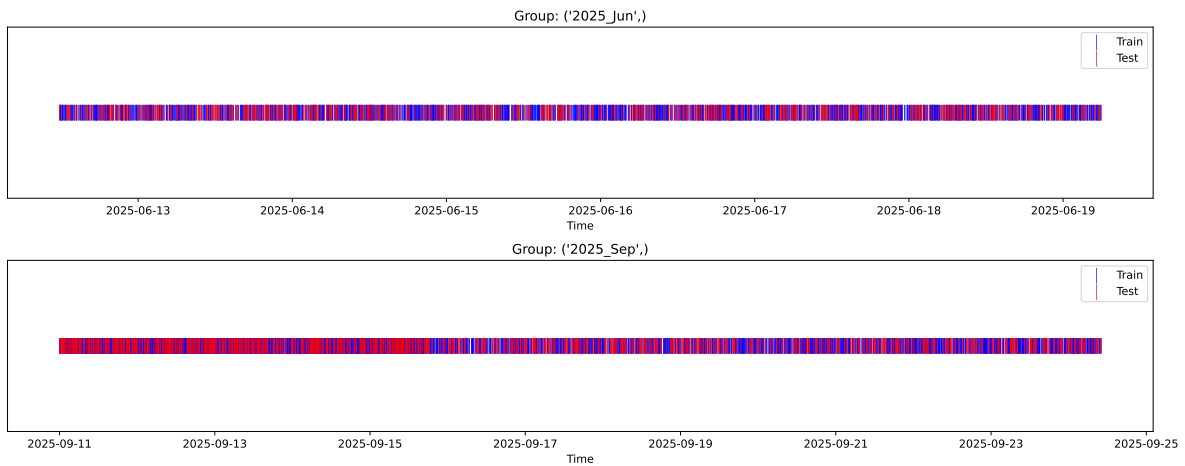
Fold 4/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



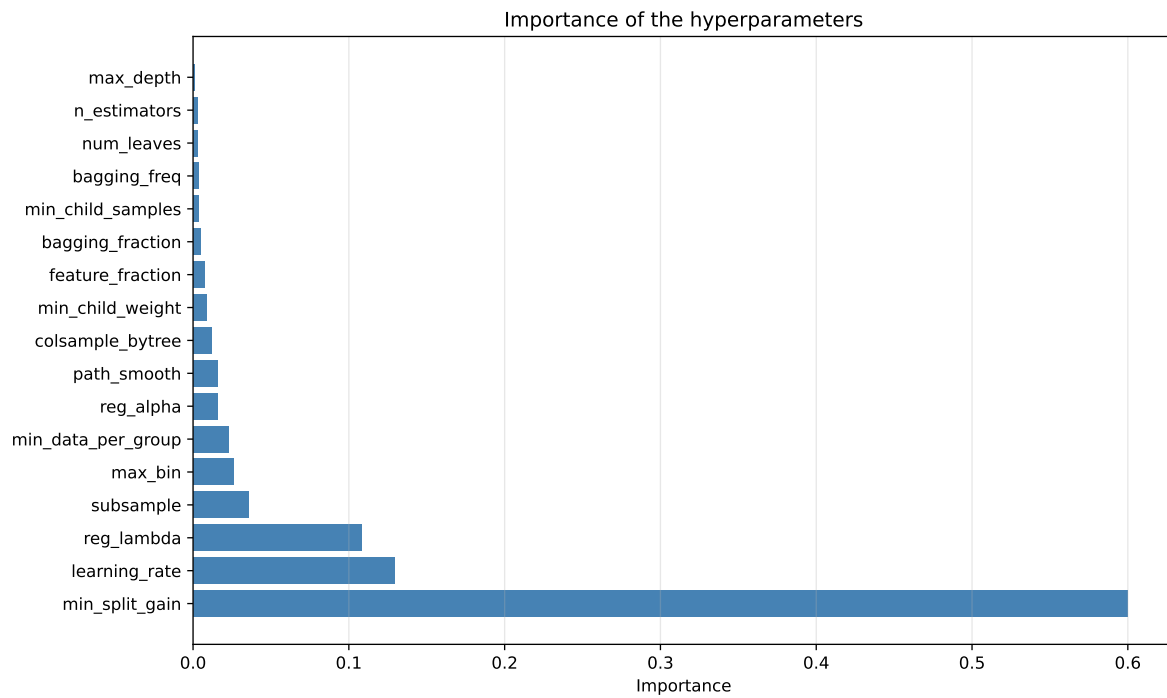
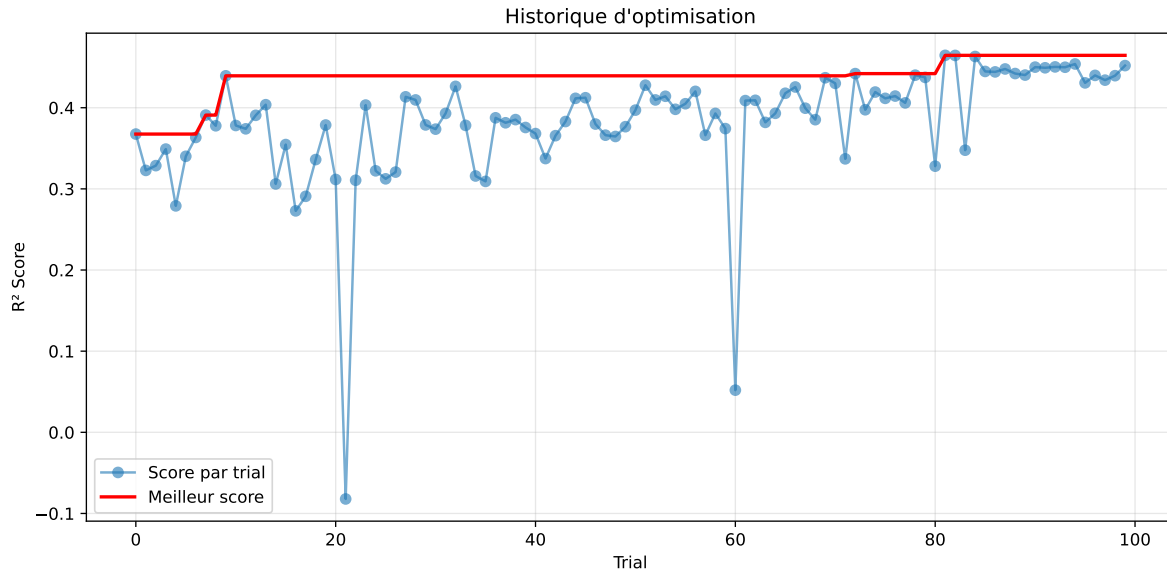
2.3 Model : LGBM

2.3.1 Gridsearch — Group: 2025_Jun

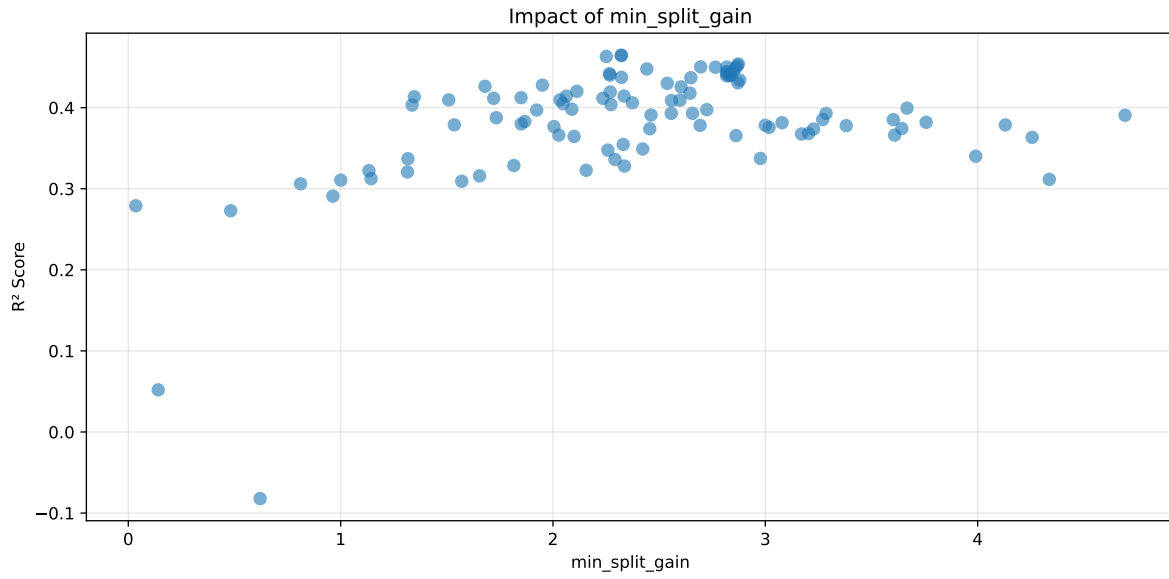
Len(group) : 3172, Len(training) : 2221, Len(test) : 951

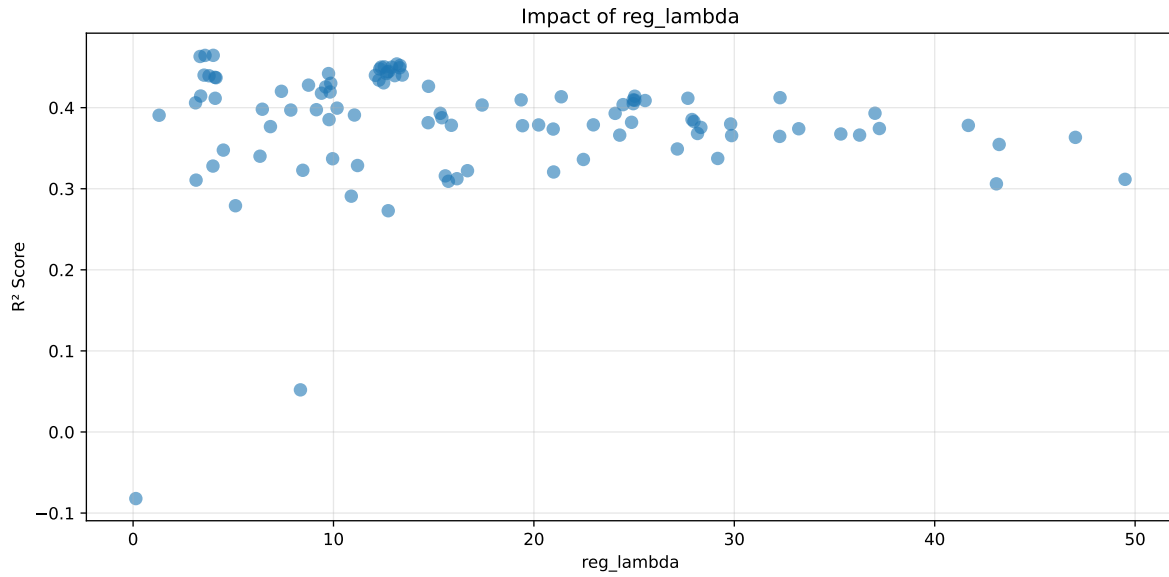
Distribution y_train vs y_test: y_train: mean=2.661, std=0.770, min=1.305, max=13.845
y_test: mean=2.621, std=0.704, min=1.253, max=5.627

===== TOP Importance FEATURES =====
feature importance r_NH3 102 r_CO2 102 r_umidity 82 r_temperature 53 r_THI 51



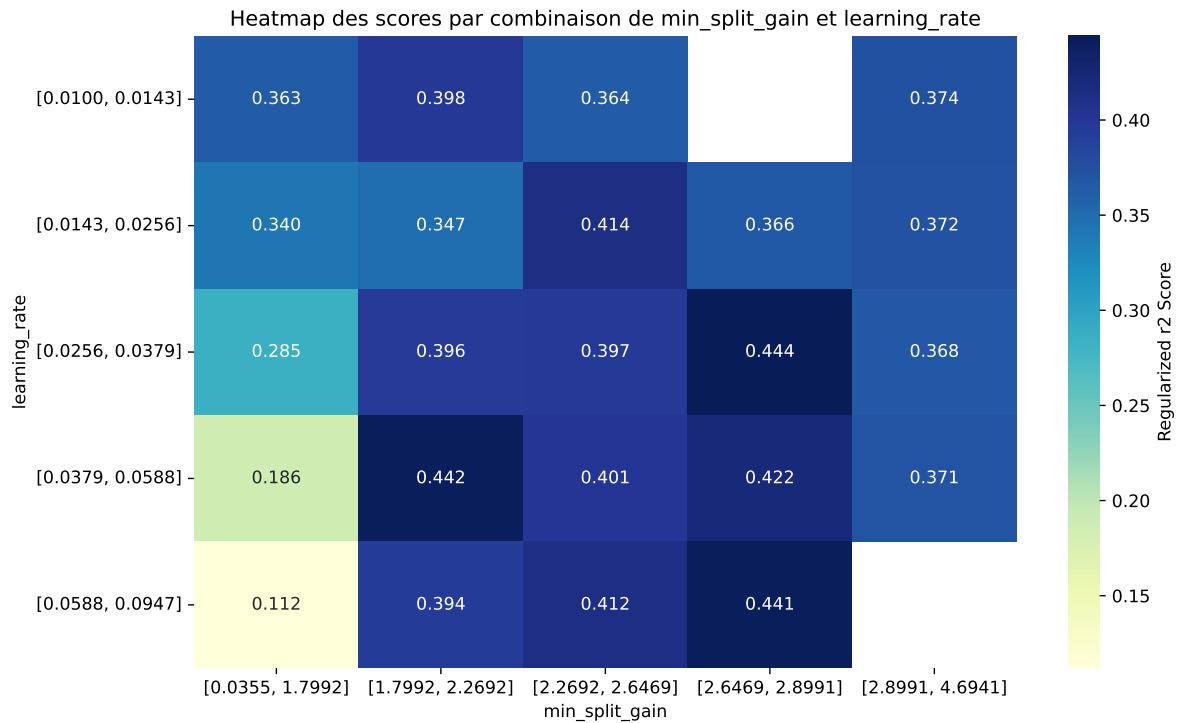
Paramètres avec >5% d'importance : ['min_split_gain', 'learning_rate', 'reg_lambda']





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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.4864	0.4645	0.5109	0.0345	0.4645	0.5377	0.0732	1.1575	1.047
2	0.4877	0.4645	0.5122	0.0365	0.4645	0.5411	0.0766	1.1649	1.05
3	0.4836	0.4631	0.5065	0.0353	0.4631	0.531	0.0679	1.1466	1.0442
4	0.4698	0.4539	0.5036	0.0362	0.4539	0.5204	0.0665	1.1464	1.0349
5	0.4692	0.4519	0.4988	0.0372	0.4519	0.52	0.0681	1.1506	1.0382
6	0.4653	0.4503	0.4946	0.037	0.4503	0.5238	0.0735	1.1632	1.0334
7	0.4649	0.45	0.497	0.0364	0.45	0.5258	0.0757	1.1683	1.0331
8	0.4671	0.4499	0.4937	0.0368	0.4499	0.5256	0.0757	1.1683	1.0382
9	0.4657	0.4492	0.4962	0.0362	0.4492	0.5256	0.0764	1.1701	1.0367
10	0.4615	0.4478	0.496	0.0364	0.4478	0.5267	0.0789	1.1763	1.0306
---	---	---	---	---	---	---	---	---	---
100	0.6382	0.5182	0.5402	0.0214	-0.0821	0.5501	0.0319	1.0616	1.2317

Hyperparameters:

Rank 1: {'reg__n_estimators': 400, 'reg__max_depth': 8, 'reg__learning_rate': 0.06475913506224305, 'reg__num_leaves': 46, 'reg__subsample': 0.6605667120054032, 'reg__colsample_bytree': 0.568618688651256, 'reg__min_child_samples': 23, 'reg__reg_alpha': 1.3329026393751244, 'reg__reg_lambda': 3.9984454576262074, 'reg__min_split_gain': 2.3228002065757924, 'reg__path_smooth': 1.5675127136306042, 'reg__min_child_weight': 0.022068811916520874, 'reg__feature_fraction': 0.846782207038257, 'reg__bagging_fraction': 0.7546405541662803, 'reg__bagging_freq': 1, 'reg__max_bin': 63, 'reg__min_data_per_group': 63}

Rank 2: {'reg__n_estimators': 400, 'reg__max_depth': 5, 'reg__learning_rate': 0.023984912529348822, 'reg__num_leaves': 47, 'reg__subsample': 0.681330406433609, 'reg__colsample_bytree': 0.5643612300925333, 'reg__min_child_samples': 24, 'reg__reg_alpha': 1.252984576207626, 'reg__reg_lambda': 3.5893758171151395, 'reg__min_split_gain': 2.320688957967912, 'reg__path_smooth': 2.7048726044200713, 'reg__min_child_weight': 0.021038779211068878, 'reg__feature_fraction': 0.8592218925119329, 'reg__bagging_fraction': 0.7531195281432471, 'reg__bagging_freq': 1, 'reg__max_bin': 103, 'reg__min_data_per_group': 65}

Rank 3: {'reg__n_estimators': 400, 'reg__max_depth': 4, 'reg__learning_rate': 0.03861415011661238, 'reg__num_leaves': 54, 'reg__subsample': 0.6511420776886405, 'reg__colsample_bytree': 0.5709897608643589, 'reg__min_child_samples': 23, 'reg__reg_alpha': 1.3889882158054332, 'reg__reg_lambda': 3.3429950441481378, 'reg__min_split_gain': 2.2511523335574695, 'reg__path_smooth': 1.4899539120503833, 'reg__min_child_weight': 0.01741839917351153, 'reg__feature_fraction': 0.8797656364427845, 'reg__bagging_fraction': 0.749464103301039, 'reg__bagging_freq': 1, 'reg__max_bin': 100, 'reg__min_data_per_group': 67}

Rank 4: {'reg__n_estimators': 300, 'reg__max_depth': 8, 'reg__learning_rate': 0.08230371086171637, 'reg__num_leaves': 54, 'reg__subsample': 0.6000971582596684, 'reg__colsample_bytree': 0.5087369017999123, 'reg__min_child_samples': 25, 'reg__reg_alpha': 0.16861473136644256, 'reg__reg_lambda': 13.158462024798075, 'reg__min_split_gain': 2.8729920699507447, 'reg__path_smooth': 1.7727975637666324, 'reg__min_child_weight': 0.014206846091530693, 'reg__feature_fraction': 0.825636351803305, 'reg__bagging_fraction': 0.7909702377151396, 'reg__bagging_freq': 1, 'reg__max_bin': 80, 'reg__min_data_per_group': 81}

Rank 5: {'reg__n_estimators': 300, 'reg__max_depth': 4, 'reg__learning_rate': 0.07736719112195078, 'reg__num_leaves': 56, 'reg__subsample': 0.5958595964900457, 'reg__colsample_bytree': 0.5642152612302639, 'reg__min_child_samples': 25, 'reg__reg_alpha': 0.22106179735854314, 'reg__reg_lambda': 13.323269708069247, 'reg__min_split_gain': 2.868719368587297, 'reg__path_smooth': 3.185090473968838, 'reg__min_child_weight': 0.004712584505723946, 'reg__feature_fraction': 0.8104650677791043, 'reg__bagging_fraction': 0.7896483696774605, 'reg__bagging_freq': 1, 'reg__max_bin': 136, 'reg__min_data_per_group': 61}

Rank 6: {'reg__n_estimators': 300, 'reg__max_depth': 8, 'reg__learning_rate': 0.07372972361688697, 'reg__num_leaves': 53, 'reg__subsample': 0.5952973444608188, 'reg__colsample_bytree': 0.5820520923038789, 'reg__min_child_samples': 22, 'reg__reg_alpha': 1.5304333581743679, 'reg__reg_lambda': 12.53727048472958, 'reg__min_split_gain': 2.6957207007826467, 'reg__path_smooth': 1.8087401921476736, 'reg__min_child_weight': 0.00453956747195751, 'reg__feature_fraction': 0.8213762221905722, 'reg__bagging_fraction': 0.7584367296673866, 'reg__bagging_freq': 1, 'reg__max_bin': 96, 'reg__min_data_per_group': 60}

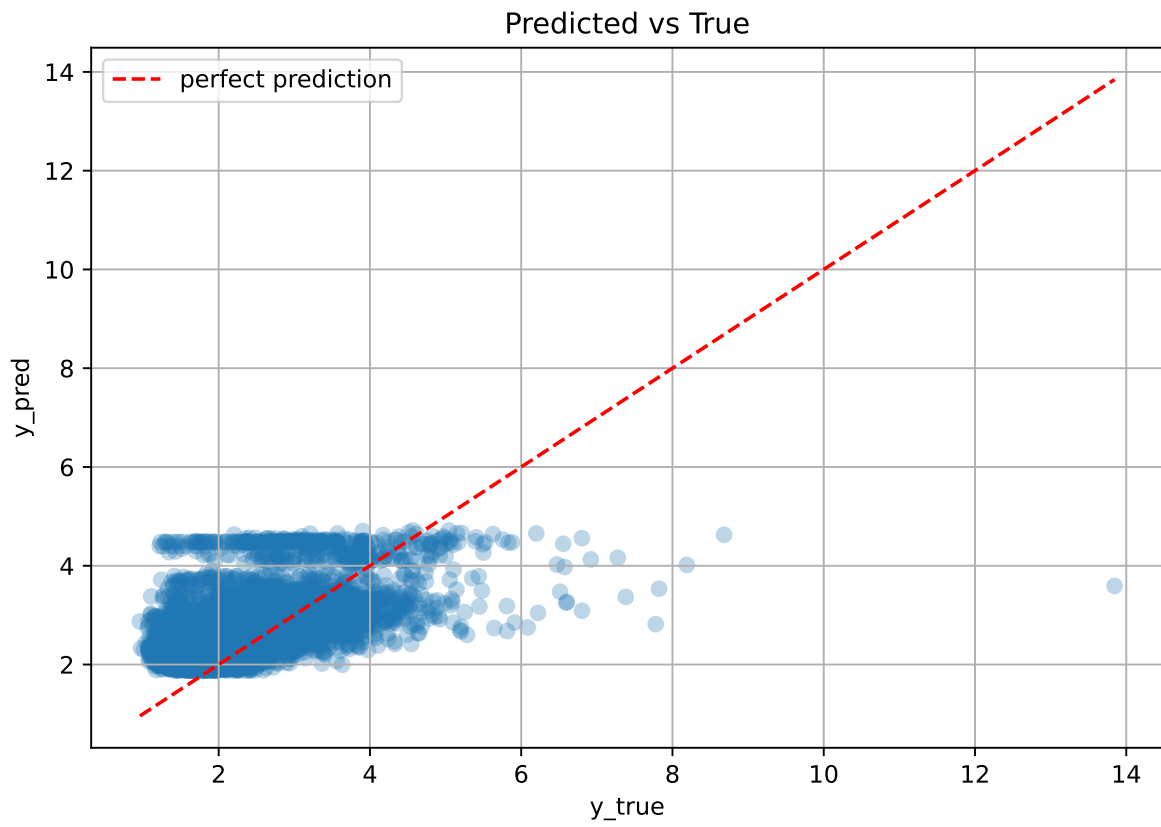
Rank 7: {'reg__n_estimators': 300, 'reg__max_depth': 8, 'reg__learning_rate': 0.07613517668707824, 'reg__num_leaves': 53, 'reg__subsample': 0.5989146970166975, 'reg__colsample_bytree': 0.5640507613373246, 'reg__min_child_samples': 25, 'reg__reg_alpha': 1.3636353574523445, 'reg__reg_lambda': 12.375687283862511, 'reg__min_split_gain': 2.818342398814676, 'reg__path_smooth': 1.8321943663011004, 'reg__min_child_weight': 0.004216134280158433, 'reg__feature_fraction': 0.8101646237648182, 'reg__bagging_fraction': 0.7600103653060383, 'reg__bagging_freq': 1, 'reg__max_bin': 138, 'reg__min_data_per_group': 64}

Rank 8: {'reg__n_estimators': 300, 'reg__max_depth': 8, 'reg__learning_rate': 0.07371545718908097, 'reg__num_leaves': 53, 'reg__subsample': 0.5984654397096989, 'reg__colsample_bytree': 0.5747382518790992, 'reg__min_child_samples': 22, 'reg__reg_alpha': 1.3924045539918035, 'reg__reg_lambda': 12.89962257390661, 'reg__min_split_gain': 2.7651979538709166, 'reg__path_smooth': 1.8503590257922995, 'reg__min_child_weight': 0.013673379884235556, 'reg__feature_fraction': 0.8123645787598447, 'reg__bagging_fraction': 0.7574168383348555, 'reg__bagging_freq': 1, 'reg__max_bin': 98, 'reg__min_data_per_group': 60}

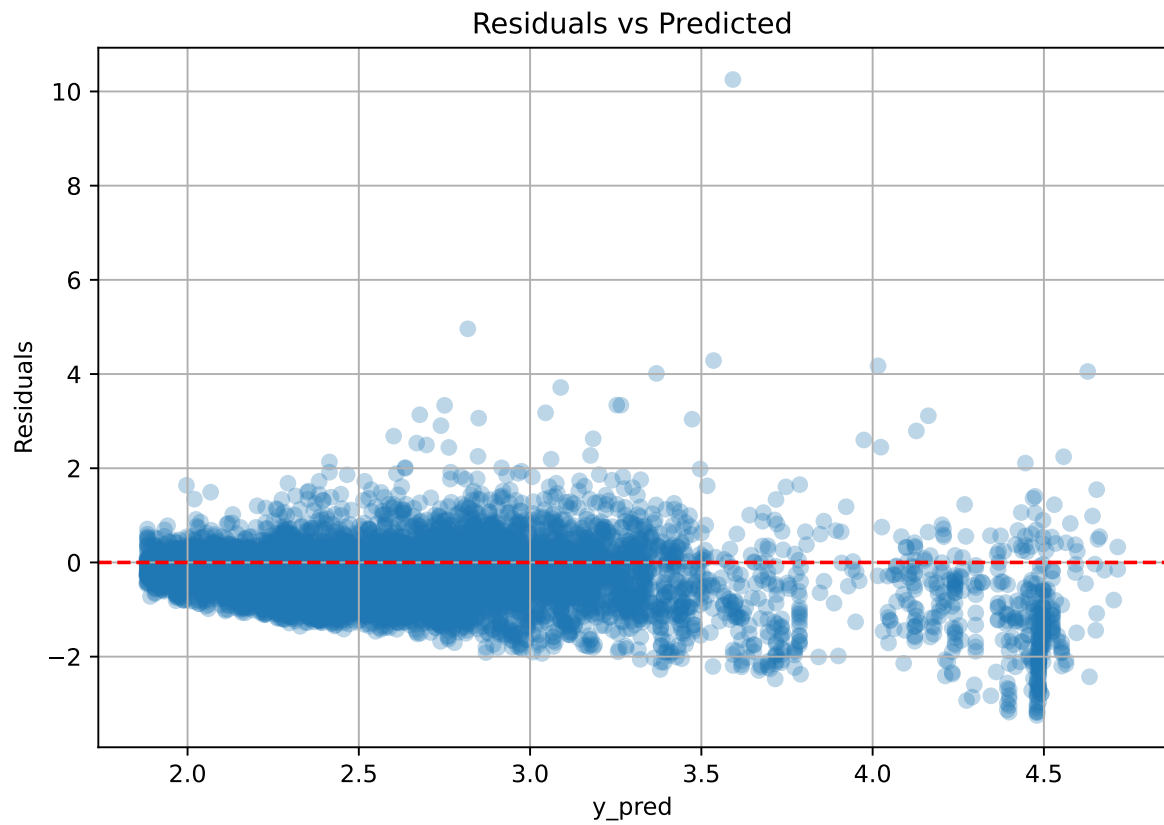
Rank 9: {'reg__n_estimators': 300, 'reg__max_depth': 8, 'reg__learning_rate': 0.08323408354298407, 'reg__num_leaves': 54, 'reg__subsample': 0.6866423209245018, 'reg__colsample_bytree': 0.566852925649022, 'reg__min_child_samples': 23, 'reg__reg_alpha': 1.4500740249216981, 'reg__reg_lambda': 13.301805525744893, 'reg__min_split_gain': 2.8606788413737987, 'reg__path_smooth': 1.8139669118920767, 'reg__min_child_weight': 0.014159288882161397, 'reg__feature_fraction': 0.8177073415311129, 'reg__bagging_fraction': 0.7653871715580096, 'reg__bagging_freq': 1, 'reg__max_bin': 139, 'reg__min_data_per_group': 60}

Rank 10: {'reg__n_estimators': 300, 'reg__max_depth': 4, 'reg__learning_rate': 0.023630051105476745, 'reg__num_leaves': 52, 'reg__subsample': 0.6905732511577143, 'reg__colsample_bytree': 0.504439811752045, 'reg__min_child_samples': 23, 'reg__reg_alpha': 0.9539343485776355, 'reg__reg_lambda': 12.306818019282153, 'reg__min_split_gain': 2.4421723459585207, 'reg__path_smooth': 2.641204191584388, 'reg__min_child_weight': 0.003565816094010682, 'reg__feature_fraction': 0.8814409079870101, 'reg__bagging_fraction': 0.657271784489807, 'reg__bagging_freq': 1, 'reg__max_bin': 139, 'reg__min_data_per_group': 64}

2.3.2 Scatter



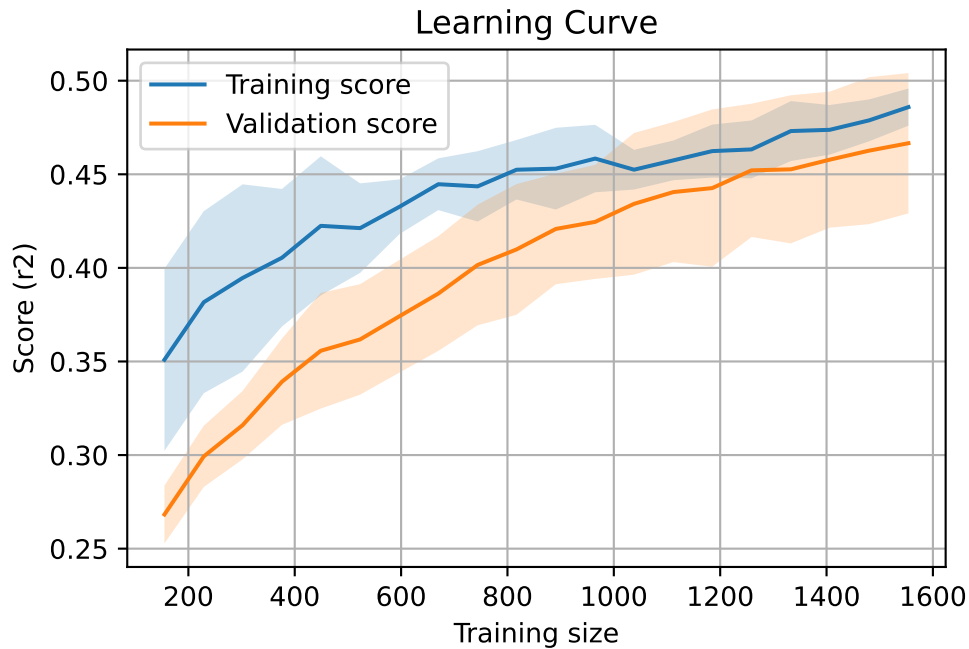
2.3.3 Scatter residuals



2.3.4 Learning curve — Group: 2025_Jun

`Len(group)` : 3172, `Len(training)` : 2221,

`Len(training_real)` : 1555, `Len(test_of_training)` : 666, `Len(test)` : 951

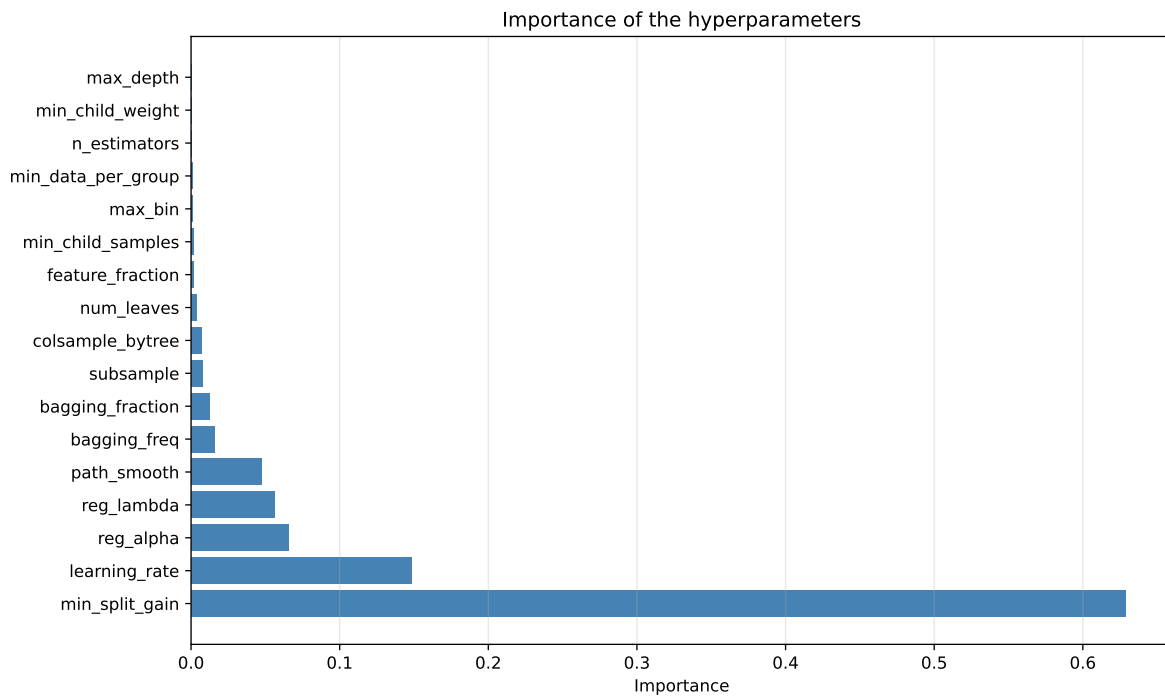
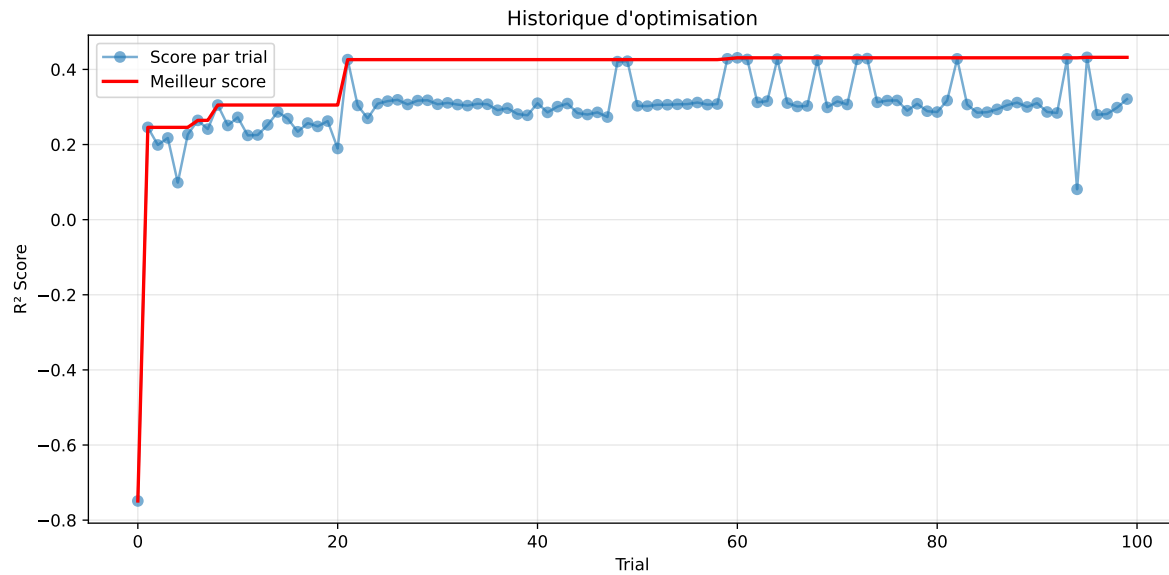


2.3.5 Gridsearch — Group: 2025_Sep

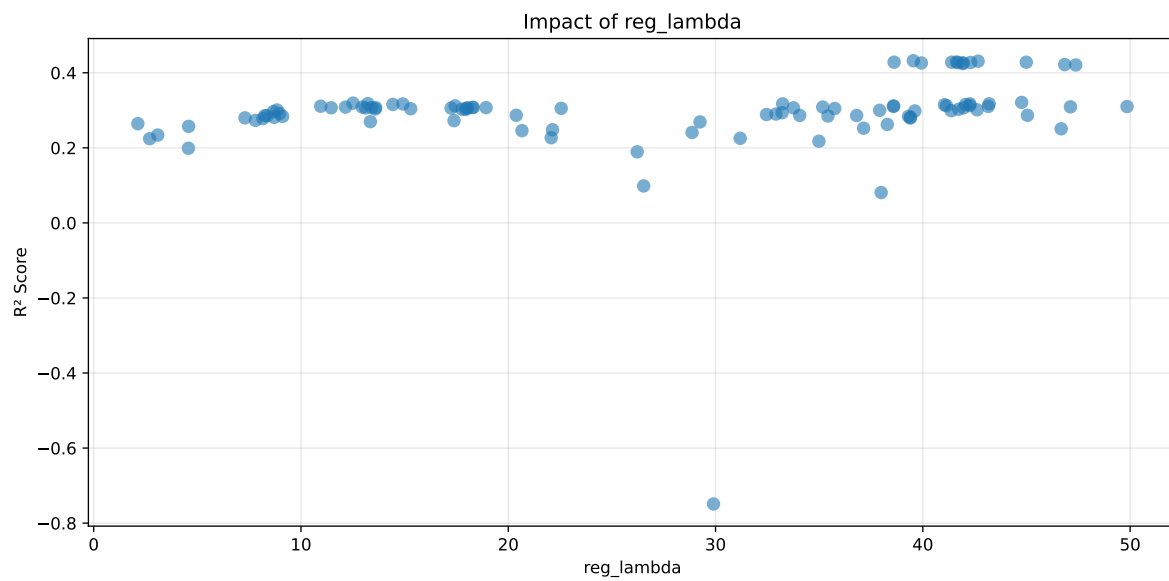
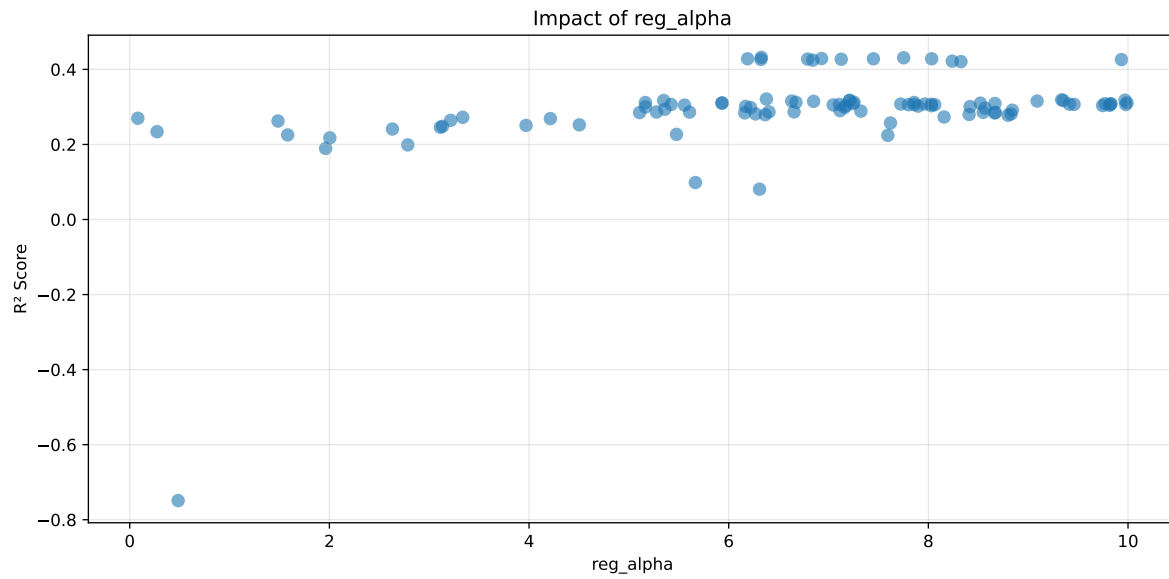
Len(group) : 4996, Len(training) : 3498, Len(test) : 1498

Distribution y_train vs y_test: y_train: mean=2.160, std=0.756, min=0.973, max=8.190
y_test: mean=2.160, std=0.777, min=0.959, max=7.779

===== TOP Importance FEATURES =====
feature importance r_umidity 141 r_NH3 85 r_CO2 57 r_THI 36 r_temperature 35

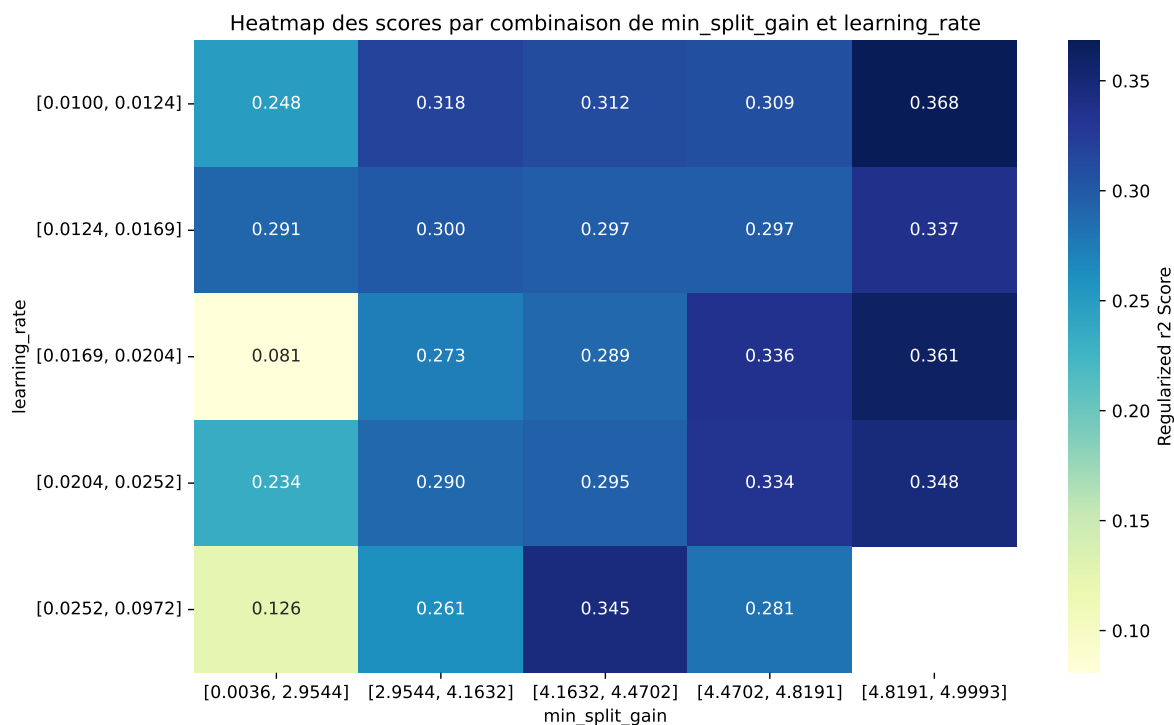


Paramètres avec >5% d'importance : ['min_split_gain', 'learning_rate', 'reg_alpha', 'reg_lambda']



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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.4532	0.4321	0.4167	0.023	0.4321	0.4347	0.0026	1.0059	1.0487
2	0.451	0.4309	0.4155	0.0248	0.4309	0.4265	-0.0044	0.9897	1.0465
3	0.4502	0.4288	0.4147	0.0202	0.4288	0.4258	-0.0031	0.9929	1.0499
4	0.4476	0.4283	0.4126	0.0227	0.4283	0.4234	-0.0049	0.9886	1.045
5	0.4485	0.4282	0.4129	0.0193	0.4282	0.4268	-0.0014	0.9967	1.0473
6	0.4493	0.4282	0.4132	0.0219	0.4282	0.4257	-0.0025	0.9941	1.0492
7	0.4477	0.4275	0.4131	0.0194	0.4275	0.4201	-0.0074	0.9827	1.0474
8	0.4483	0.427	0.413	0.02	0.427	0.4275	0.0005	1.0012	1.0497
9	0.4459	0.4266	0.4105	0.0239	0.4266	0.4255	-0.0011	0.9974	1.0453
10	0.4455	0.4261	0.4105	0.0208	0.4261	0.4283	0.0022	1.0053	1.0457
---	---	---	---	---	---	---	---	---	---
100	0.8085	0.5489	0.5646	0.0158	-0.749	0.5748	0.0259	1.0472	1.4729

Hyperparameters:

Rank 1: {'reg__n_estimators': 500, 'reg__max_depth': 7, 'reg__learning_rate': 0.025262942467231188, 'reg__num_leaves': 22, 'reg__subsample': 0.5667115825744952, 'reg__colsample_bytree': 0.8747351023104203, 'reg__min_child_samples': 18, 'reg__reg_alpha': 6.327478788941173, 'reg__reg_lambda': 39.54194721709057, 'reg__min_split_gain': 4.31139597098672, 'reg__path_smooth': 3.3403712854766767, 'reg__min_child_weight':

0.08684471756048902, 'reg__feature_fraction': 0.8718944346896709, 'reg__bagging_fraction': 0.5165414836799199, 'reg__bagging_freq': 7, 'reg__max_bin': 159, 'reg__min_data_per_group': 63}

Rank 2: {'reg__n_estimators': 600, 'reg__max_depth': 7, 'reg__learning_rate': 0.01763735354440072, 'reg__num_leaves': 29, 'reg__subsample': 0.5023054343254034, 'reg__colsample_bytree': 0.7184521600448047, 'reg__min_child_samples': 20, 'reg__reg_alpha': 7.752463343609523, 'reg__reg_lambda': 42.667390973136804, 'reg__min_split_gain': 4.785310718207181, 'reg__path_smooth': 4.0996370178124195, 'reg__min_child_weight': 0.26054994392960124, 'reg__feature_fraction': 0.7609321916338202, 'reg__bagging_fraction': 0.5600885867506759, 'reg__bagging_freq': 7, 'reg__max_bin': 129, 'reg__min_data_per_group': 76}

Rank 3: {'reg__n_estimators': 600, 'reg__max_depth': 8, 'reg__learning_rate': 0.02211730814527345, 'reg__num_leaves': 29, 'reg__subsample': 0.5317215074317577, 'reg__colsample_bytree': 0.8299792863812434, 'reg__min_child_samples': 15, 'reg__reg_alpha': 6.9303314741699875, 'reg__reg_lambda': 41.62690971833305, 'reg__min_split_gain': 4.963303617457054, 'reg__path_smooth': 3.8083815512345183, 'reg__min_child_weight': 0.3030412650329742, 'reg__feature_fraction': 0.7174217355764644, 'reg__bagging_fraction': 0.5821668679379506, 'reg__bagging_freq': 7, 'reg__max_bin': 127, 'reg__min_data_per_group': 74}

Rank 4: {'reg__n_estimators': 600, 'reg__max_depth': 7, 'reg__learning_rate': 0.018738981604161285, 'reg__num_leaves': 60, 'reg__subsample': 0.7270756155781871, 'reg__colsample_bytree': 0.8261036194650995, 'reg__min_child_samples': 21, 'reg__reg_alpha': 8.033208102701533, 'reg__reg_lambda': 41.392382933847685, 'reg__min_split_gain': 4.967815334950907, 'reg__path_smooth': 4.171088001084497, 'reg__min_child_weight': 0.2729928934677018, 'reg__feature_fraction': 0.7612988321349802, 'reg__bagging_fraction': 0.5643976420870348, 'reg__bagging_freq': 7, 'reg__max_bin': 105, 'reg__min_data_per_group': 121}

Rank 5: {'reg__n_estimators': 500, 'reg__max_depth': 7, 'reg__learning_rate': 0.019512367836652486, 'reg__num_leaves': 52, 'reg__subsample': 0.5523833574947542, 'reg__colsample_bytree': 0.873086258015852, 'reg__min_child_samples': 12, 'reg__reg_alpha': 6.189255446519626, 'reg__reg_lambda': 38.6146523029963, 'reg__min_split_gain': 4.8271558130186865, 'reg__path_smooth': 4.293915362257807, 'reg__min_child_weight': 0.10305588373545446, 'reg__feature_fraction': 0.815560267120122, 'reg__bagging_fraction': 0.5266377312896322, 'reg__bagging_freq': 7, 'reg__max_bin': 118, 'reg__min_data_per_group': 63}

Rank 6: {'reg__n_estimators': 500, 'reg__max_depth': 7, 'reg__learning_rate': 0.022622494221606816, 'reg__num_leaves': 25, 'reg__subsample': 0.5294521665625112, 'reg__colsample_bytree': 0.8419911496204229, 'reg__min_child_samples': 36, 'reg__reg_alpha': 7.449719074451861, 'reg__reg_lambda': 44.99165673695443, 'reg__min_split_gain': 4.512262730237249, 'reg__path_smooth': 3.5444964233549405, 'reg__min_child_weight':

0.6845502449800427, 'reg__feature_fraction': 0.8388723862950089, 'reg__bagging_fraction': 0.5409715393129176, 'reg__bagging_freq': 7, 'reg__max_bin': 143, 'reg__min_data_per_group': 134}

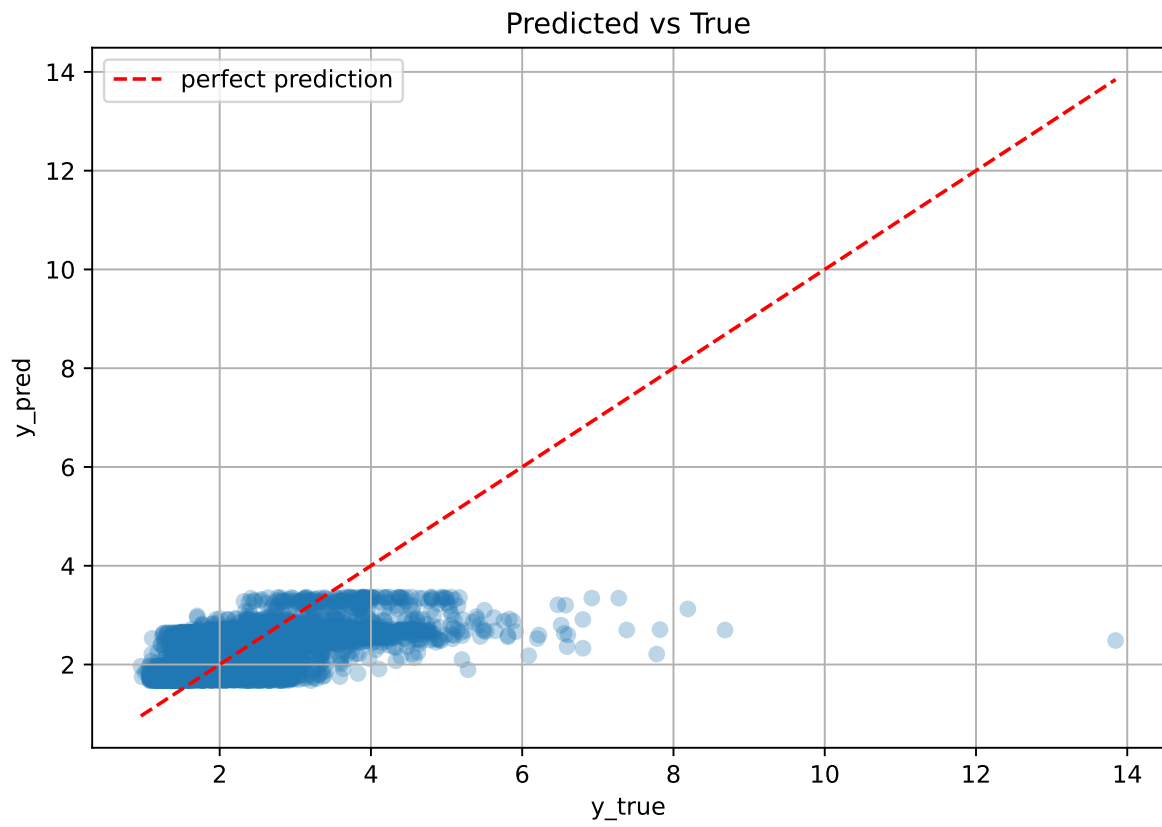
Rank 7: {'reg__n_estimators': 600, 'reg__max_depth': 8, 'reg__learning_rate': 0.012359730518360682, 'reg__num_leaves': 60, 'reg__subsample': 0.5046077007191683, 'reg__colsample_bytree': 0.8182316838701359, 'reg__min_child_samples': 21, 'reg__reg_alpha': 6.791127075509913, 'reg__reg_lambda': 42.30567781089999, 'reg__min_split_gain': 4.9010708934895195, 'reg__path_smooth': 4.133610142501154, 'reg__min_child_weight': 5.687978593425851, 'reg__feature_fraction': 0.7104611619636224, 'reg__bagging_fraction': 0.539137582898578, 'reg__bagging_freq': 7, 'reg__max_bin': 129, 'reg__min_data_per_group': 121}

Rank 8: {'reg__n_estimators': 600, 'reg__max_depth': 8, 'reg__learning_rate': 0.021958191567668543, 'reg__num_leaves': 27, 'reg__subsample': 0.5045246042565203, 'reg__colsample_bytree': 0.8294005444876781, 'reg__min_child_samples': 18, 'reg__reg_alpha': 7.127380835339795, 'reg__reg_lambda': 41.69644364592366, 'reg__min_split_gain': 4.9938202246052175, 'reg__path_smooth': 3.826897548158837, 'reg__min_child_weight': 0.33146332880380186, 'reg__feature_fraction': 0.8485666072549568, 'reg__bagging_fraction': 0.579985419562551, 'reg__bagging_freq': 7, 'reg__max_bin': 124, 'reg__min_data_per_group': 76}

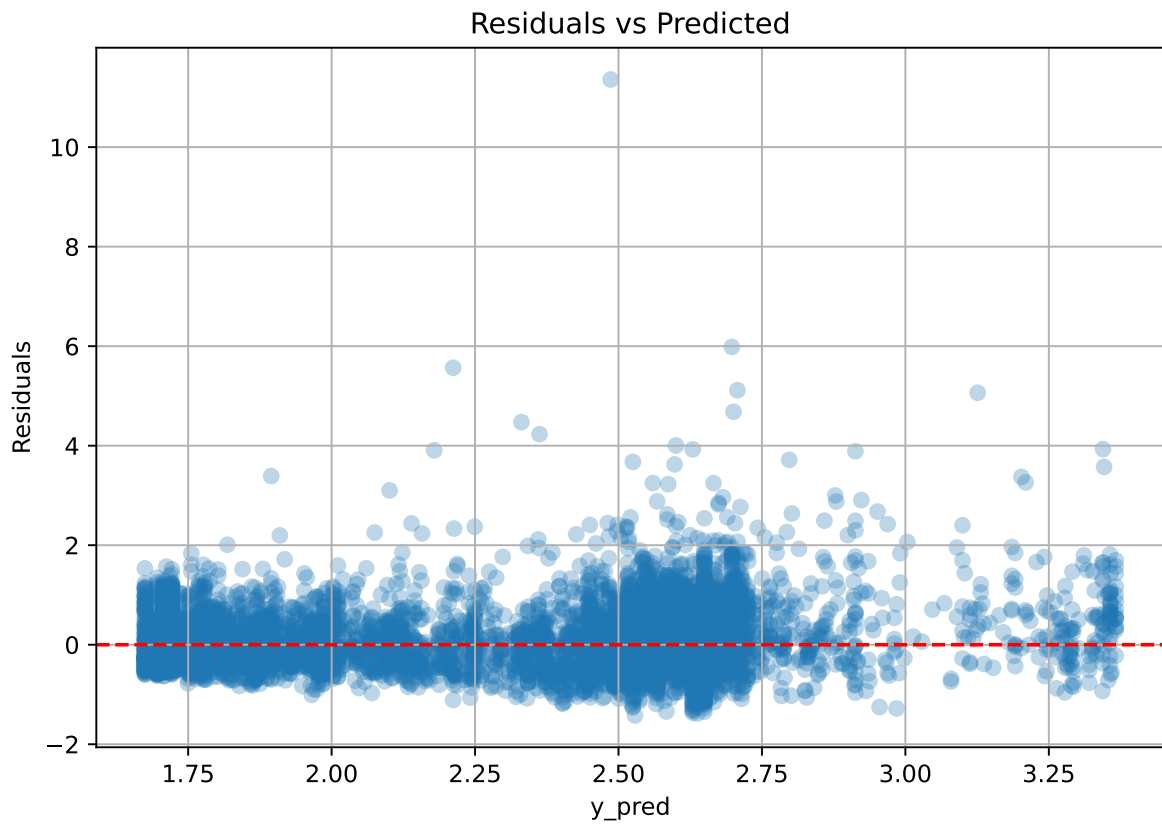
Rank 9: {'reg__n_estimators': 600, 'reg__max_depth': 8, 'reg__learning_rate': 0.018136900851613864, 'reg__num_leaves': 60, 'reg__subsample': 0.6508707346832967, 'reg__colsample_bytree': 0.8260409261028605, 'reg__min_child_samples': 34, 'reg__reg_alpha': 6.322374652422061, 'reg__reg_lambda': 41.94095907857718, 'reg__min_split_gain': 4.999347058842909, 'reg__path_smooth': 4.165791701250229, 'reg__min_child_weight': 0.2736383726813054, 'reg__feature_fraction': 0.7560990987024667, 'reg__bagging_fraction': 0.5609785318682999, 'reg__bagging_freq': 7, 'reg__max_bin': 129, 'reg__min_data_per_group': 124}

Rank 10: {'reg__n_estimators': 800, 'reg__max_depth': 3, 'reg__learning_rate': 0.010092460464849115, 'reg__num_leaves': 63, 'reg__subsample': 0.8798363802789787, 'reg__colsample_bytree': 0.5042152394233772, 'reg__min_child_samples': 22, 'reg__reg_alpha': 9.934395707557826, 'reg__reg_lambda': 39.93045120170348, 'reg__min_split_gain': 3.94357212787697, 'reg__path_smooth': 4.740058907700259, 'reg__min_child_weight': 0.0010186875909021737, 'reg__feature_fraction': 0.7978287713418877, 'reg__bagging_fraction': 0.5090241723925121, 'reg__bagging_freq': 7, 'reg__max_bin': 78, 'reg__min_data_per_group': 100}

2.3.6 Scatter



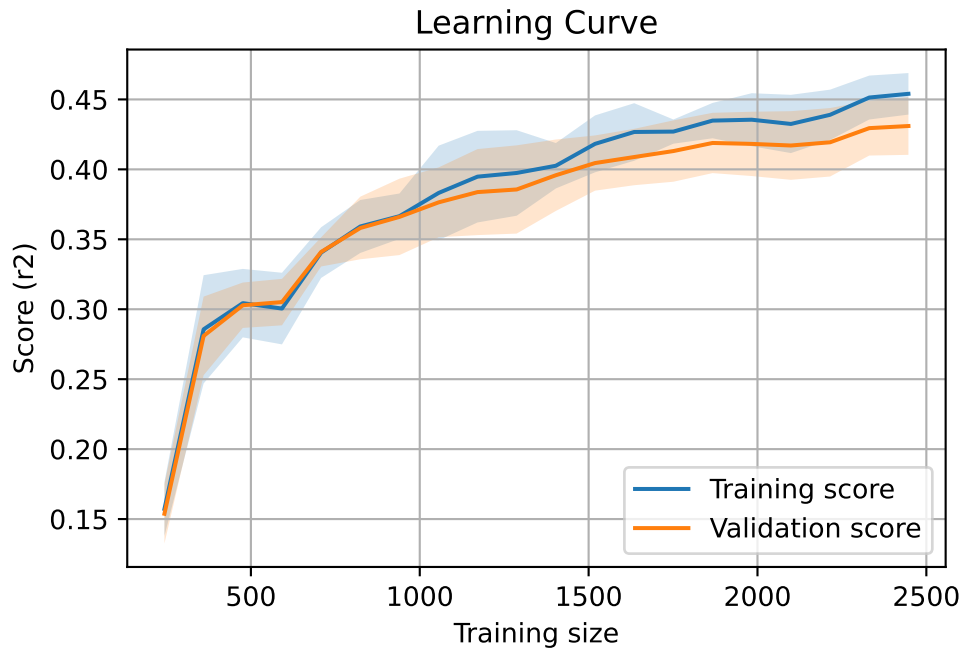
2.3.7 Scatter residuals



2.3.8 Learning curve — Group: 2025_Sep

`Len(group)` : 4996, `Len(training)` : 3498,

`Len(training_real)` : 2449, `Len(test_of_training)` : 1049, `Len(test)` : 1498



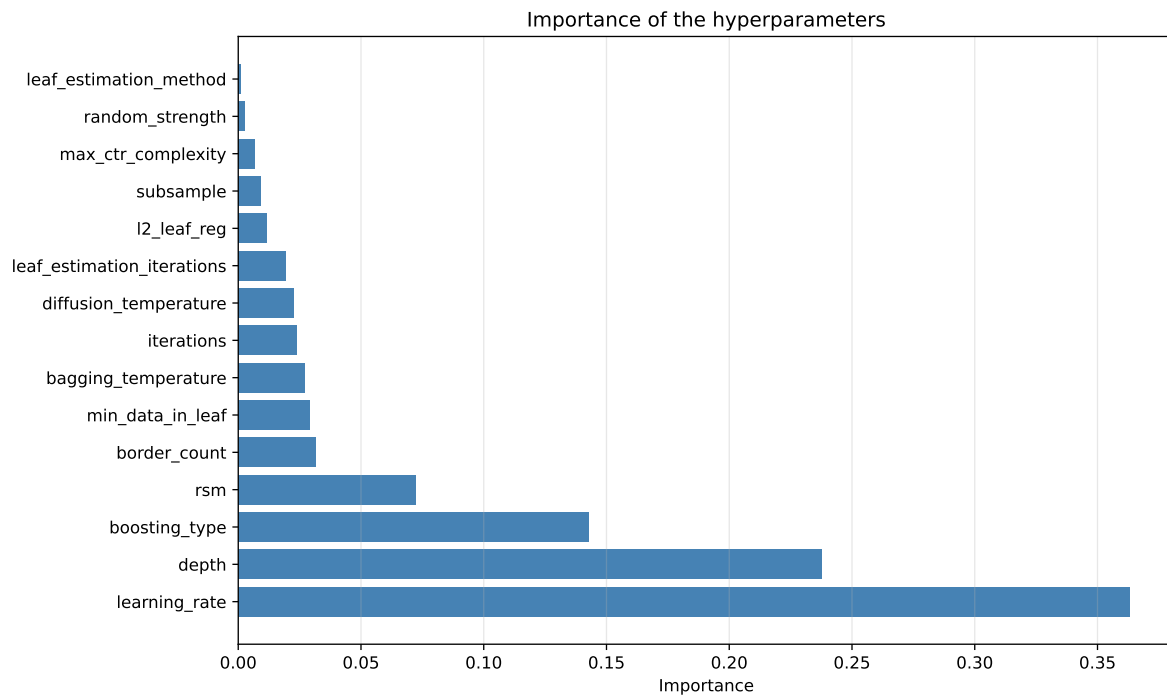
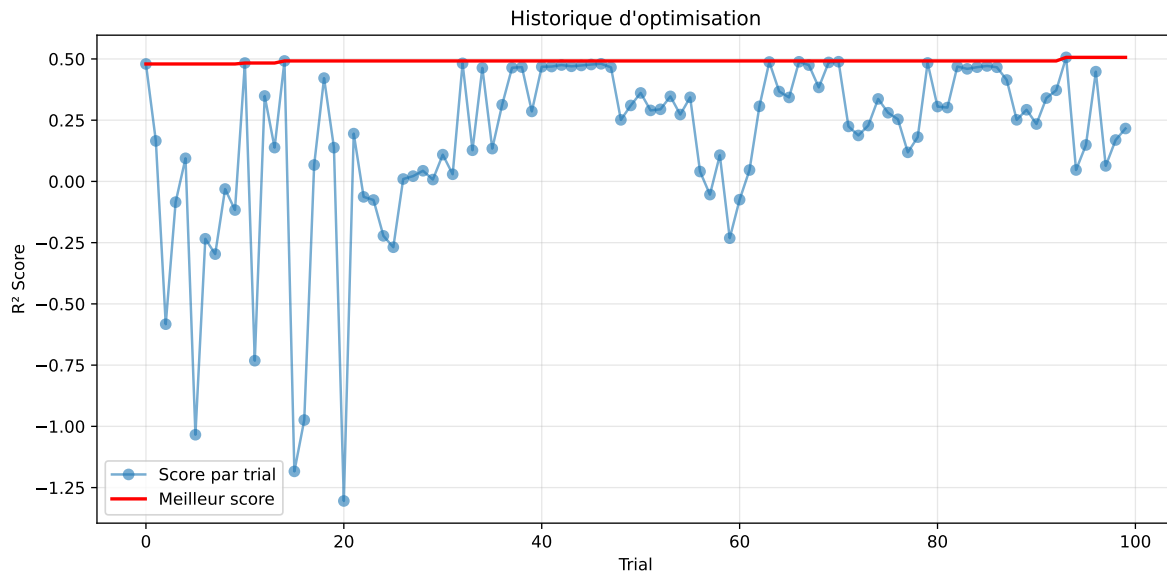
2.4 Model : CatBoost

2.4.1 Gridsearch — Group: 2025_Jun

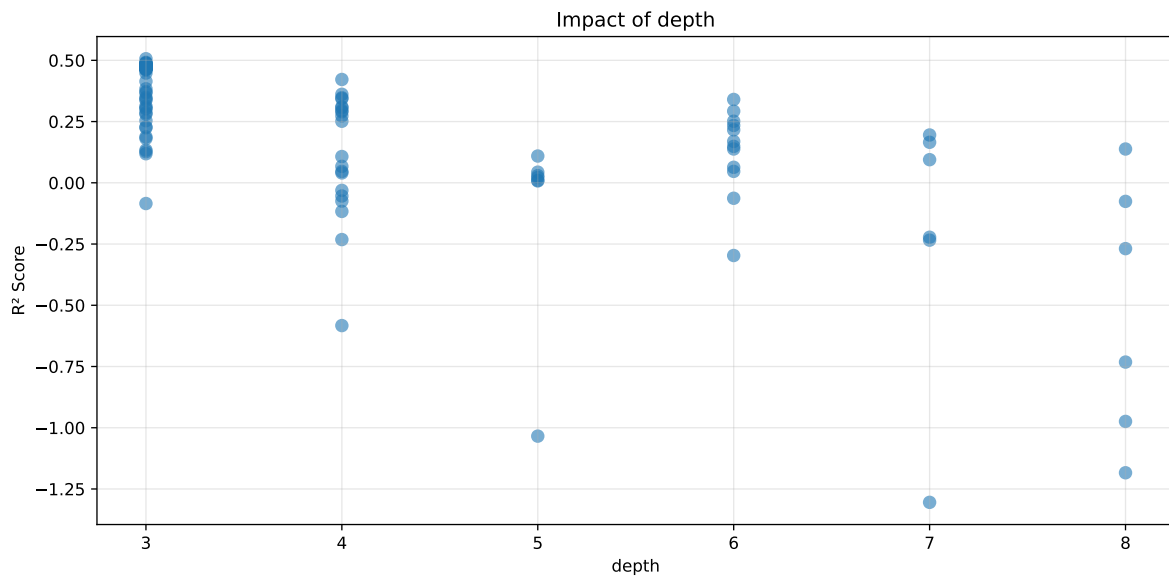
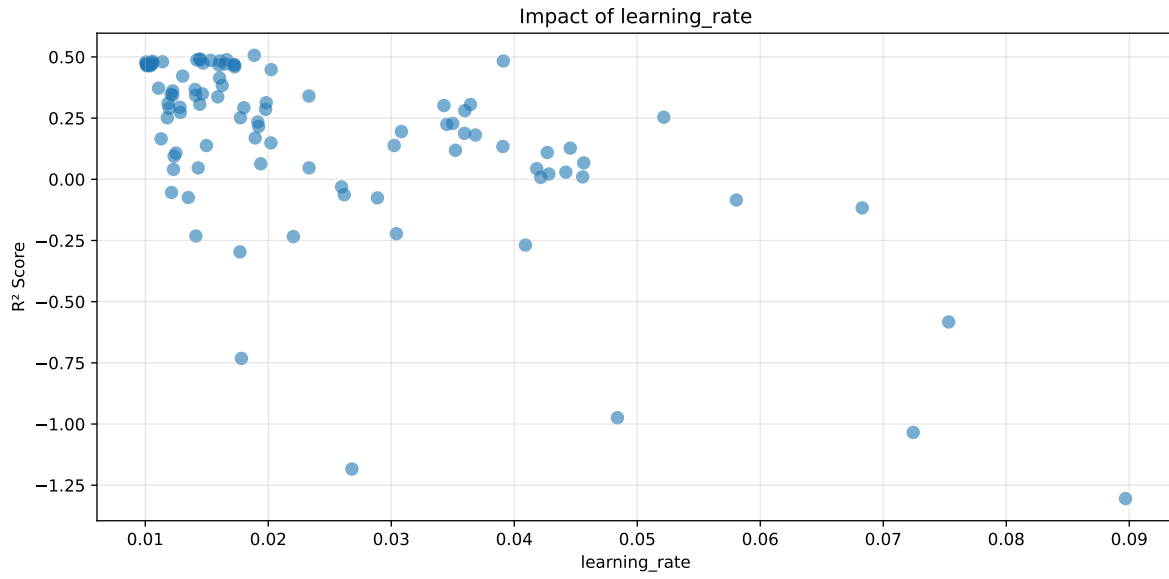
Len(group) : 3172, Len(training) : 2221, Len(test) : 951

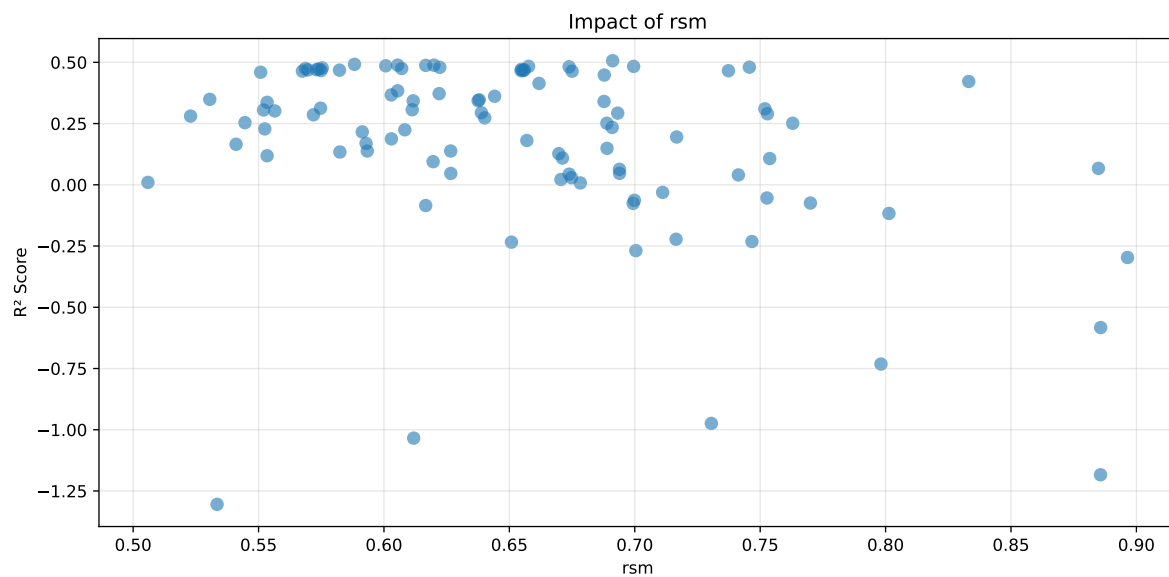
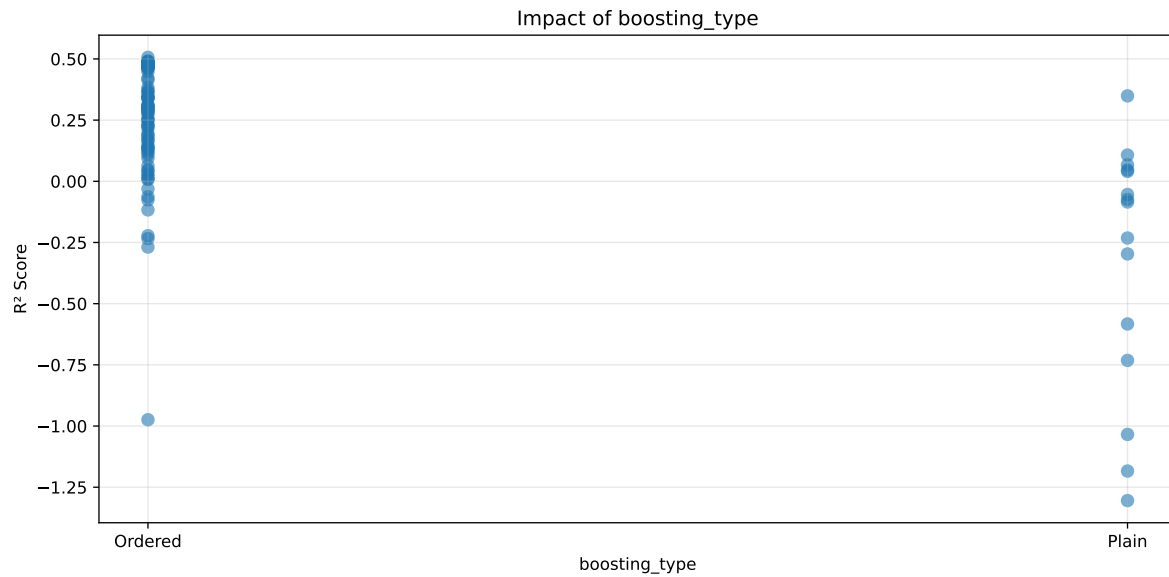
Distribution y_train vs y_test: y_train: mean=2.661, std=0.770, min=1.305, max=13.845
y_test: mean=2.621, std=0.704, min=1.253, max=5.627

===== TOP Importance FEATURES =====
feature importance r_umidity 20.921864 r_NH3 20.803980 r_CO2 20.168037 r_temperature
19.482174 r_THI 18.623945



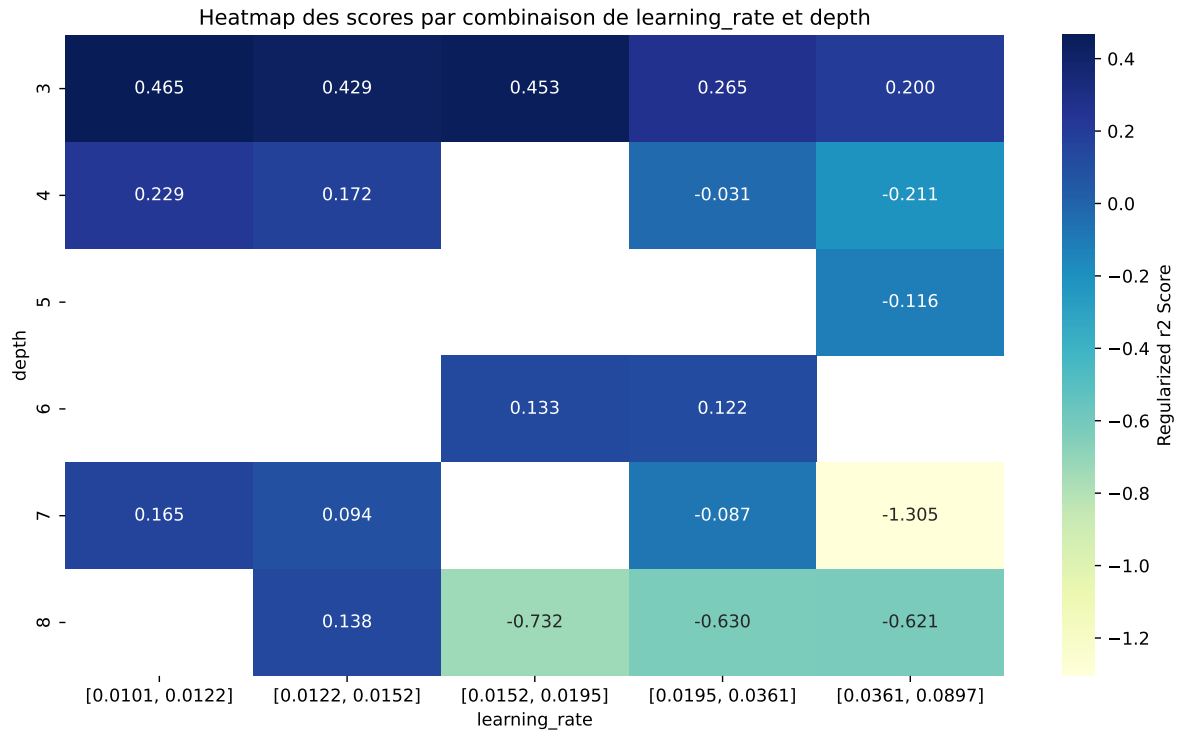
Paramètres avec >5% d'importance : ['learning_rate', 'depth', 'boosting_type', 'rsm']





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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5306	0.5065	0.5327	0.0322	0.5065	0.549	0.0425	1.0839	1.0477
2	0.5137	0.4918	0.5178	0.0354	0.4918	0.5301	0.0383	1.0778	1.0445
3	0.5107	0.4887	0.5208	0.0404	0.4887	0.5368	0.0481	1.0985	1.045
4	0.5067	0.4884	0.517	0.0383	0.4884	0.5335	0.0451	1.0923	1.0375
5	0.5053	0.4875	0.518	0.0395	0.4875	0.5343	0.0468	1.0959	1.0365
6	0.5076	0.4858	0.5103	0.036	0.4858	0.5276	0.0418	1.0859	1.0448
7	0.501	0.4836	0.518	0.0392	0.4836	0.5353	0.0517	1.1068	1.0359
8	0.5025	0.4834	0.515	0.0396	0.4834	0.5316	0.0483	1.0998	1.0396
9	0.4963	0.482	0.5114	0.0317	0.482	0.5223	0.0404	1.0837	1.0297
10	0.4919	0.4802	0.51	0.029	0.4802	0.5194	0.0392	1.0817	1.0244
---	---	---	---	---	---	---	---	---	---
100	0.9444	0.5696	0.5577	0.0231	-1.3046	0.5806	0.011	1.0193	1.6581

Hyperparameters:

Rank 1: {'reg__iterations': 800, 'reg__depth': 3, 'reg__learning_rate': 0.018857855355868185, 'reg__l2_leaf_reg': 9.4636282783914, 'reg__bagging_temperature': 0.897563993039988, 'reg__random_strength': 0.6727667802115895, 'reg__subsample': 0.782946767880635, 'reg__min_data_in_leaf': 31, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.6911679347975777, 'reg__border_count': 39, 'reg__leaf_estimation_method': 'Gradient',

'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 1225.1224443682222}

Rank 2: {'reg__iterations': 900, 'reg__depth': 3, 'reg__learning_rate': 0.014414507997873824, 'reg__l2_leaf_reg': 42.783082592306066, 'reg__bagging_temperature': 2.899337300245608, 'reg__random_strength': 3.688042288698316, 'reg__subsample': 0.6382045243366601, 'reg__min_data_in_leaf': 27, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.5882455368556917, 'reg__border_count': 78, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 2032.949688206368}

Rank 3: {'reg__iterations': 700, 'reg__depth': 3, 'reg__learning_rate': 0.016617861400332647, 'reg__l2_leaf_reg': 6.957254522756713, 'reg__bagging_temperature': 1.5382754220179973, 'reg__random_strength': 1.1747961135168572, 'reg__subsample': 0.6959008003420895, 'reg__min_data_in_leaf': 30, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.6054646831096034, 'reg__border_count': 34, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 984.2134935307213}

Rank 4: {'reg__iterations': 900, 'reg__depth': 3, 'reg__learning_rate': 0.01449623889261608, 'reg__l2_leaf_reg': 10.89131685802679, 'reg__bagging_temperature': 3.579320366178186, 'reg__random_strength': 3.189856483698514, 'reg__subsample': 0.827461921785512, 'reg__min_data_in_leaf': 30, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.6199422195107831, 'reg__border_count': 32, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 952.9060338871503}

Rank 5: {'reg__iterations': 900, 'reg__depth': 3, 'reg__learning_rate': 0.01419396994313518, 'reg__l2_leaf_reg': 11.001596059091126, 'reg__bagging_temperature': 0.5644007956539927, 'reg__random_strength': 3.0649914956304443, 'reg__subsample': 0.7691808929200953, 'reg__min_data_in_leaf': 30, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.6166674778395571, 'reg__border_count': 32, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 935.8055467559766}

Rank 6: {'reg__iterations': 700, 'reg__depth': 3, 'reg__learning_rate': 0.015317609228009, 'reg__l2_leaf_reg': 11.824510944738266, 'reg__bagging_temperature': 0.6523214153424494, 'reg__random_strength': 1.9054066414036317, 'reg__subsample': 0.623395326450543, 'reg__min_data_in_leaf': 20, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.600694869531277, 'reg__border_count': 122, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 765.5223989262081}

Rank 7: {'reg__iterations': 700, 'reg__depth': 3, 'reg__learning_rate': 0.016065057162055522, 'reg__l2_leaf_reg': 7.553990828858469, 'reg__bagging_temperature': 1.3965384461118948, 'reg__random_strength': 1.0984237017889675, 'reg__subsample': 0.7003258323737076,

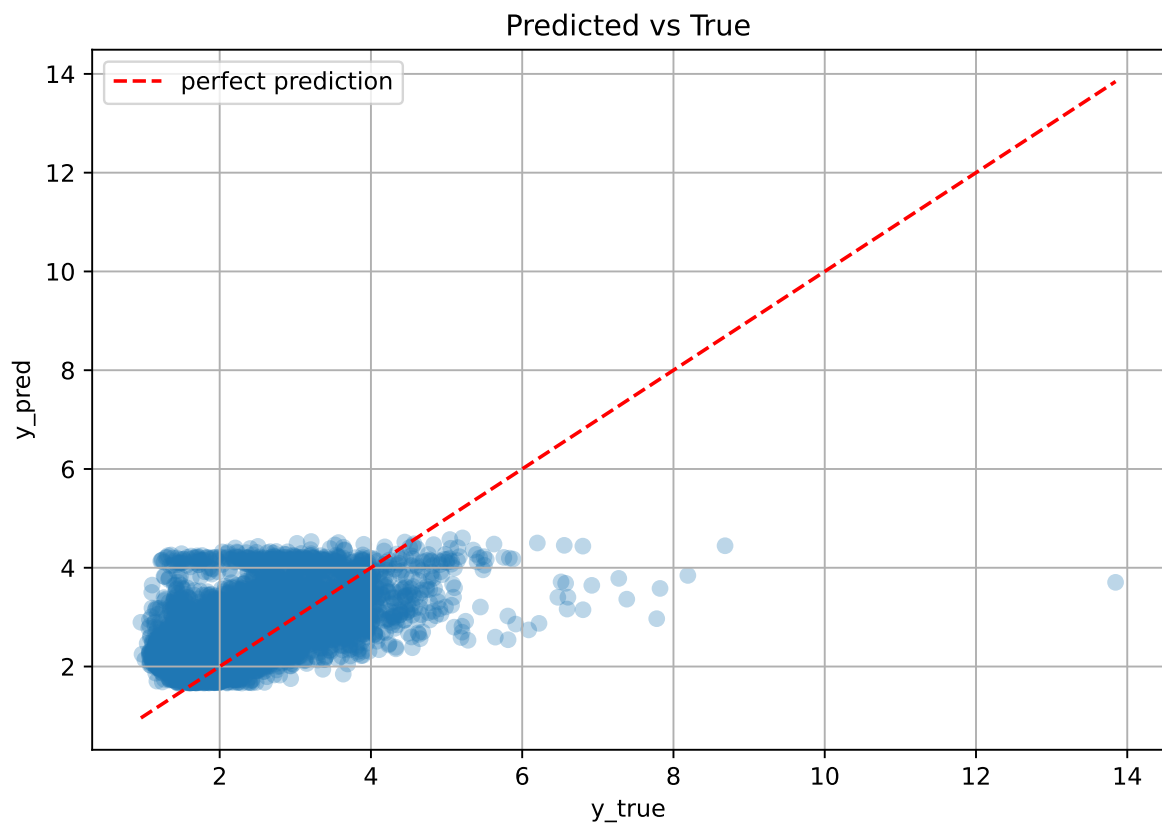
'reg__min_data_in_leaf': 31, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.6576709146795434, 'reg__border_count': 34, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 6141.051040068187}

Rank 8: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.03911113734001622, 'reg__l2_leaf_reg': 48.564343878513036, 'reg__bagging_temperature': 2.796090115483885, 'reg__random_strength': 2.090470190148073, 'reg__subsample': 0.6340472162583791, 'reg__min_data_in_leaf': 36, 'reg__leaf_estimation_iterations': 9, 'reg__rsm': 0.6995406551659292, 'reg__border_count': 37, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 316.32709956841086}

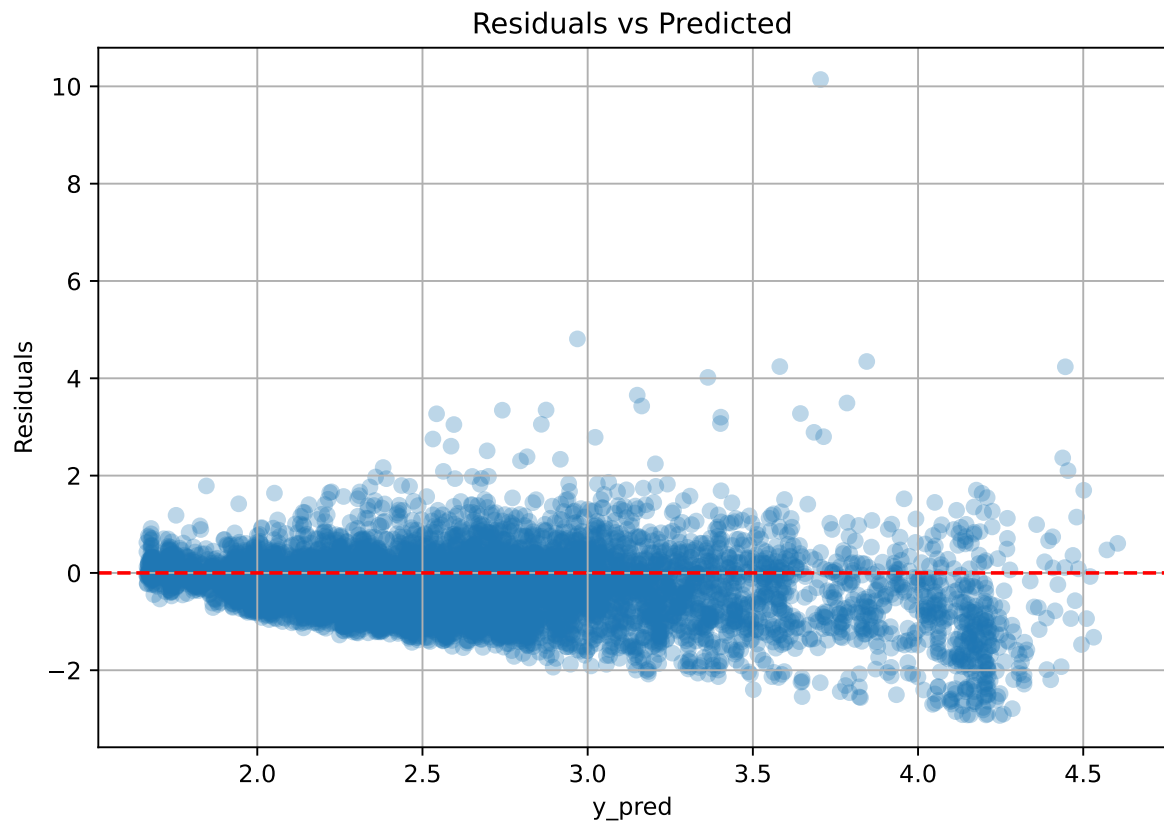
Rank 9: {'reg__iterations': 1000, 'reg__depth': 3, 'reg__learning_rate': 0.010590368228776046, 'reg__l2_leaf_reg': 1.2756079751696259, 'reg__bagging_temperature': 1.3223256033837194, 'reg__random_strength': 2.0786605685757875, 'reg__subsample': 0.8604457593500274, 'reg__min_data_in_leaf': 25, 'reg__leaf_estimation_iterations': 3, 'reg__rsm': 0.6738260293482045, 'reg__border_count': 48, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 1377.3727793331907}

Rank 10: {'reg__iterations': 900, 'reg__depth': 3, 'reg__learning_rate': 0.011401212393694774, 'reg__l2_leaf_reg': 3.36838000714271, 'reg__bagging_temperature': 2.777299524771923, 'reg__random_strength': 2.6059970034156206, 'reg__subsample': 0.8581548199219773, 'reg__min_data_in_leaf': 43, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.7456889598542791, 'reg__border_count': 43, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 1883.5091704693039}

2.4.2 Scatter



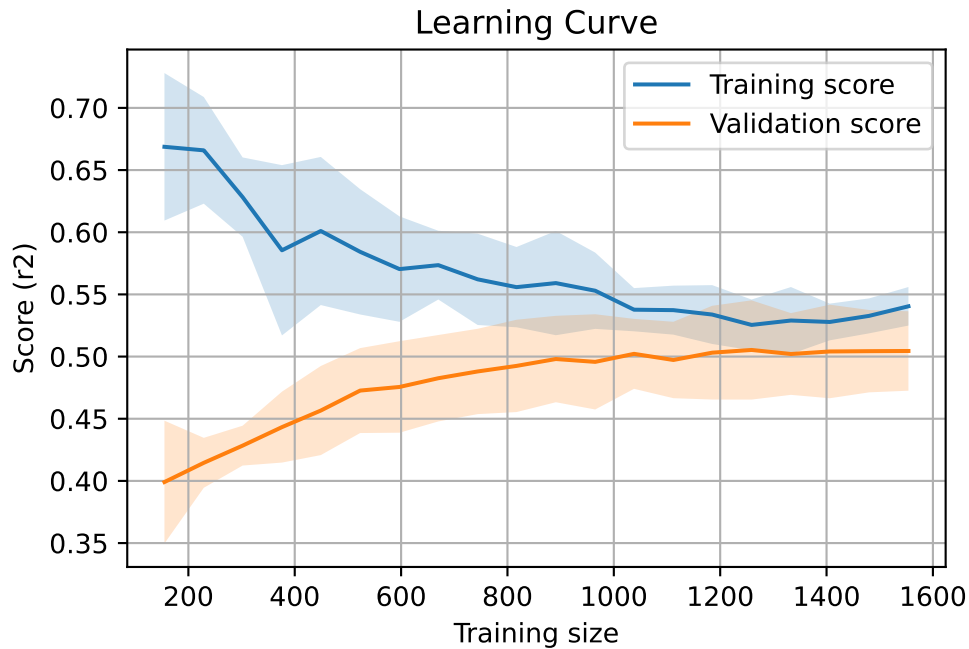
2.4.3 Scatter residuals



2.4.4 Learning curve — Group: 2025_Jun

`Len(group)` : 3172, `Len(training)` : 2221,

`Len(training_real)` : 1555, `Len(test_of_training)` : 666, `Len(test)` : 951

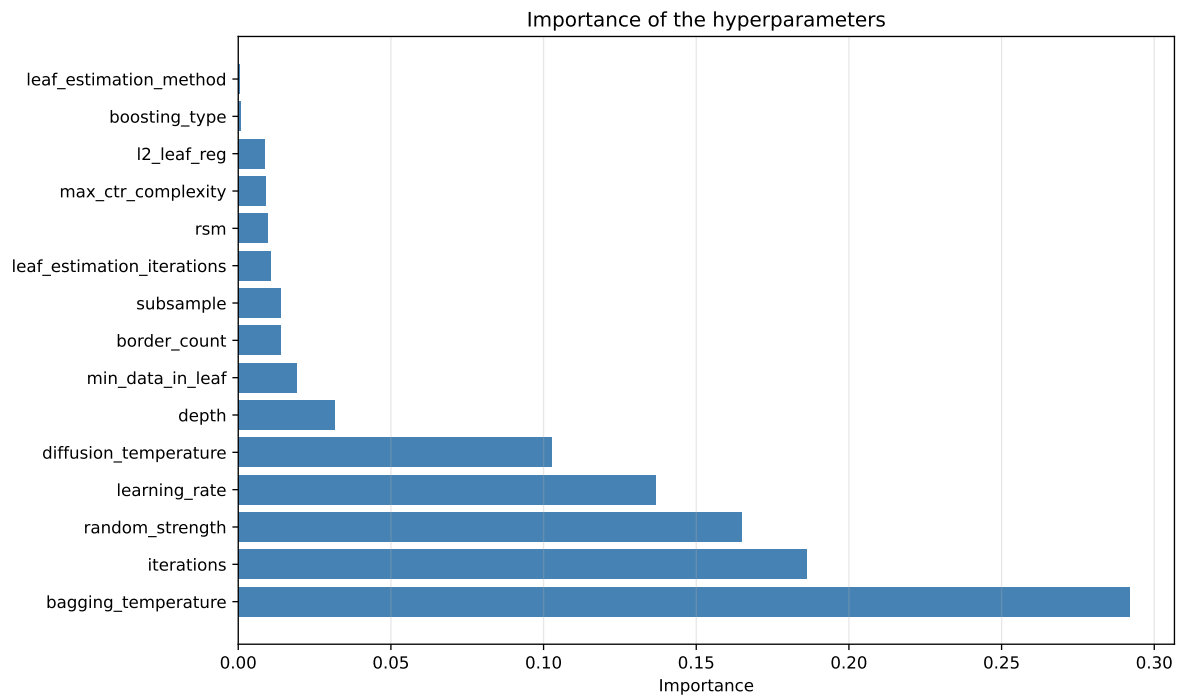
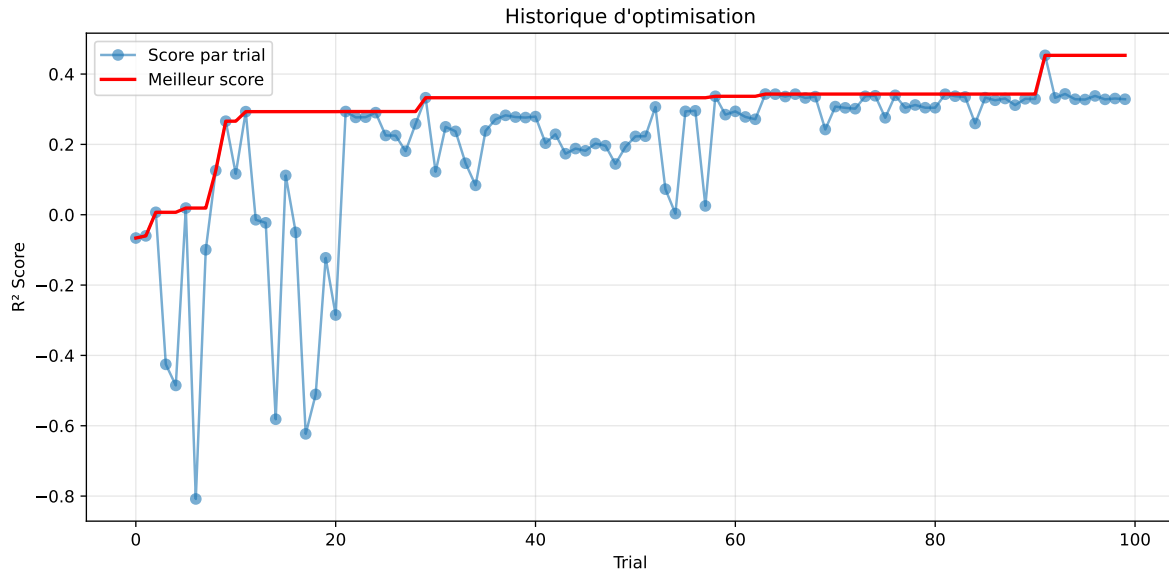


2.4.5 Gridsearch — Group: 2025_Sep

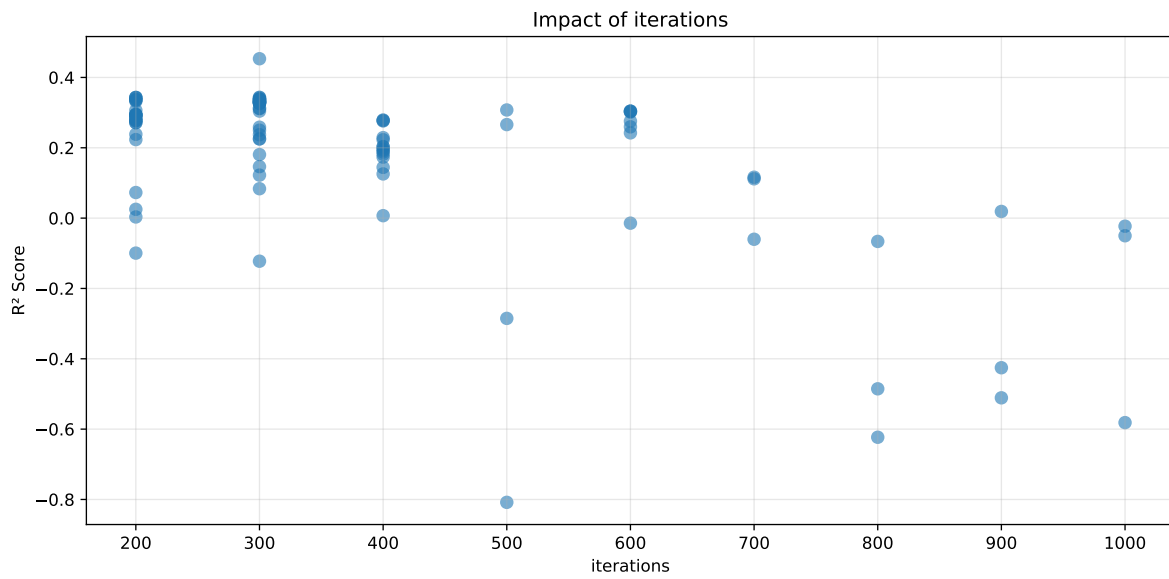
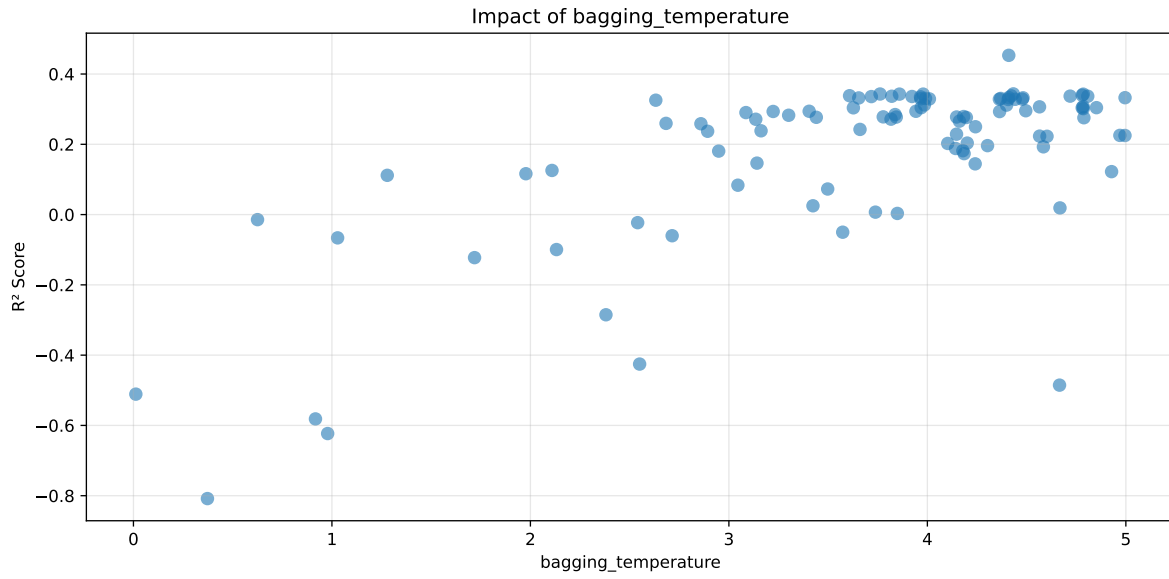
Len(group) : 4996, Len(training) : 3498, Len(test) : 1498

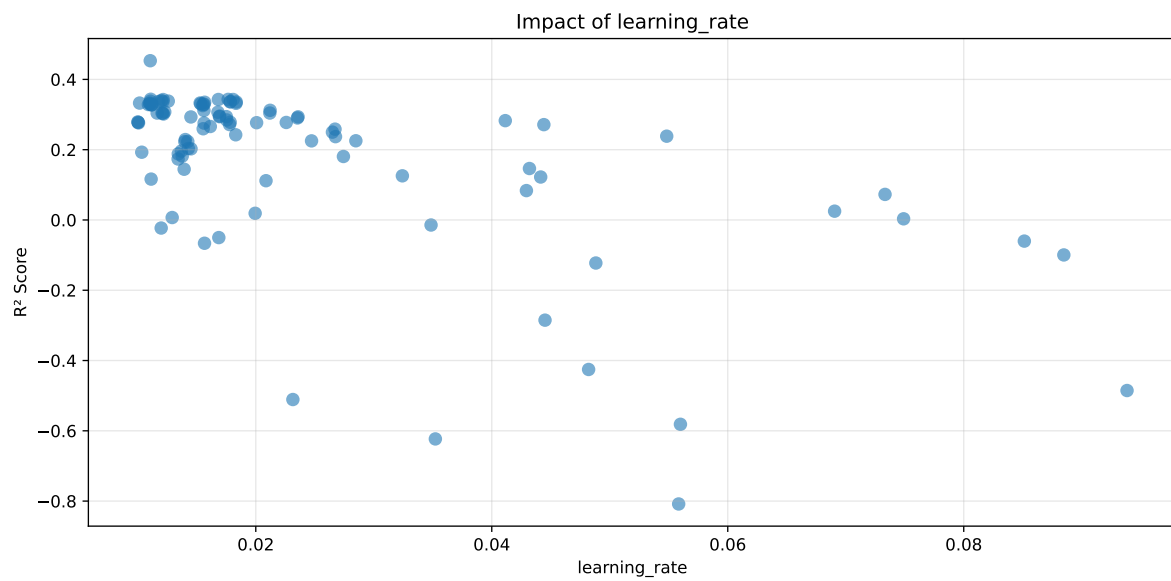
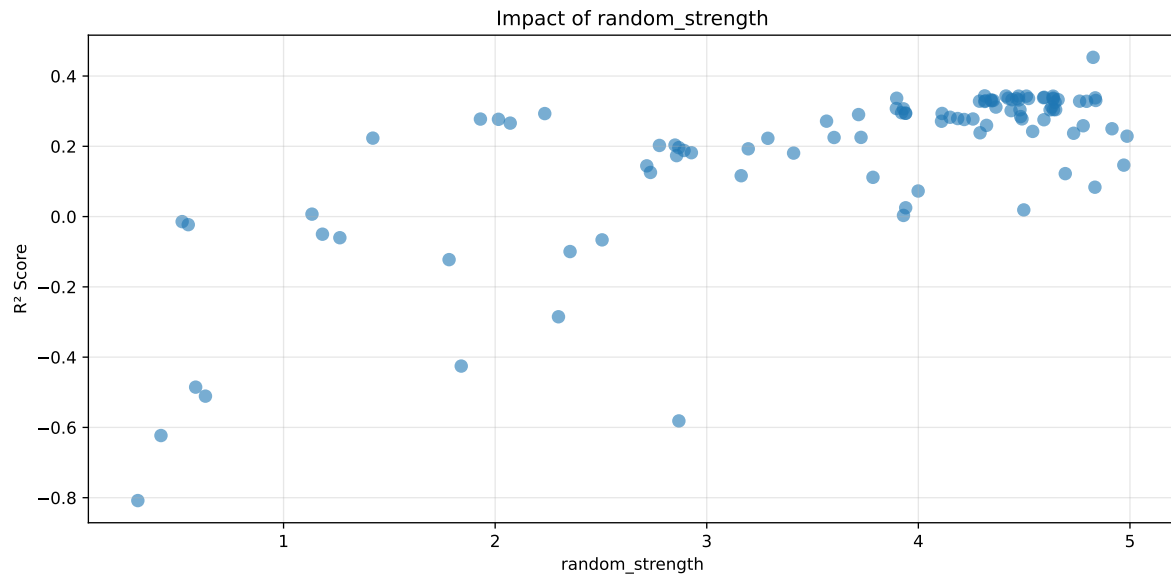
Distribution y_train vs y_test: y_train: mean=2.160, std=0.756, min=0.973, max=8.190
y_test: mean=2.160, std=0.777, min=0.959, max=7.779

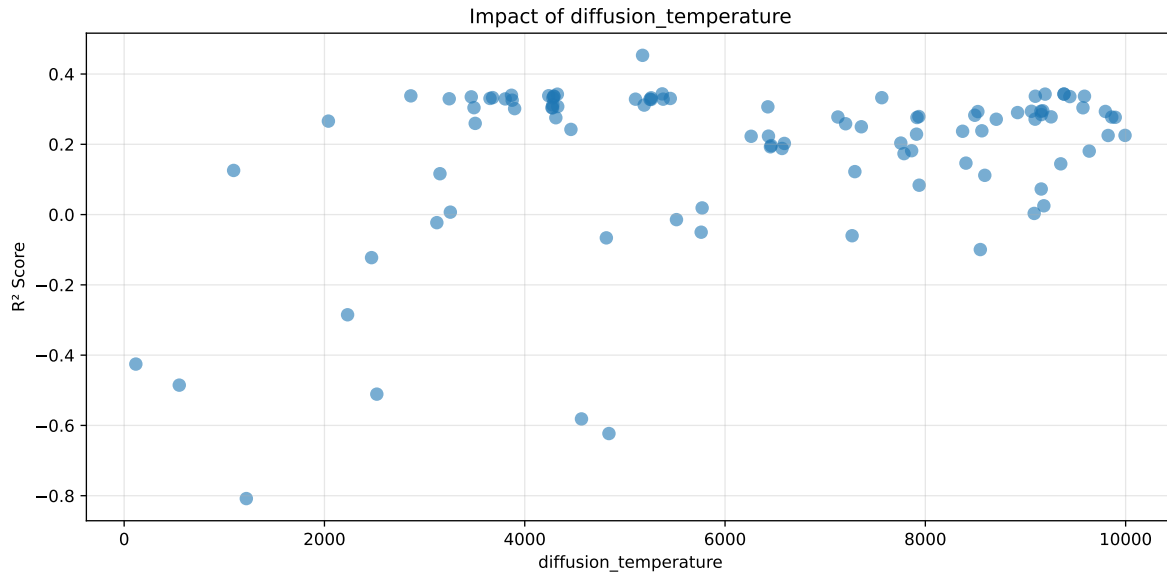
===== TOP Importance FEATURES =====
feature importance r_umidity 42.163507 r_NH3 27.148382 r_temperature 13.429298 r_CO2
8.800211 r_THI 8.458601



Paramètres avec >5% d'importance : ['bagging_temperature', 'iterations', 'random_strength', 'learning_rate', 'diffusion_temperature']

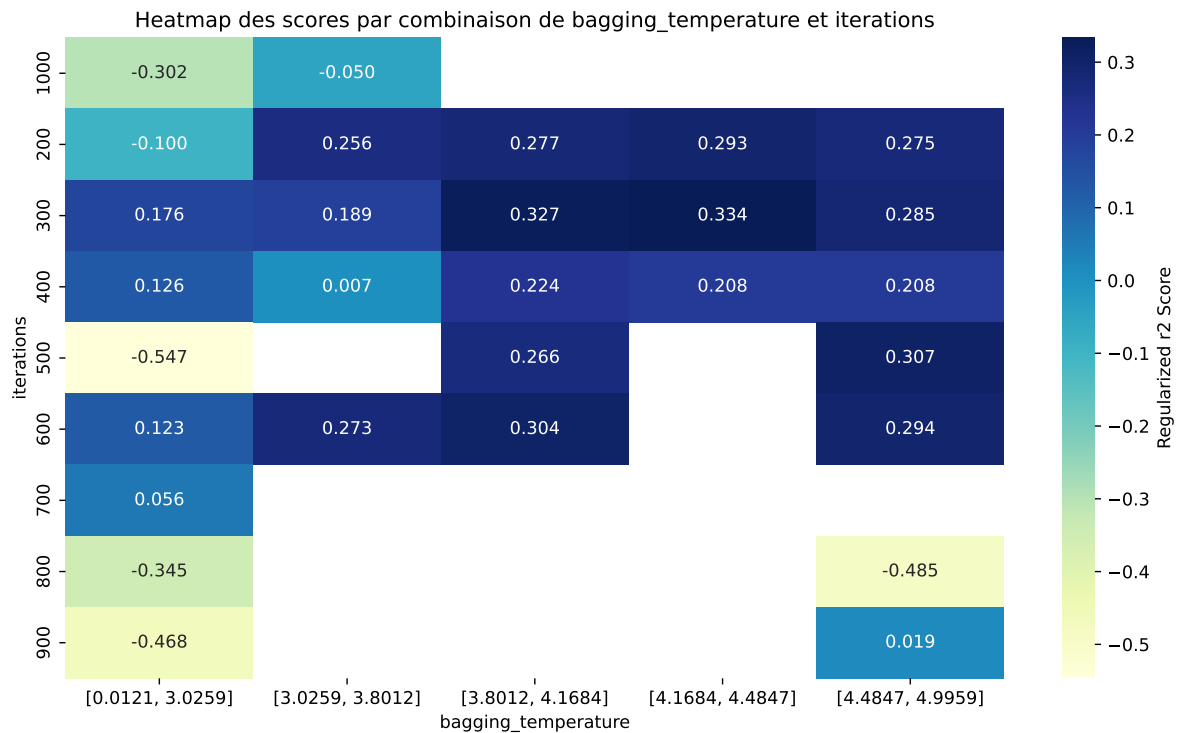






C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.4737	0.4532	0.4384	0.0207	0.4532	0.4396	-0.0136	0.97	1.0452
2	0.4853	0.4616	0.4487	0.0185	0.3435	0.45	-0.0117	0.9747	1.0512
3	0.4899	0.4654	0.4523	0.0195	0.3431	0.4503	-0.0152	0.9674	1.0526
4	0.4867	0.4627	0.4506	0.0193	0.3428	0.4494	-0.0133	0.9712	1.0518
5	0.4883	0.464	0.4517	0.02	0.3428	0.4522	-0.0118	0.9745	1.0522
6	0.4895	0.465	0.4522	0.02	0.3427	0.4522	-0.0129	0.9724	1.0526
7	0.4853	0.461	0.4478	0.0203	0.3396	0.4466	-0.0144	0.9687	1.0527
8	0.488	0.4631	0.4502	0.0203	0.3383	0.4498	-0.0132	0.9715	1.0539
9	0.5035	0.4759	0.4634	0.0205	0.3378	0.4638	-0.0121	0.9746	1.058
10	0.482	0.4579	0.4461	0.0208	0.3371	0.4457	-0.0122	0.9734	1.0528
---	---	---	---	---	---	---	---	---	---
100	0.863	0.5845	0.5881	0.0166	-0.8082	0.6003	0.0158	1.027	1.4765

Hyperparameters:

Rank 1: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.011056226112266608, 'reg__l2_leaf_reg': 29.585059701355192, 'reg__bagging_temperature': 4.410117973452438, 'reg__random_strength': 4.825729597020976, 'reg__subsample': 0.7461257017104317, 'reg__min_data_in_leaf': 29, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.5527455137616413, 'reg__border_count': 33, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 5176.373213714412}

Rank 2: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.011093227282194616, 'reg__l2_leaf_reg': 20.969556869073692, 'reg__bagging_temperature': 4.43300516581941, 'reg__random_strength': 4.313487188617137, 'reg__subsample': 0.7238168936281939, 'reg__min_data_in_leaf': 30, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.5979841310120633, 'reg__border_count': 64, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 5372.702576979247}

Rank 3: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.017657682181603938, 'reg__l2_leaf_reg': 20.6917579034668, 'reg__bagging_temperature': 3.761574465213344, 'reg__random_strength': 4.414036909657183, 'reg__subsample': 0.6389643026829359, 'reg__min_data_in_leaf': 28, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.6289177032553512, 'reg__border_count': 114, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 9384.197008409514}

Rank 4: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.016830051826235256, 'reg__l2_leaf_reg': 7.228080996634942, 'reg__bagging_temperature': 3.9787570233389014, 'reg__random_strength': 4.473690590638895, 'reg__subsample': 0.640390183744346, 'reg__min_data_in_leaf': 27, 'reg__leaf_estimation_iterations': 6, 'reg__rsm':

0.6152340067702717, 'reg__border_count': 117, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 9384.257955336214}

Rank 5: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.018079368204753865, 'reg__l2_leaf_reg': 20.770966406777948, 'reg__bagging_temperature': 3.85932096916834, 'reg__random_strength': 4.5105484824500115, 'reg__subsample': 0.6382495023503963, 'reg__min_data_in_leaf': 28, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.5665937741618416, 'reg__border_count': 65, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 9195.904421009542}

Rank 6: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.01214488425433768, 'reg__l2_leaf_reg': 26.388990933534345, 'reg__bagging_temperature': 4.785382688848787, 'reg__random_strength': 4.636580330357951, 'reg__subsample': 0.6922500247128488, 'reg__min_data_in_leaf': 33, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.6557141898261284, 'reg__border_count': 128, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 4325.672251523801}

Rank 7: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.012002364020875306, 'reg__l2_leaf_reg': 7.572740698277395, 'reg__bagging_temperature': 4.779971529426677, 'reg__random_strength': 4.5948086196500535, 'reg__subsample': 0.6980008908366492, 'reg__min_data_in_leaf': 33, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.5751068023888057, 'reg__border_count': 120, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 3867.5634243604186}

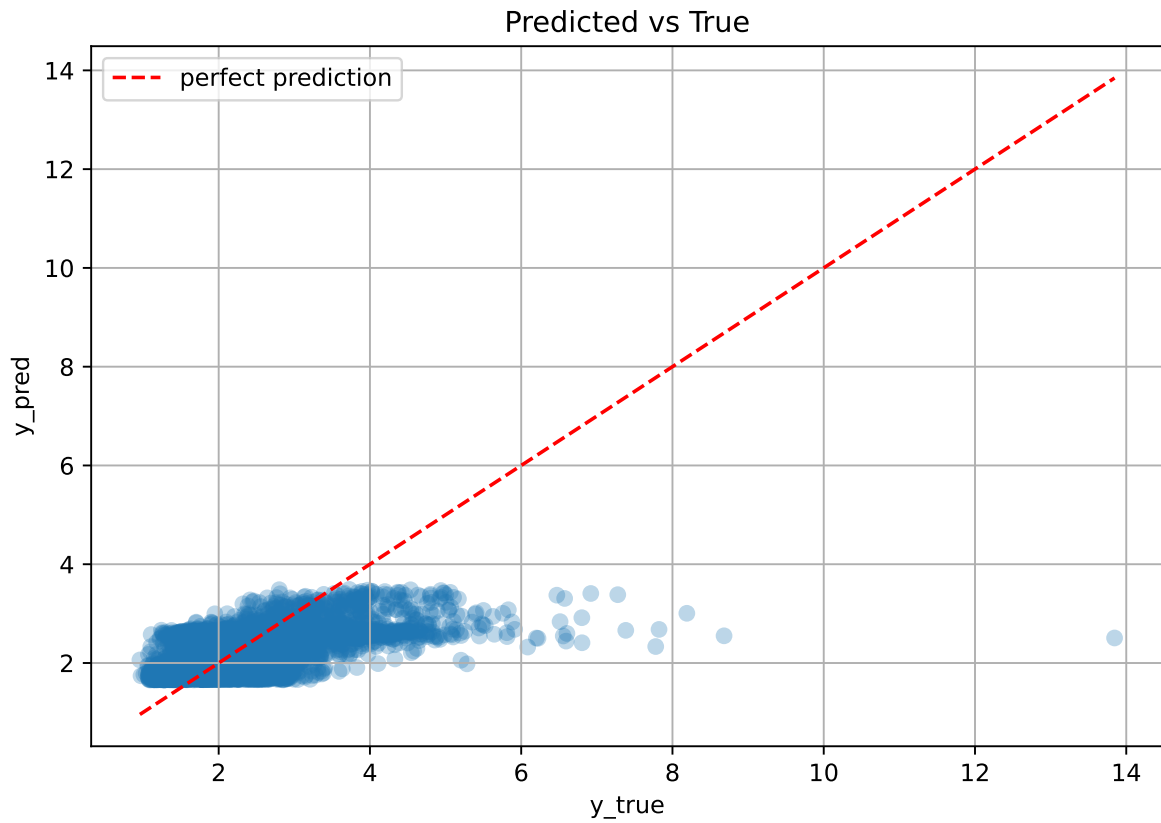
Rank 8: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.012583504306801416, 'reg__l2_leaf_reg': 26.524169880671856, 'reg__bagging_temperature': 3.60802052931802, 'reg__random_strength': 4.591534621540362, 'reg__subsample': 0.6915319397170008, 'reg__min_data_in_leaf': 33, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.5747858294649186, 'reg__border_count': 122, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 4239.518270119886}

Rank 9: {'reg__iterations': 300, 'reg__depth': 4, 'reg__learning_rate': 0.011068956288580668, 'reg__l2_leaf_reg': 28.999699674356087, 'reg__bagging_temperature': 4.422701613570807, 'reg__random_strength': 4.835868296894309, 'reg__subsample': 0.7355922633914196, 'reg__min_data_in_leaf': 30, 'reg__leaf_estimation_iterations': 9, 'reg__rsm': 0.5492041290376927, 'reg__border_count': 124, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 2862.6574019702393}

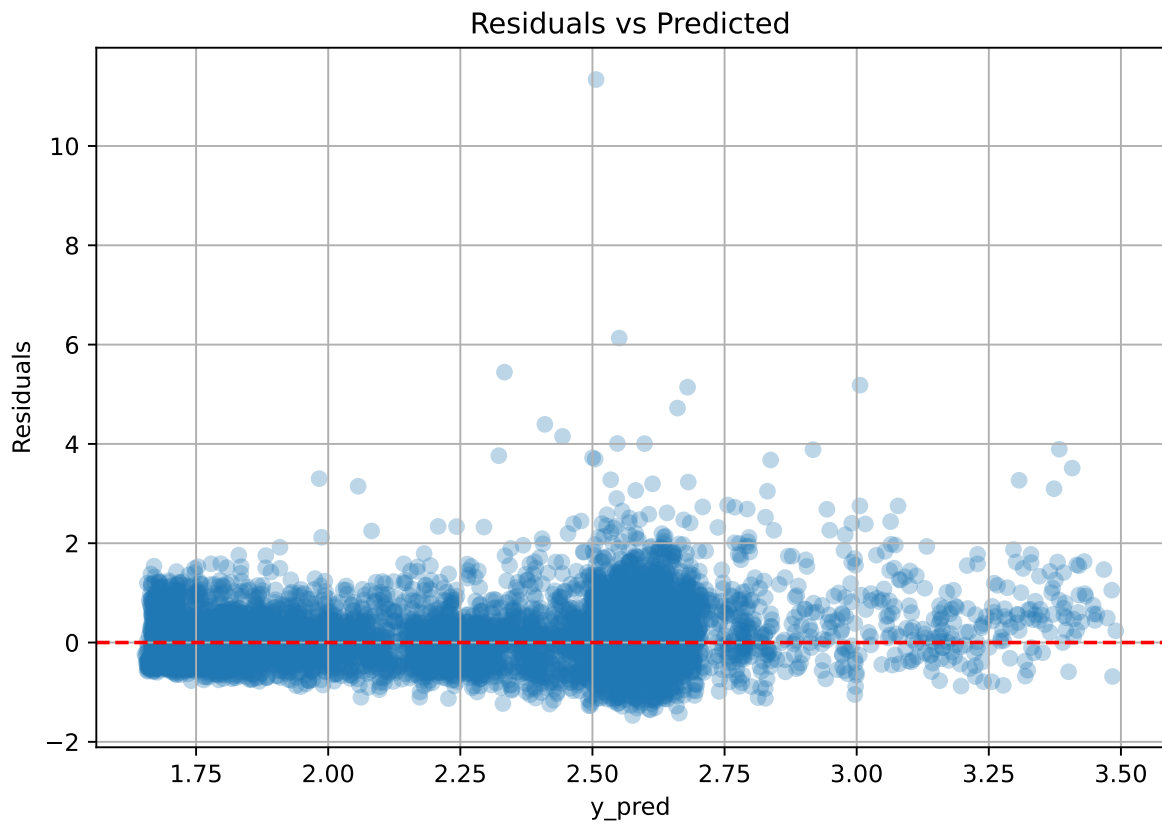
Rank 10: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.011751632579964056, 'reg__l2_leaf_reg': 26.976034806270658, 'reg__bagging_temperature': 4.719373032617989,

'reg__random_strength': 4.637457251912009, 'reg__subsample': 0.6906323187531036,
'reg__min_data_in_leaf': 34, 'reg__leaf_estimation_iterations': 7, 'reg__rsm':
0.6458761947615714, 'reg__border_count': 126, 'reg__leaf_estimation_method': 'Gradient',
'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature':
4289.655276297027}

2.4.6 Scatter



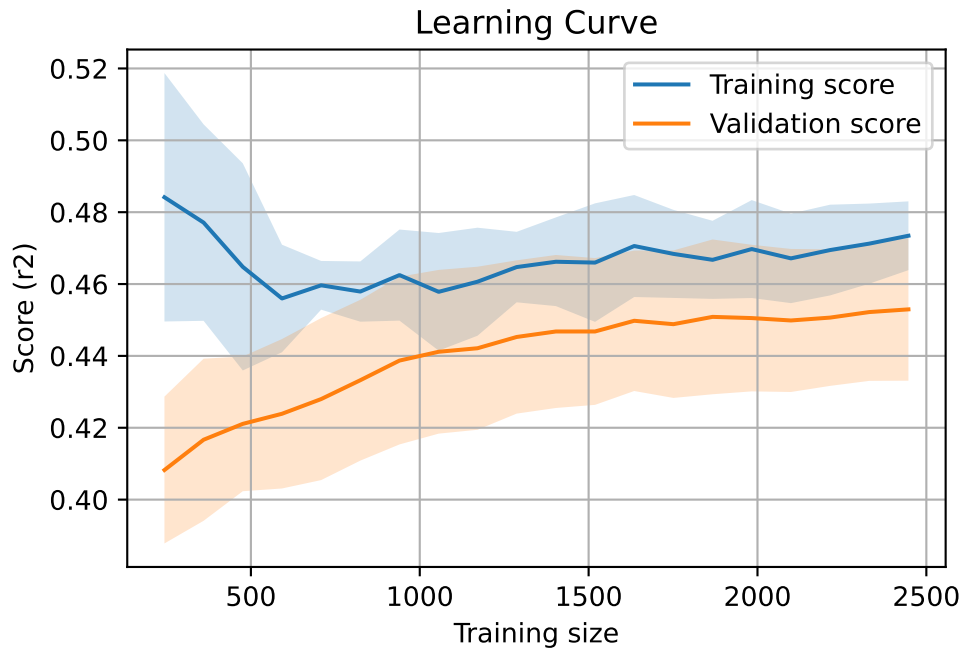
2.4.7 Scatter residuals



2.4.8 Learning curve — Group: 2025_Sep

`Len(group)` : 4996, `Len(training)` : 3498,

`Len(training_real)` : 2449, `Len(test_of_training)` : 1049, `Len(test)` : 1498



2.5 Sumup-table

model	group	Len training	Len test	Len total	mean_train_cv	mean_test_cv	test_cv	std_test_cv	pen_score	test	diff	r_test	r_train
LGBM	2025_Jun	2221	951	3172	0.486	0.464	0.511	0.034	0.4645	0.538	0.073	1.158	1.047
LGBM	2025_Sep	3498	1498	4996	0.453	0.432	0.417	0.023	0.4321	0.435	0.003	1.006	1.049
LGBM	MEAN	5719	2449	8168	0.47	0.448	0.464	0.029	0.448	0.486	0.038	1.082	1.048
CatBoost	2025_Jun	2221	951	3172	0.531	0.506	0.533	0.032	0.5065	0.549	0.042	1.084	1.048
CatBoost	2025_Sep	3498	1498	4996	0.474	0.453	0.438	0.021	0.4532	0.44	-0.014	0.97	1.045
CatBoost	MEAN	5719	2449	8168	0.502	0.48	0.486	0.026	0.48	0.494	0.014	1.027	1.046

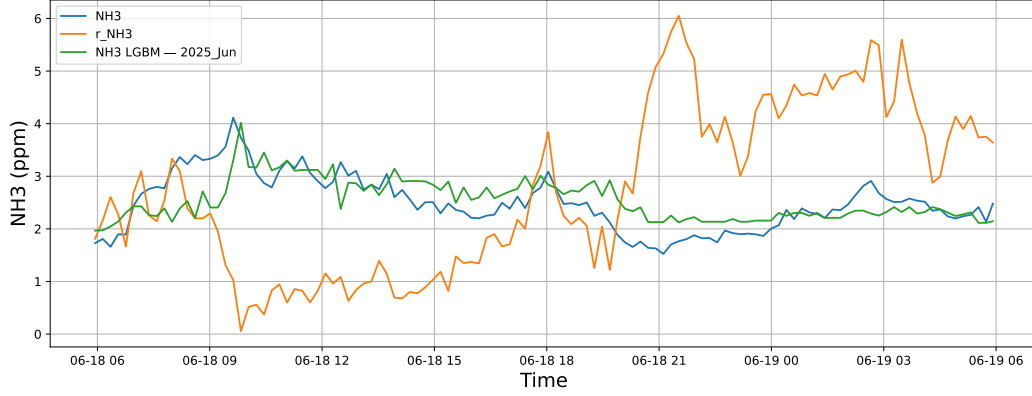
3 Plot

3.1 Standalone plots

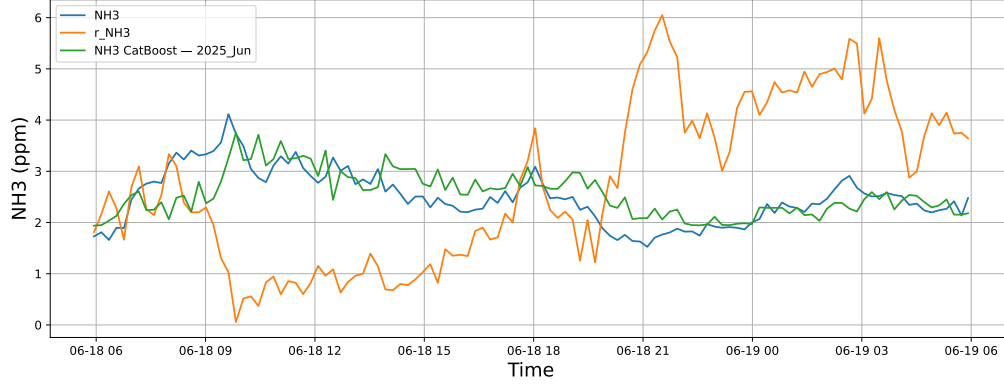
3.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

3.1.1.1 Time window : 24

NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

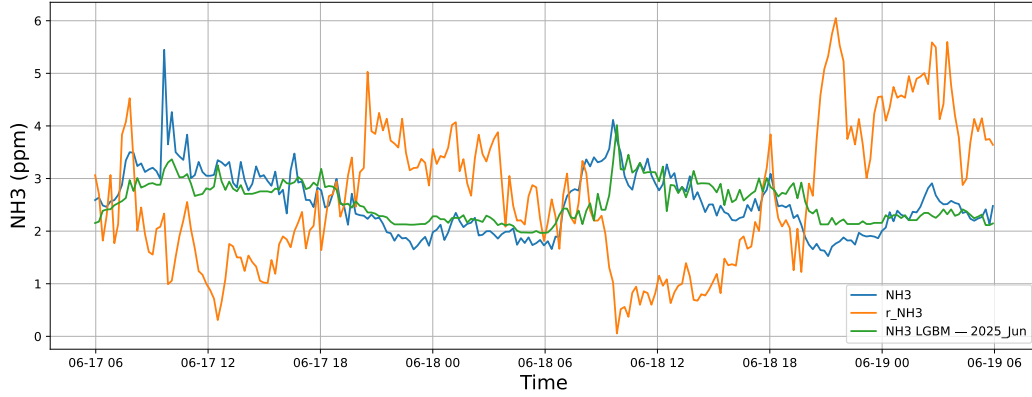


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

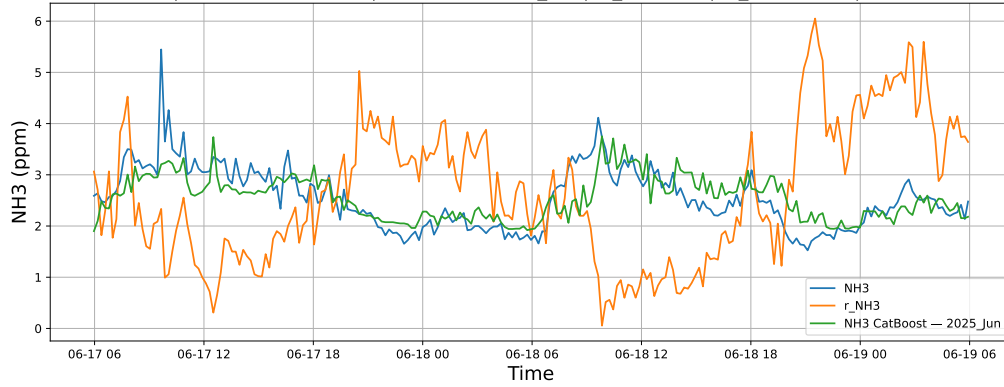


3.1.1.2 Time window : 48

NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

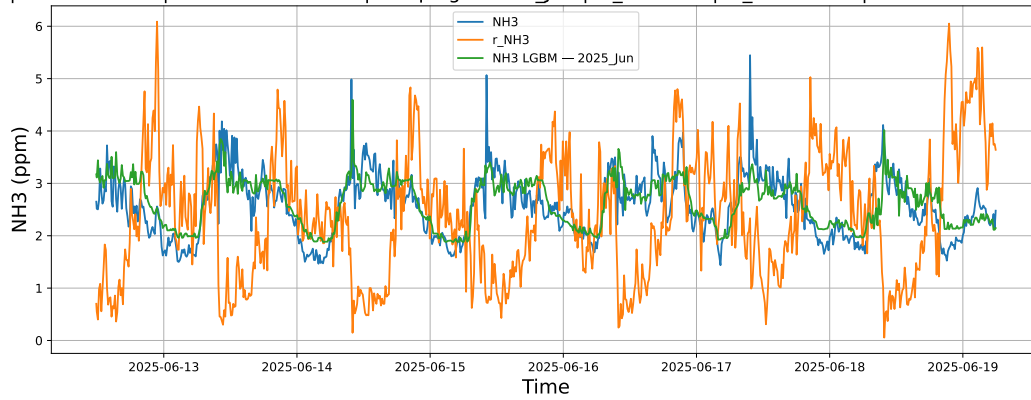


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

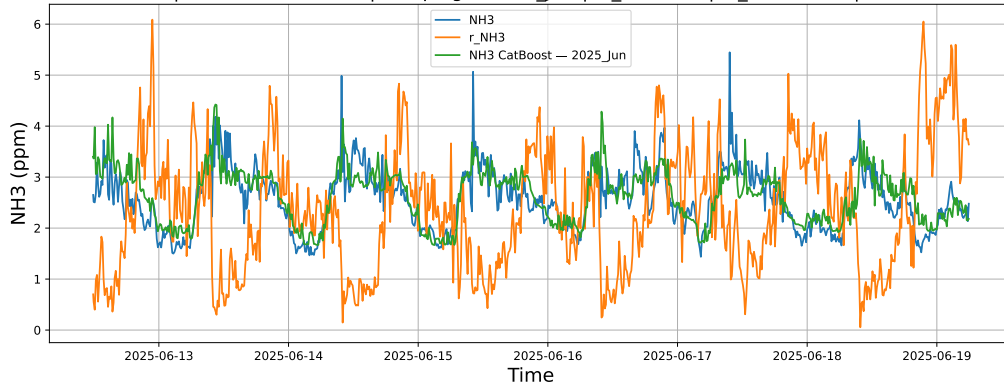


3.1.1.3 Time window : ALL

NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



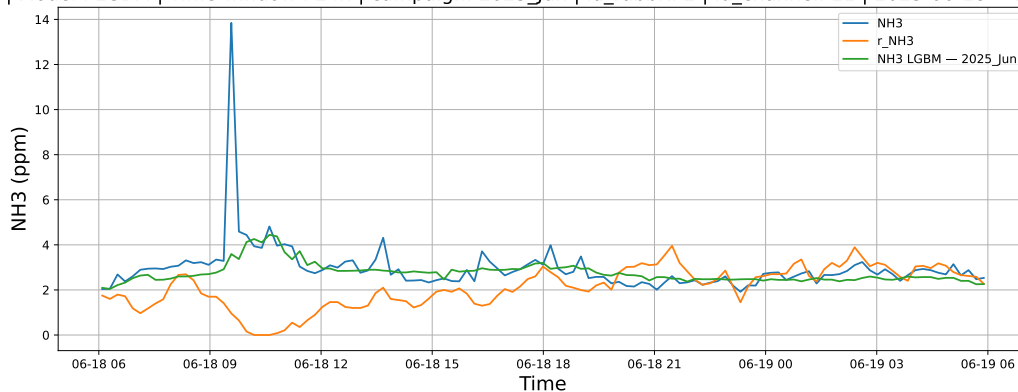
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



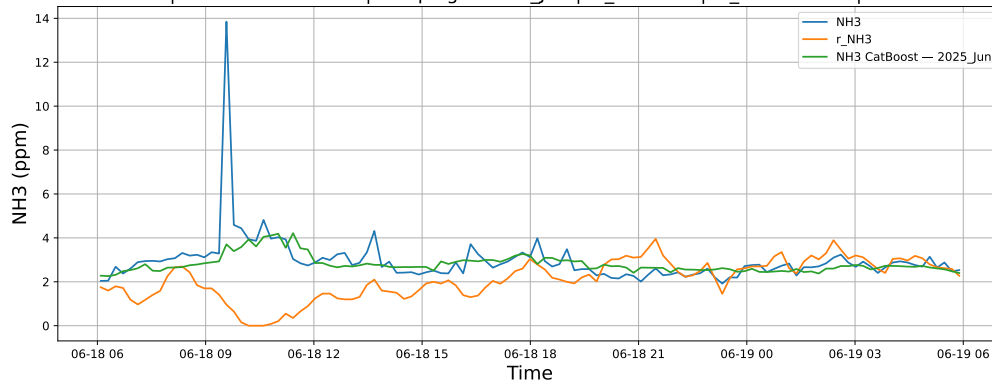
3.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

3.1.2.1 Time window : 24

NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

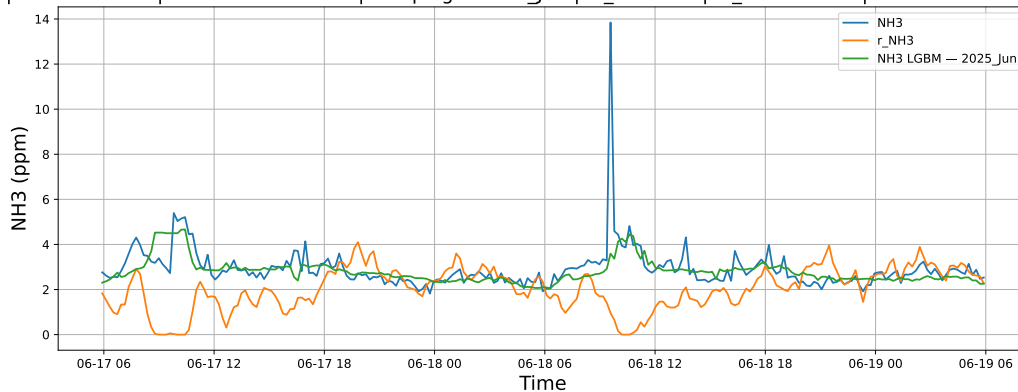


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

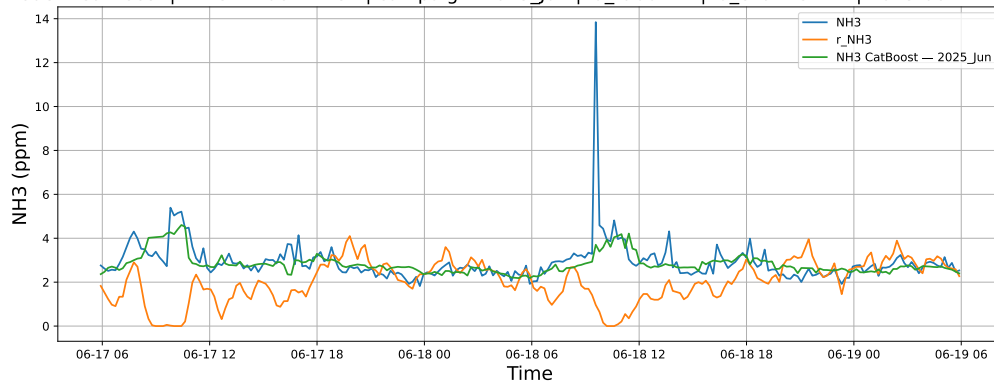


3.1.2.2 Time window : 48

NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

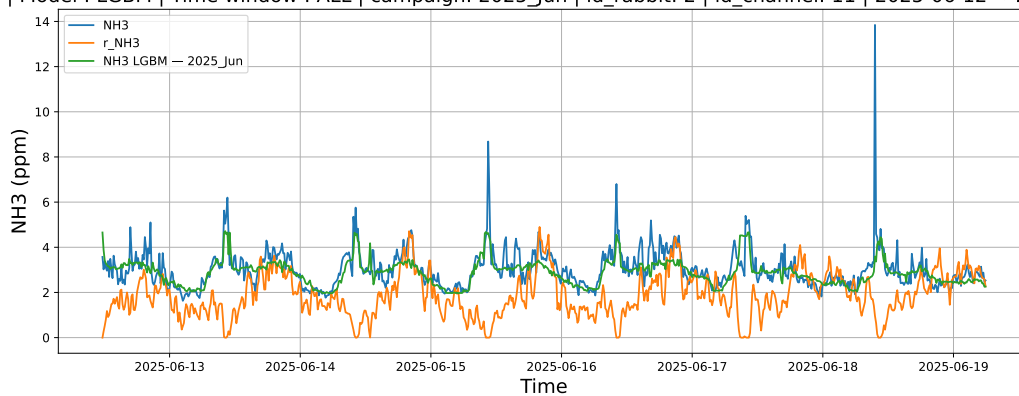


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

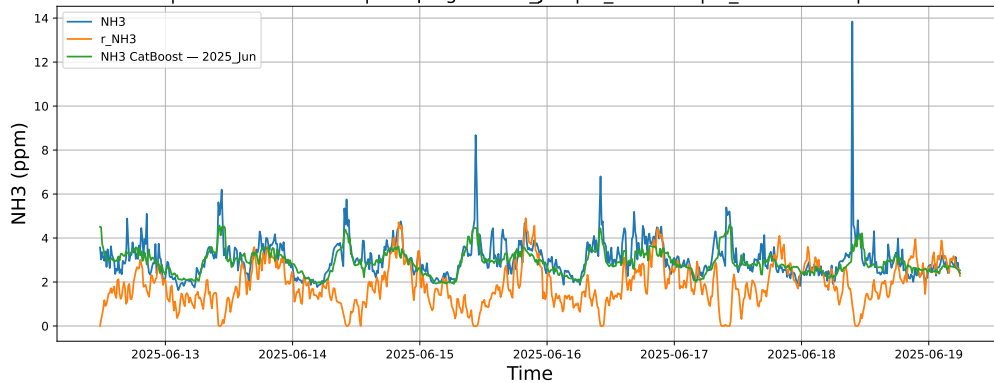


3.1.2.3 Time window : ALL

NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



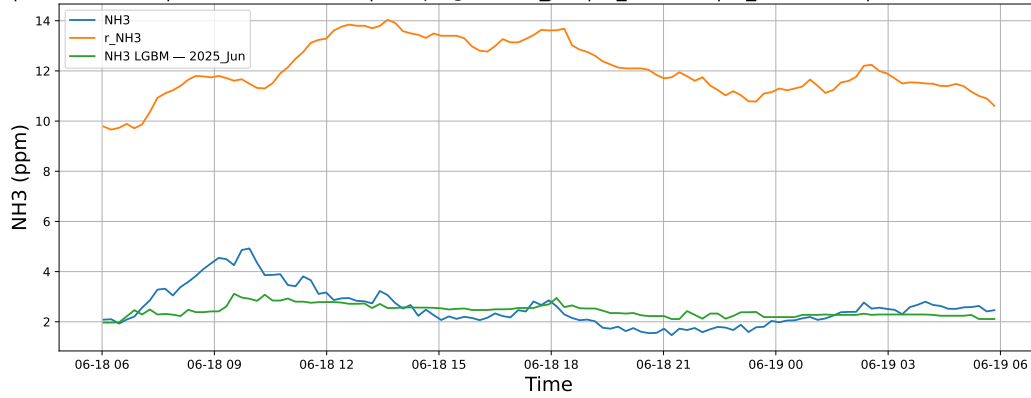
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



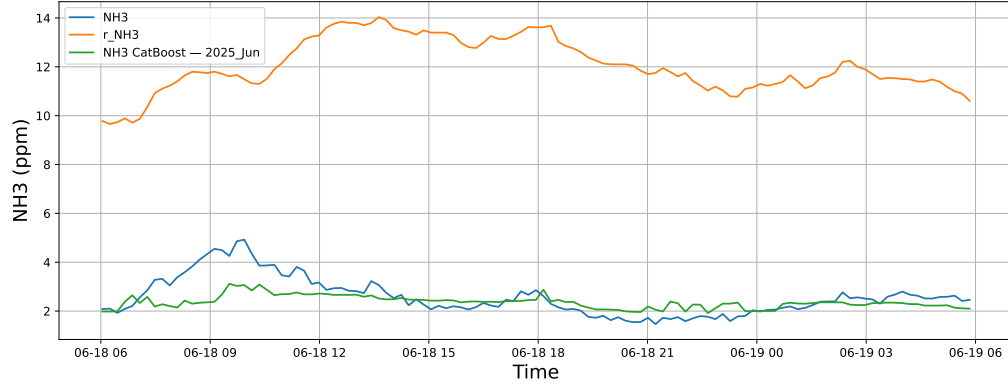
3.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

3.1.3.1 Time window : 24

NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

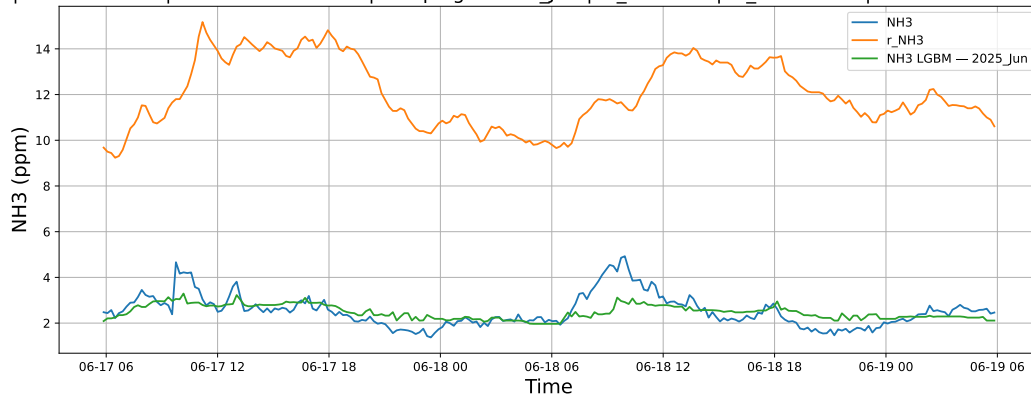


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

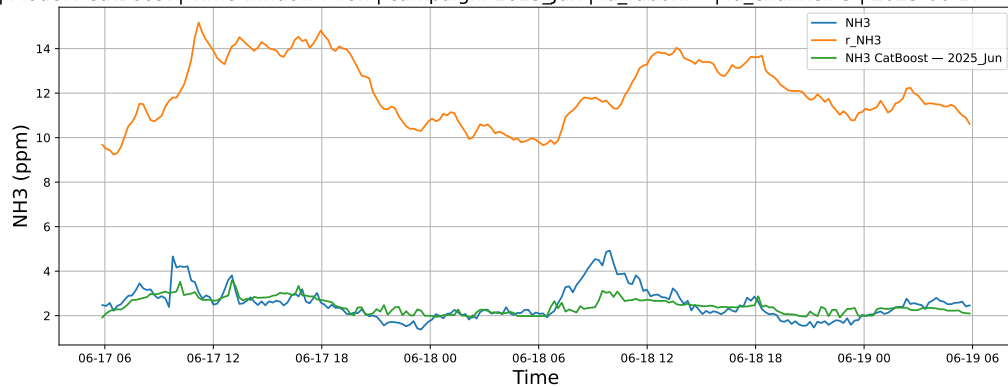


3.1.3.2 Time window : 48

NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

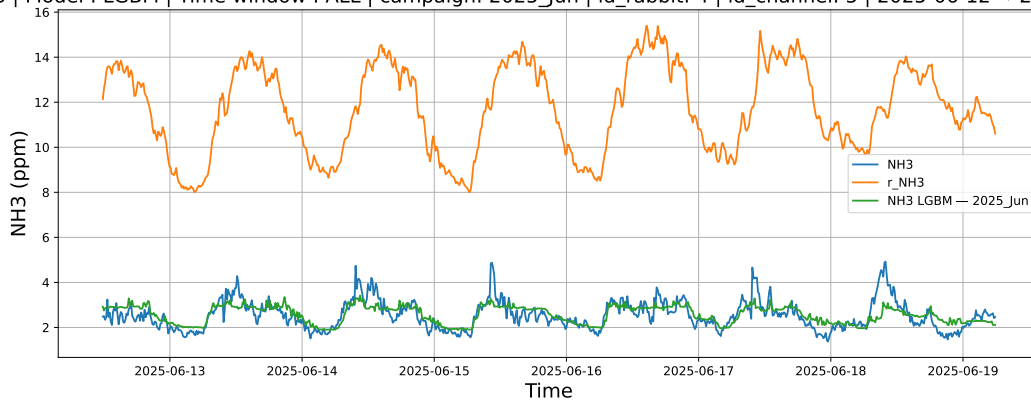


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

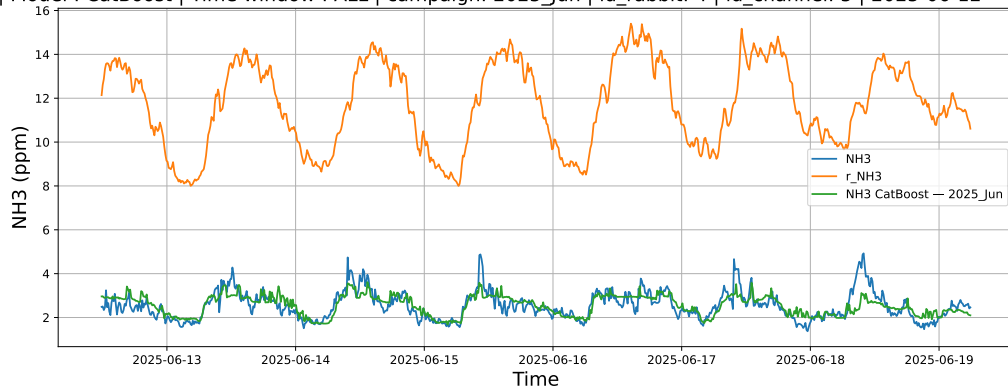


3.1.3.3 Time window : ALL

NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



3.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

3.1.4.1 Time window : 24

NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

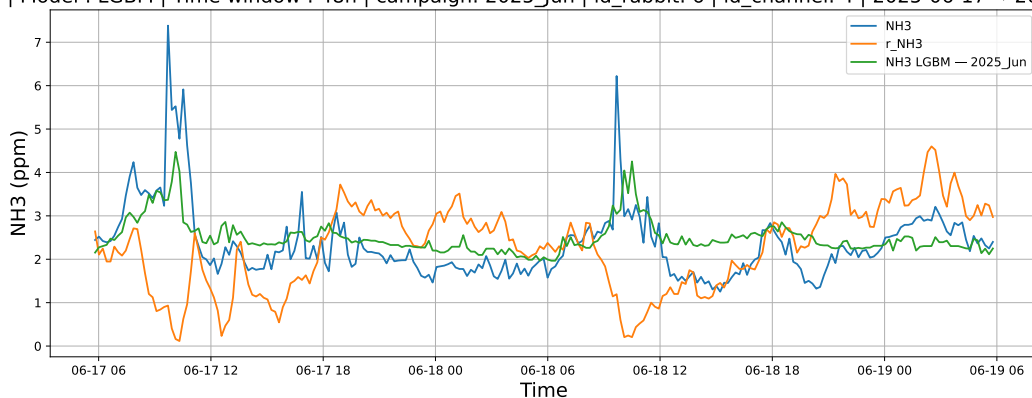


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

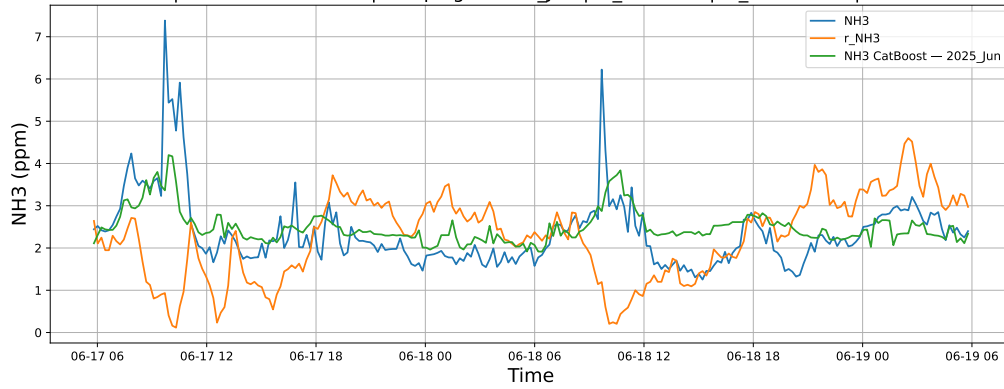


3.1.4.2 Time window : 48

NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

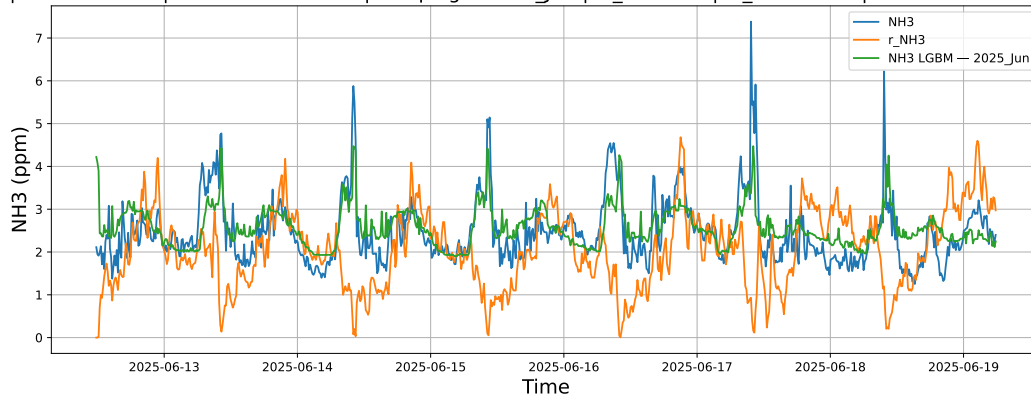


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

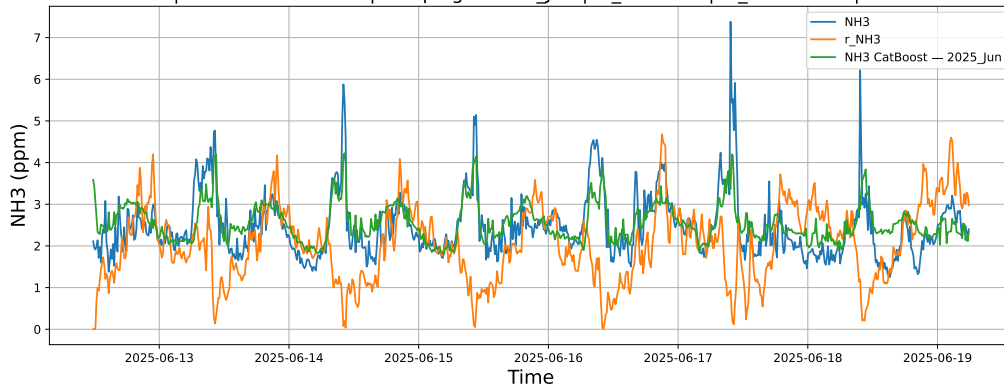


3.1.4.3 Time window : ALL

NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



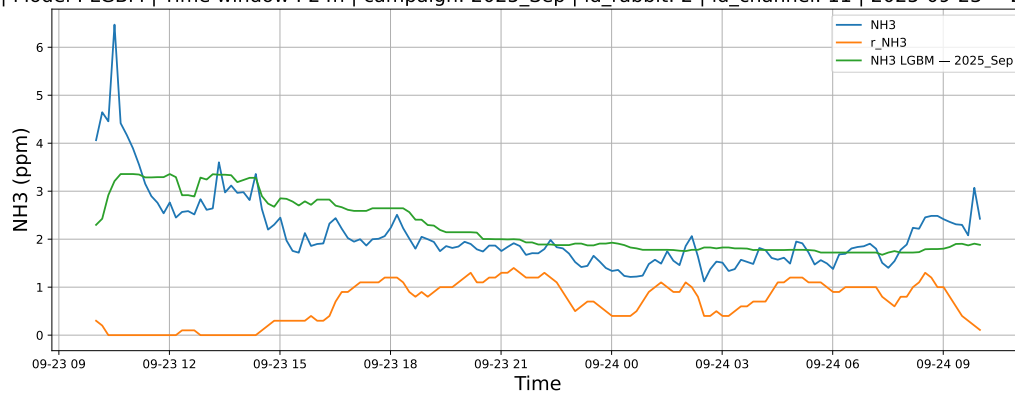
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



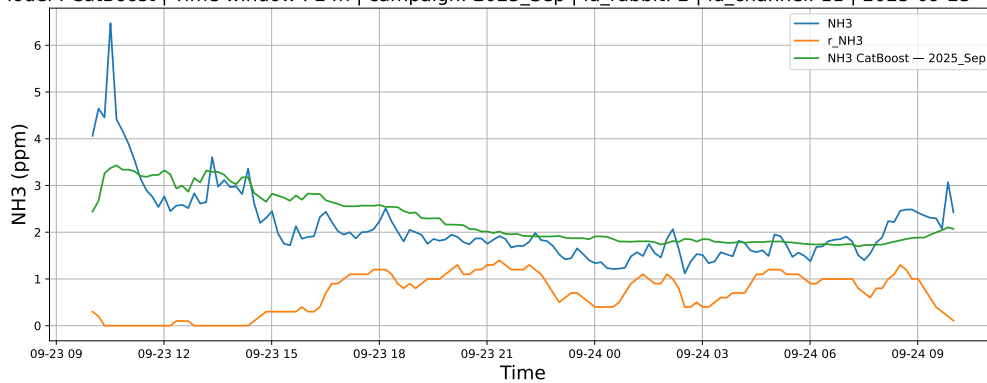
3.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

3.1.5.1 Time window : 24

NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

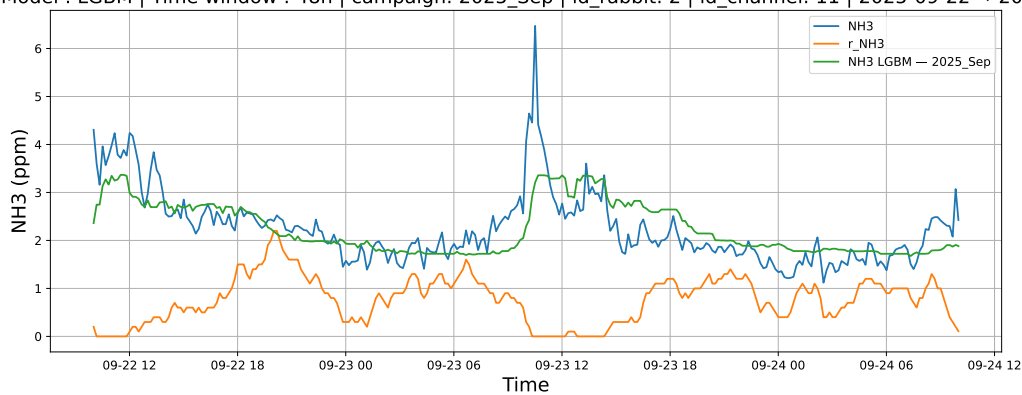


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

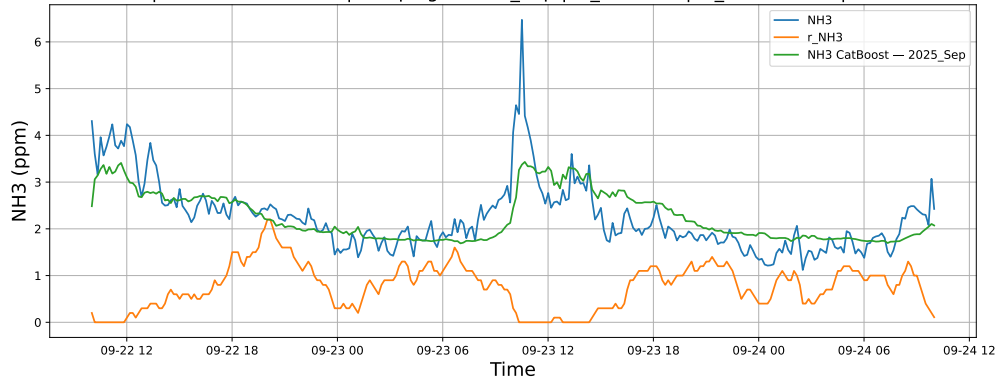


3.1.5.2 Time window : 48

NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

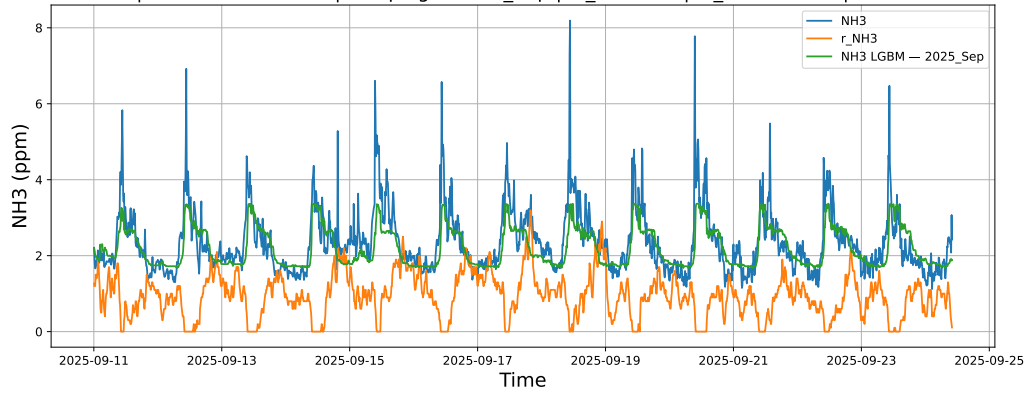


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

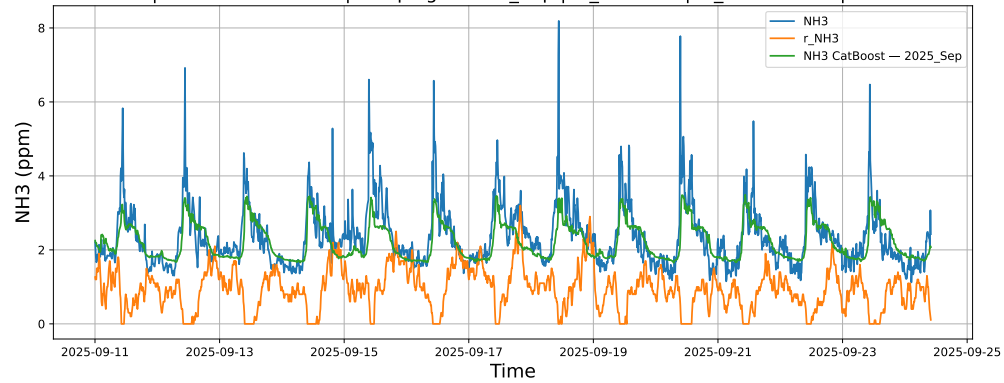


3.1.5.3 Time window : ALL

NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



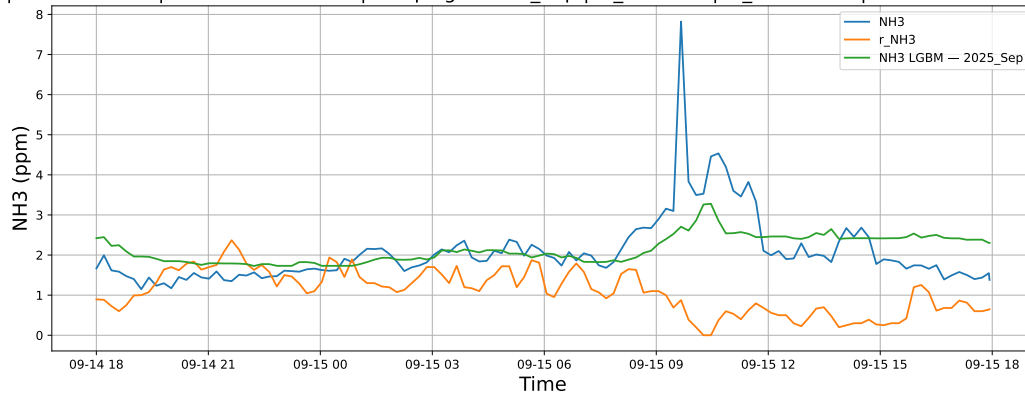
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



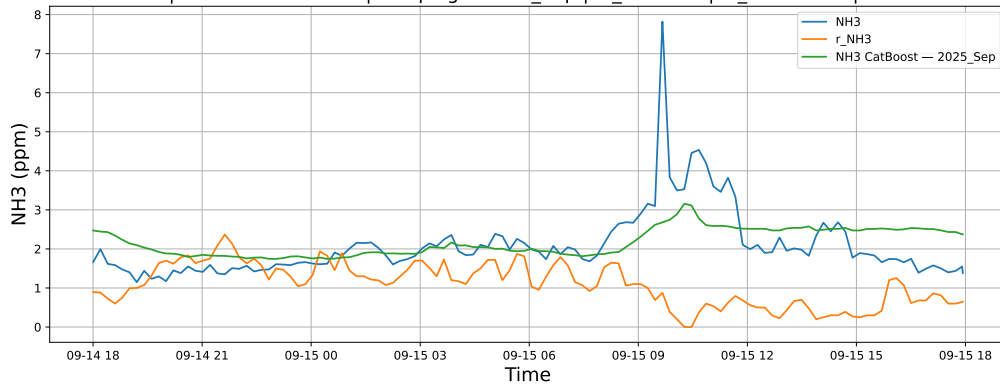
3.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

3.1.6.1 Time window : 24

NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

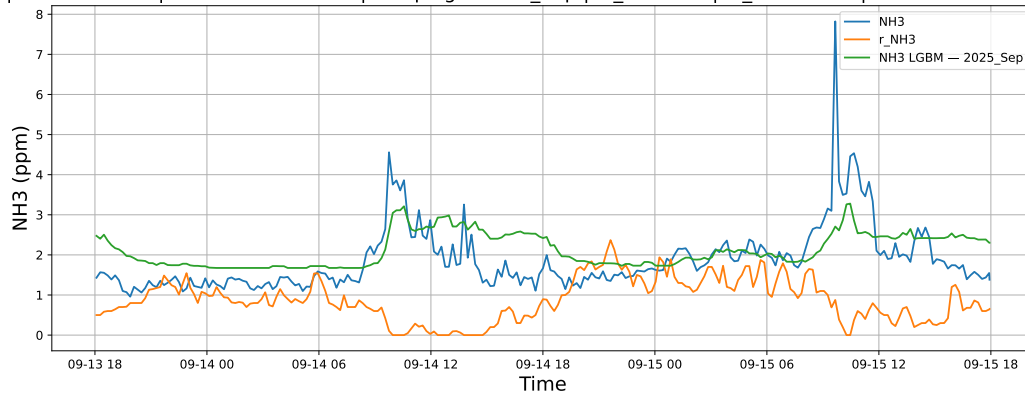


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

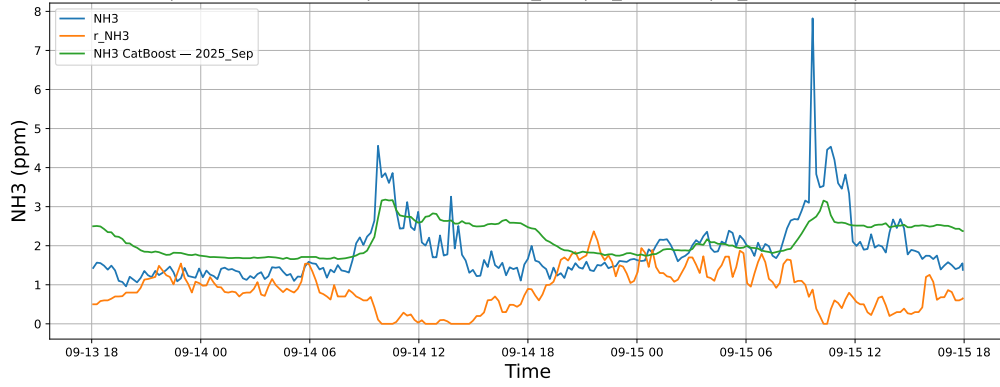


3.1.6.2 Time window : 48

NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

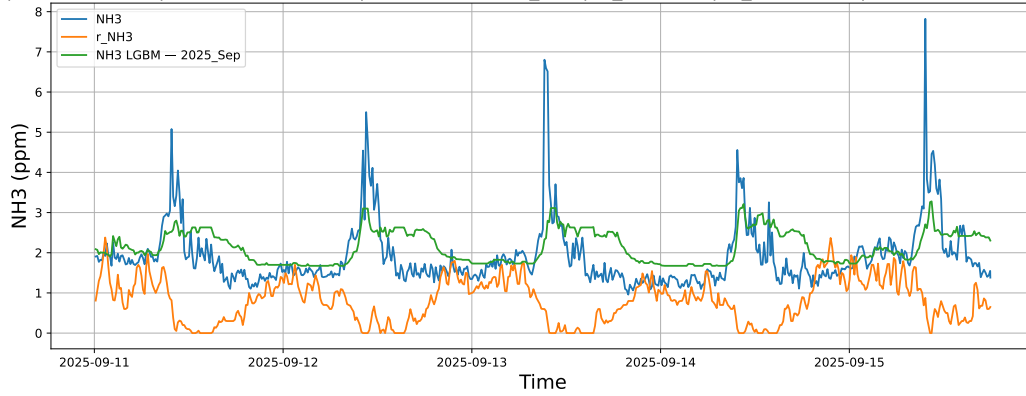


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

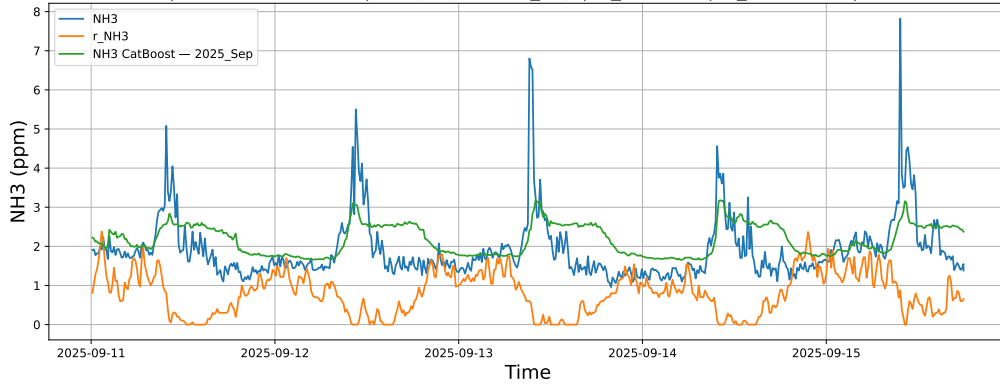


3.1.6.3 Time window : ALL

NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



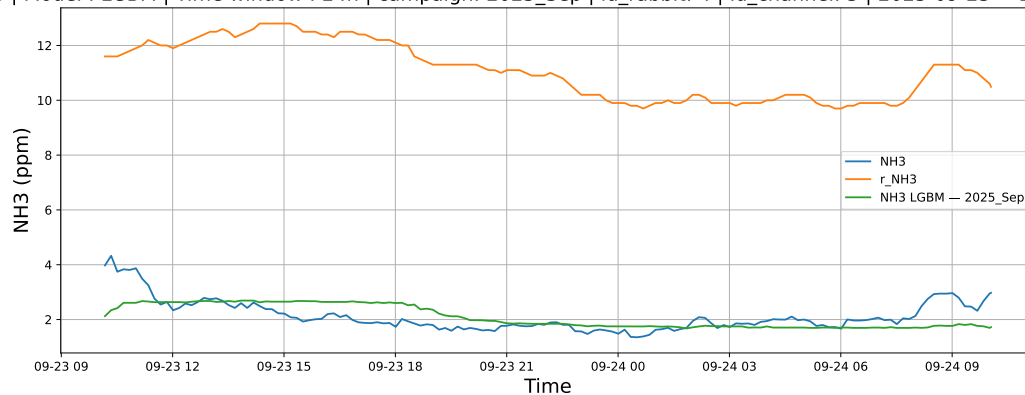
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



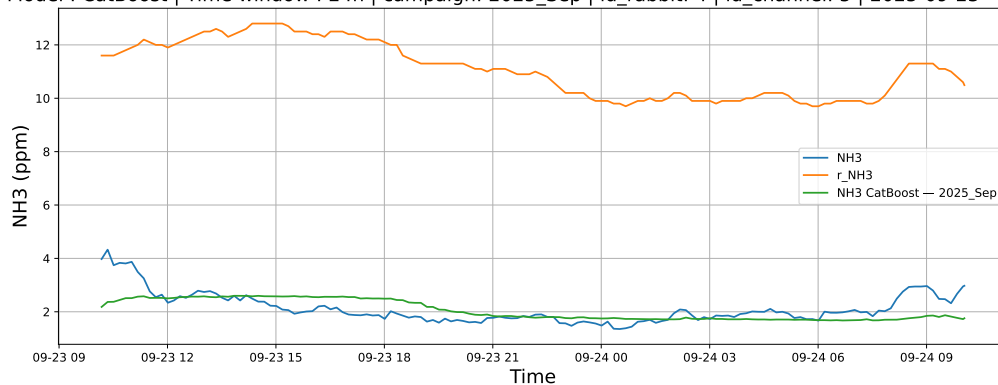
3.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

3.1.7.1 Time window : 24

NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

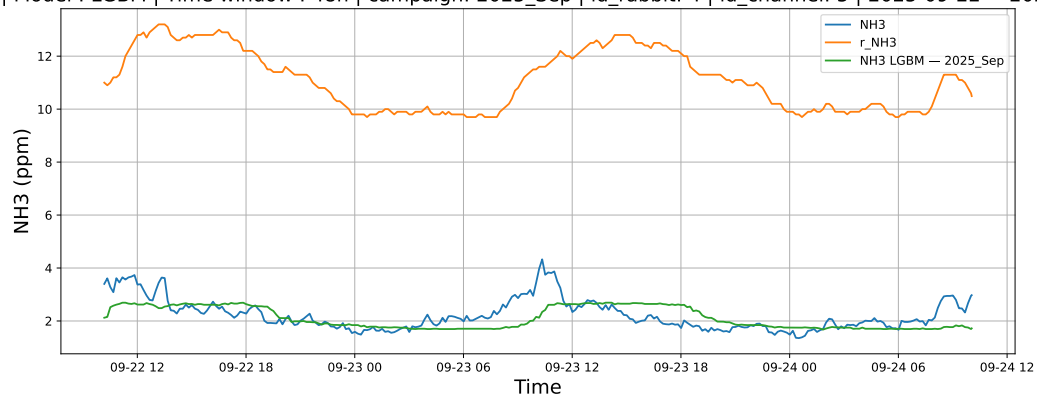


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

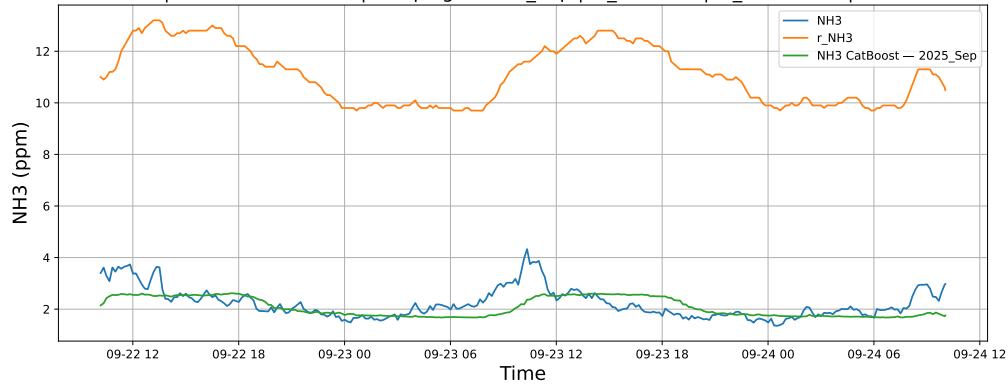


3.1.7.2 Time window : 48

NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

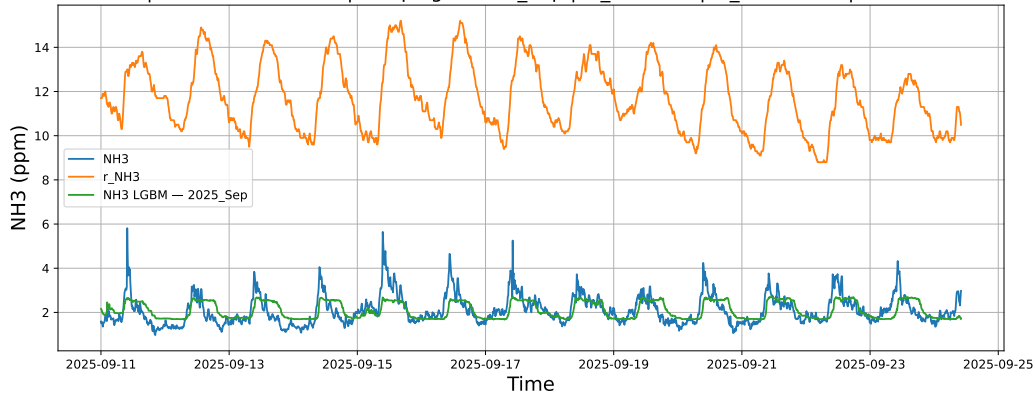


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

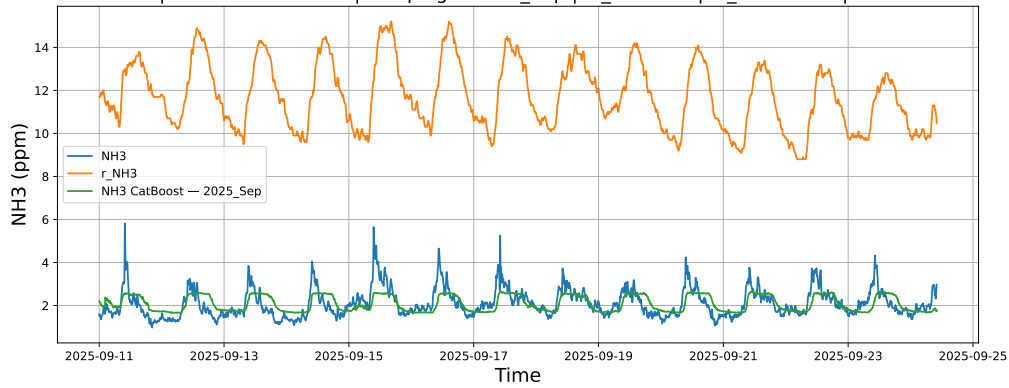


3.1.7.3 Time window : ALL

NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



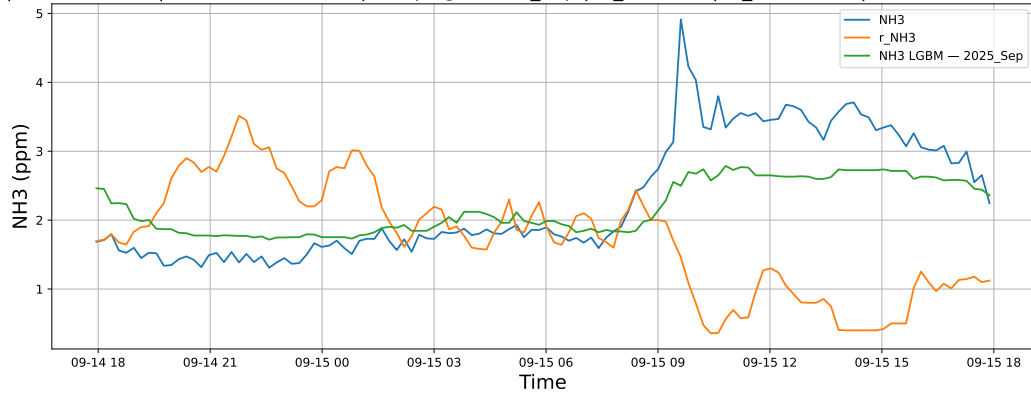
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



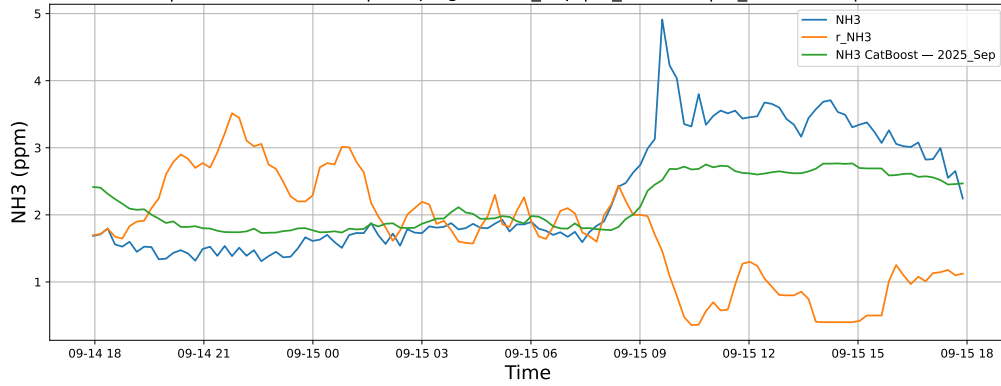
3.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

3.1.8.1 Time window : 24

NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

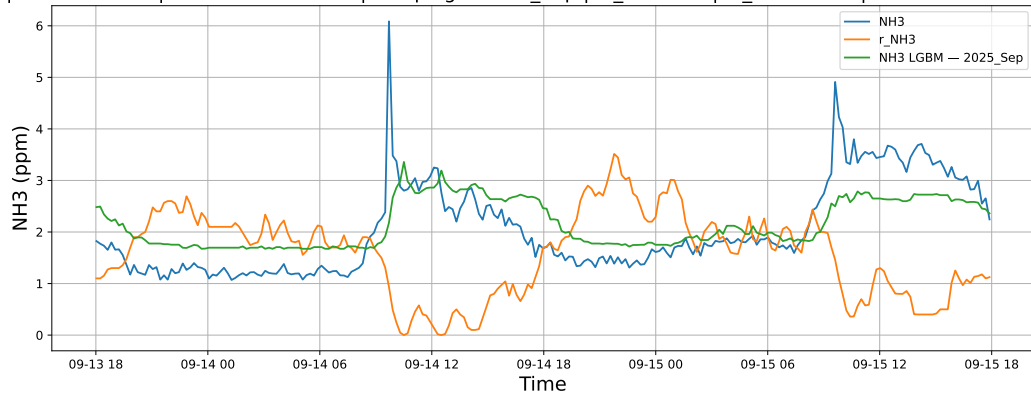


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

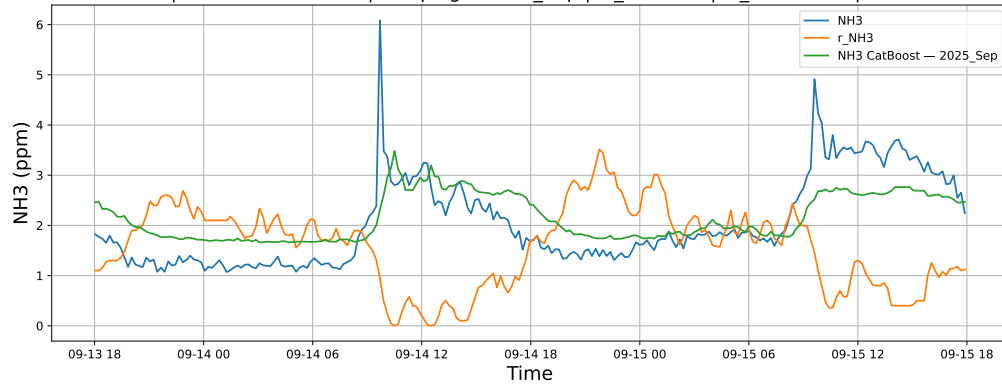


3.1.8.2 Time window : 48

NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

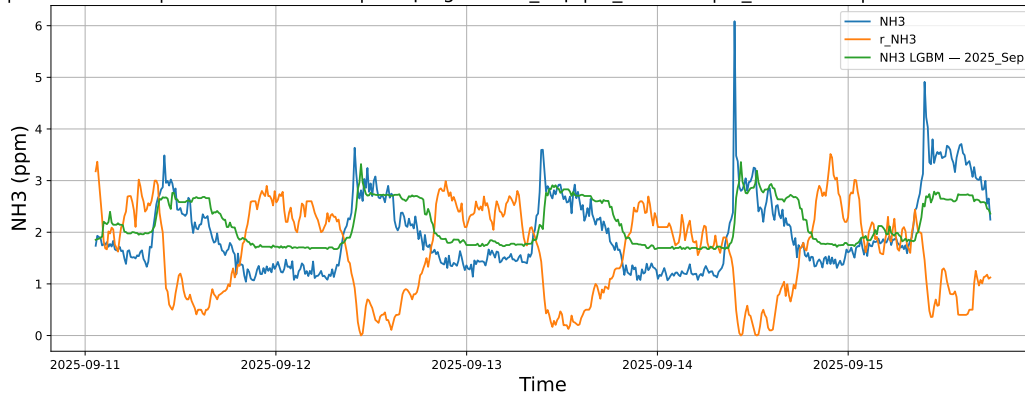


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

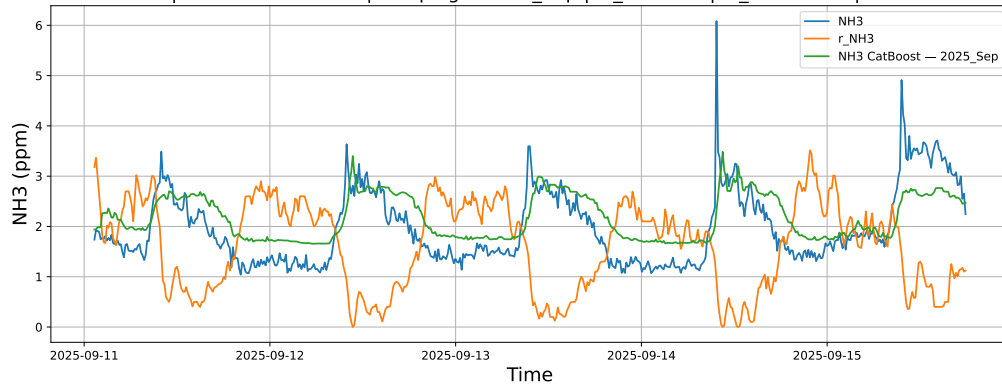


3.1.8.3 Time window : ALL

NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

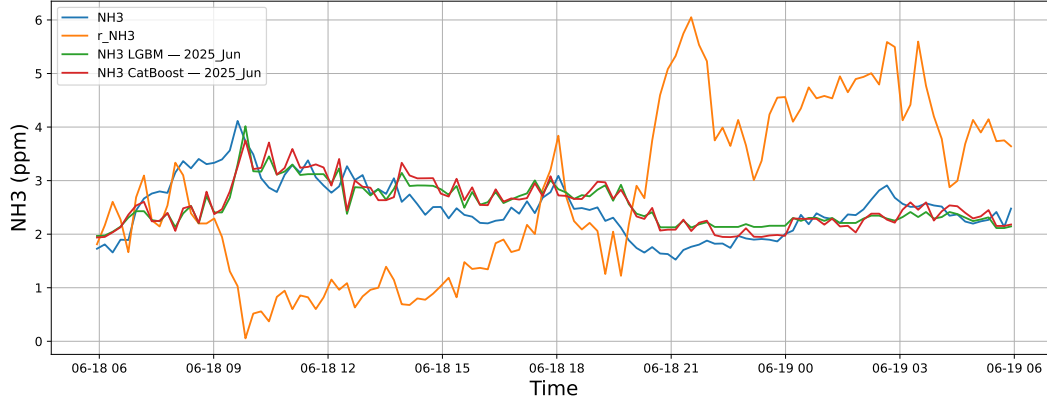


3.2 Compare plots

3.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

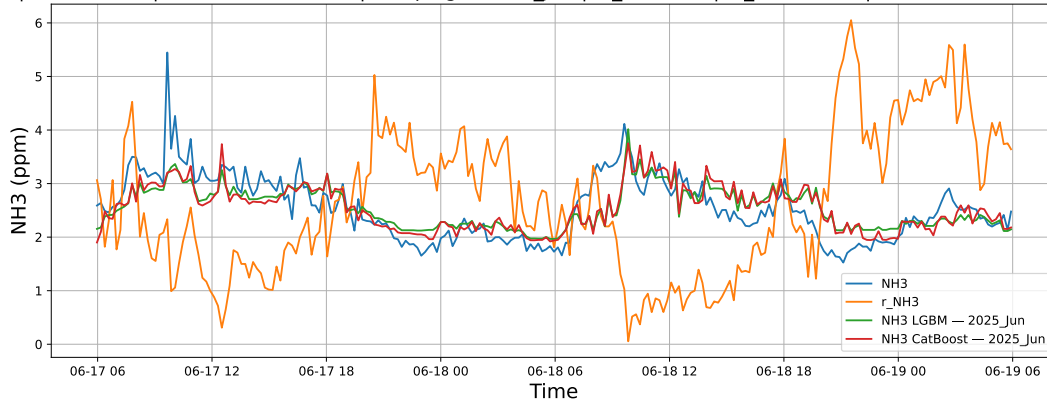
3.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



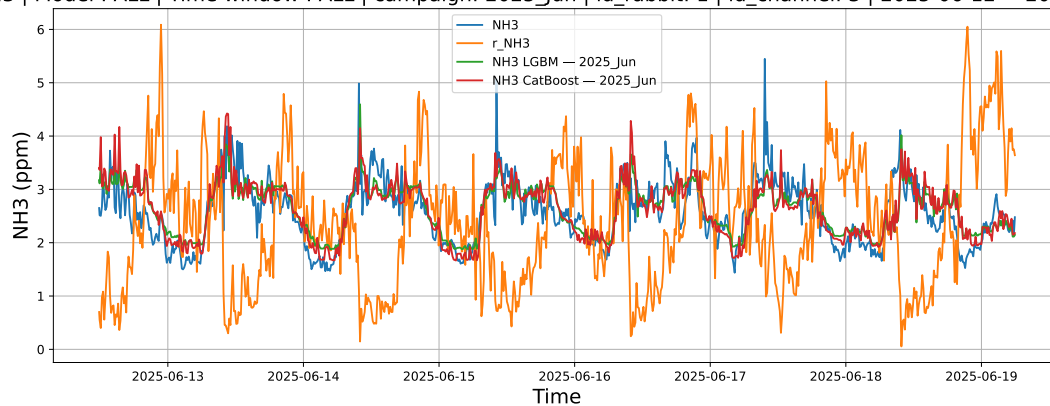
3.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



3.2.1.3 Time window : ALL

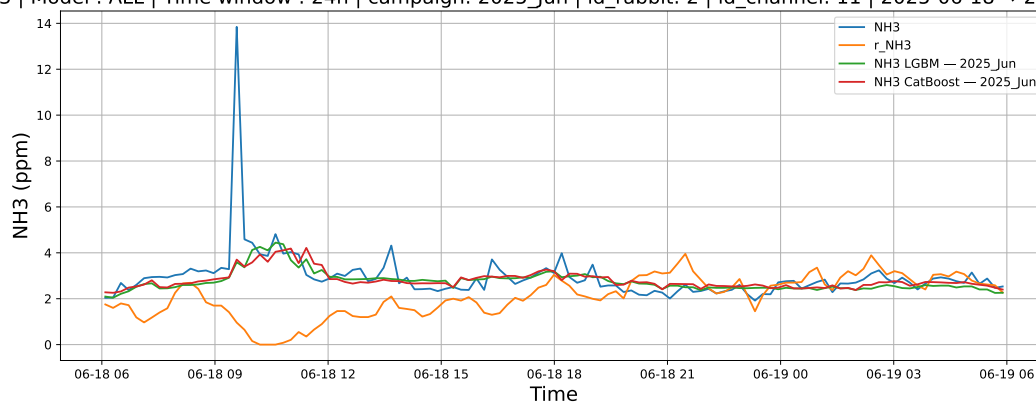
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



3.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

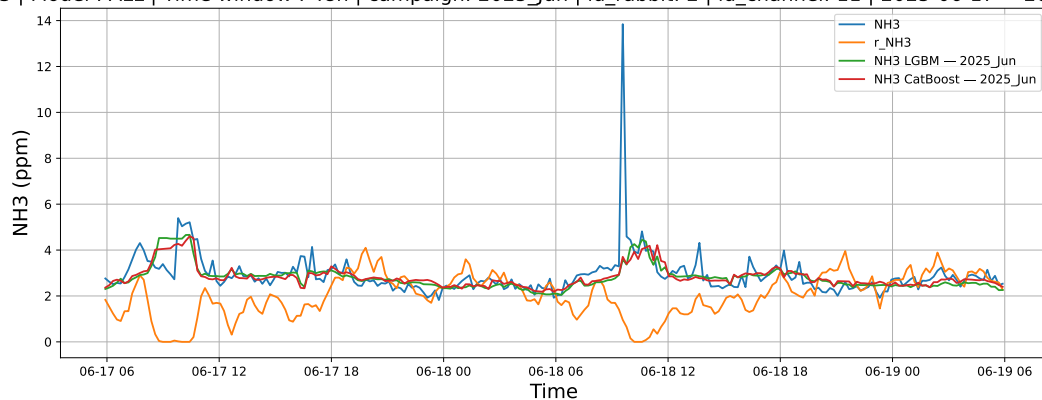
3.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



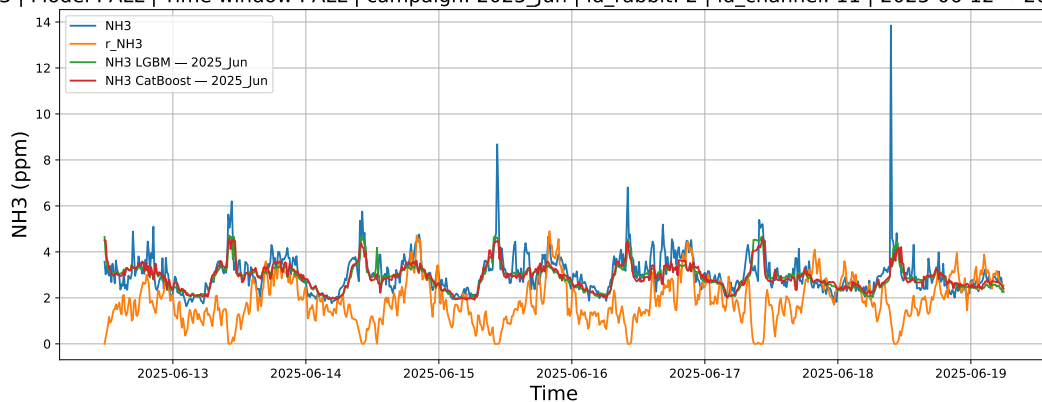
3.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



3.2.2.3 Time window : ALL

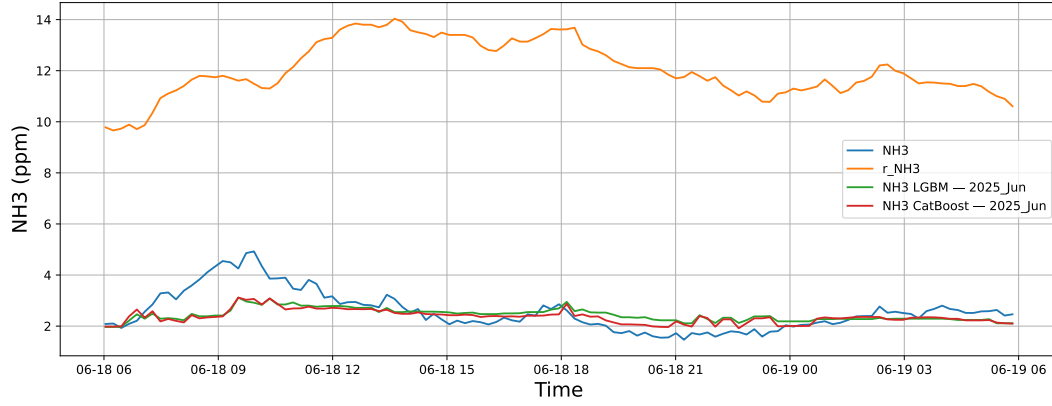
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



3.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

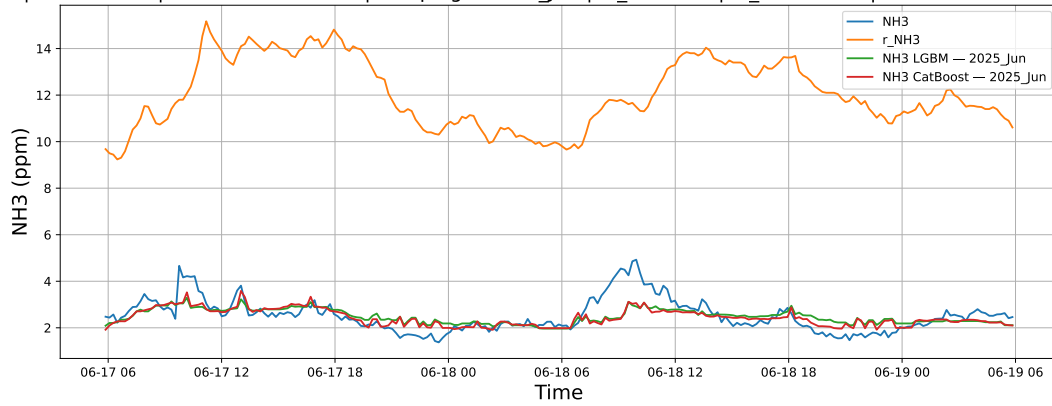
3.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



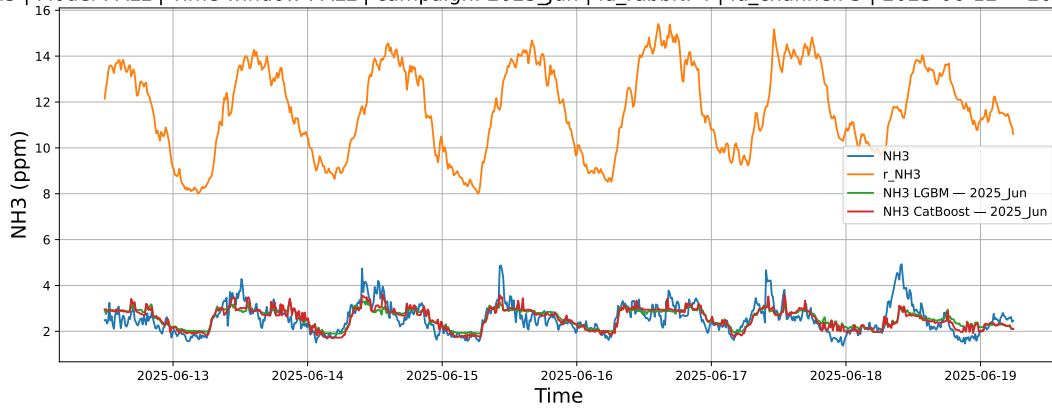
3.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



3.2.3.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



3.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

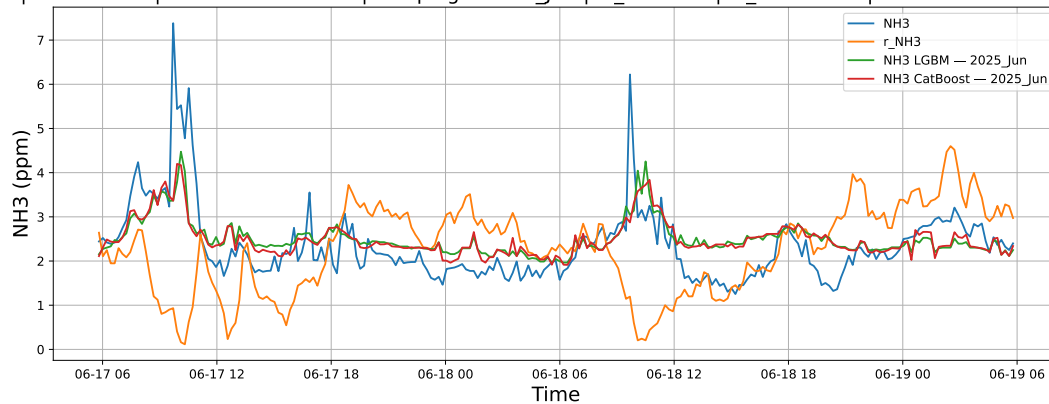
3.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



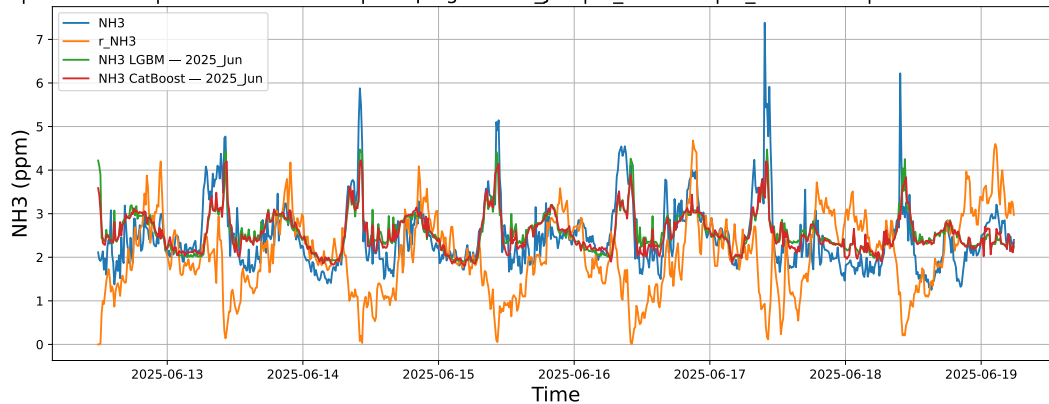
3.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



3.2.4.3 Time window : ALL

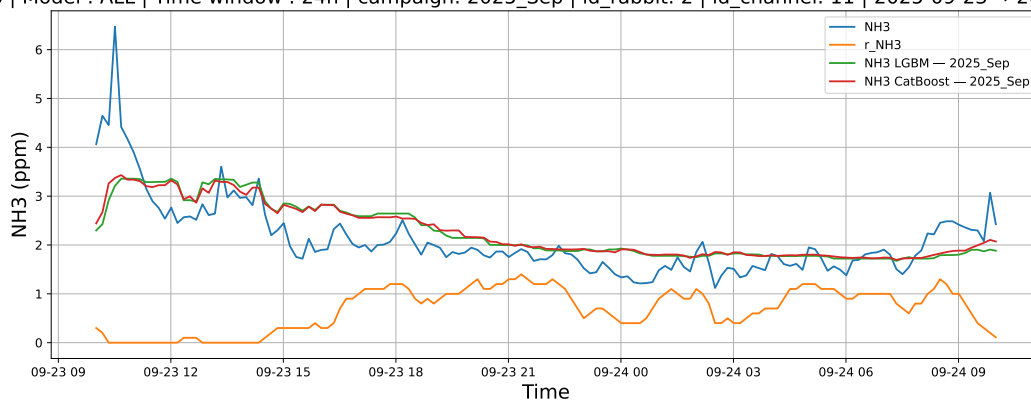
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



3.2.5 Group: campaign: 2025_Sep, id_rabbit: 2, id_channel: 11

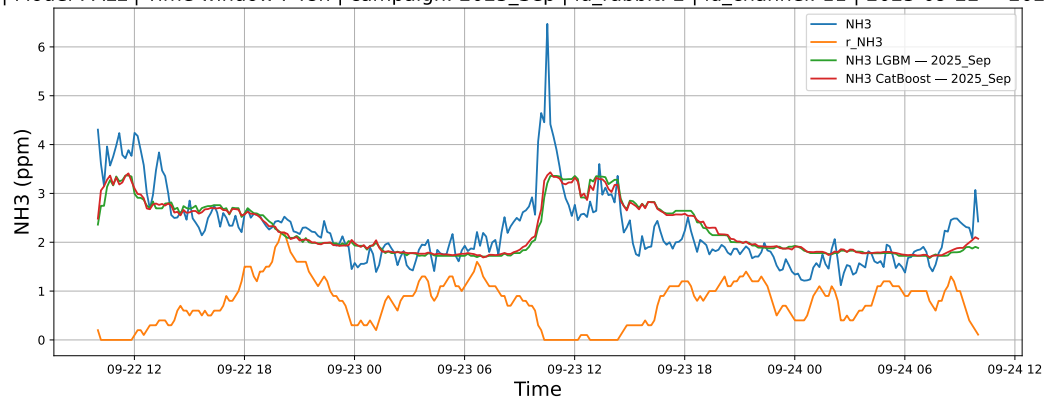
3.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



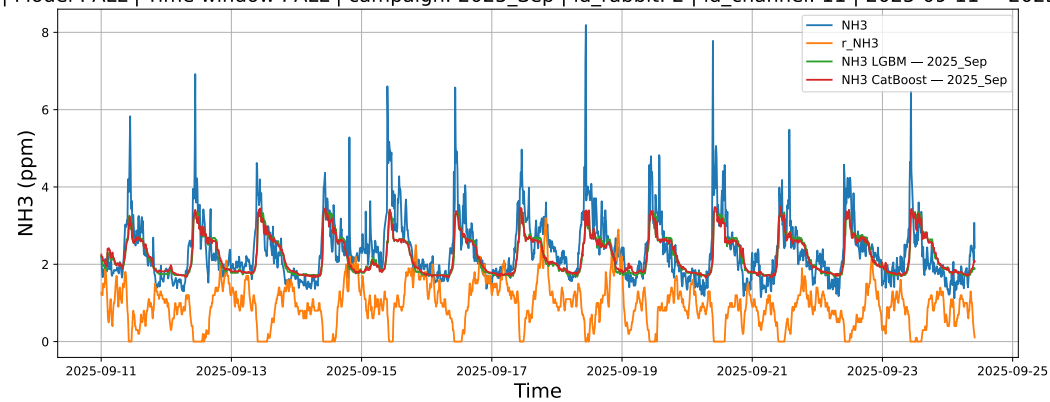
3.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



3.2.5.3 Time window : ALL

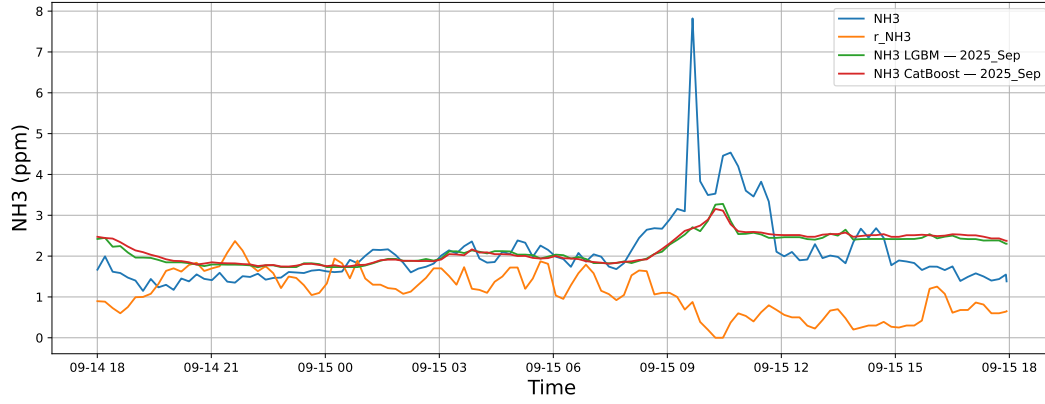
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



3.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

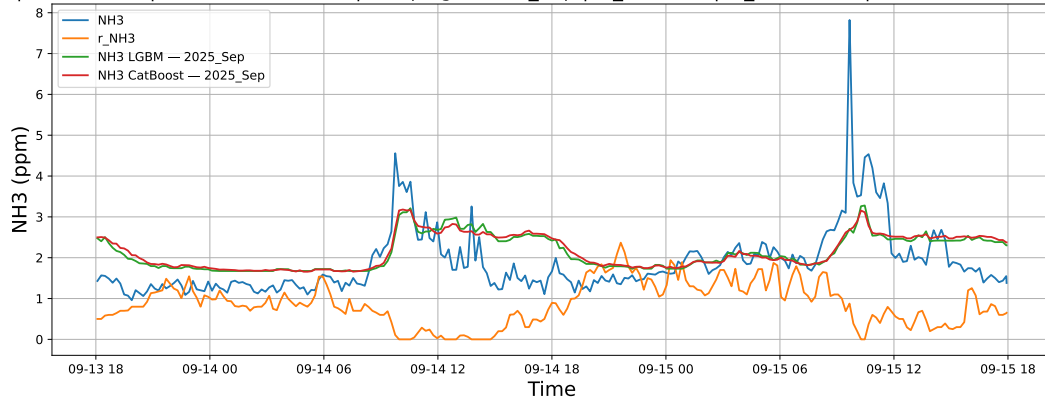
3.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



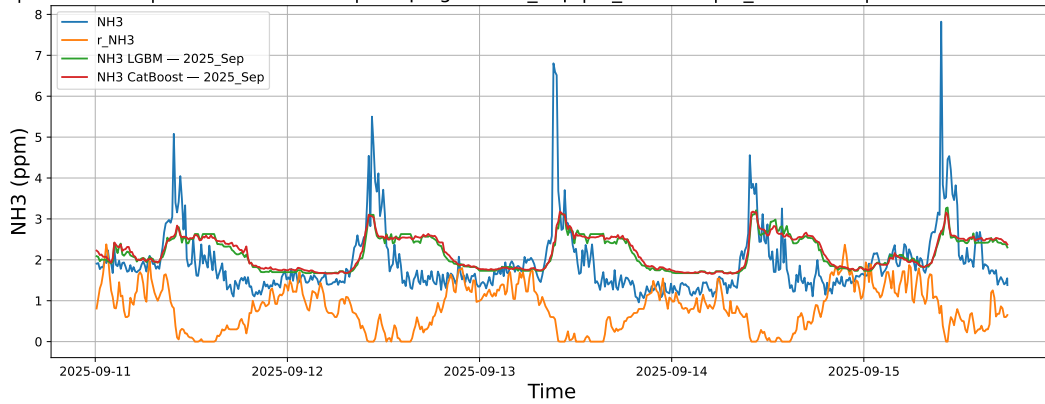
3.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



3.2.6.3 Time window : ALL

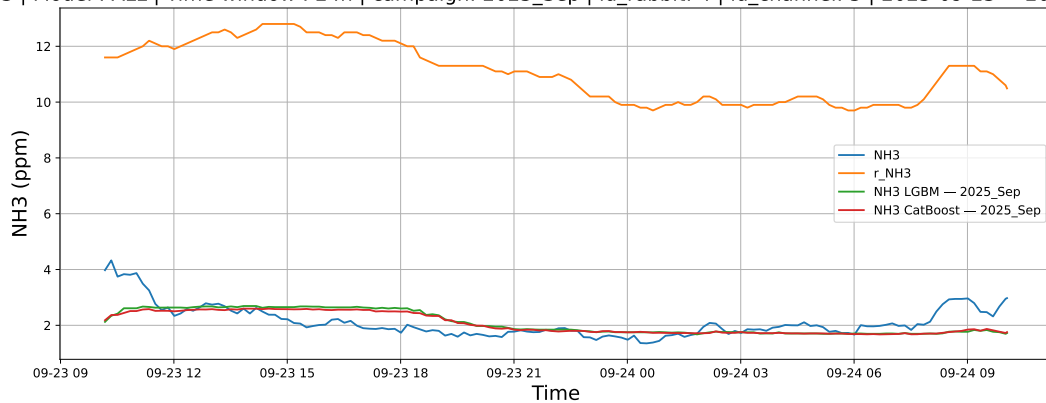
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



3.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

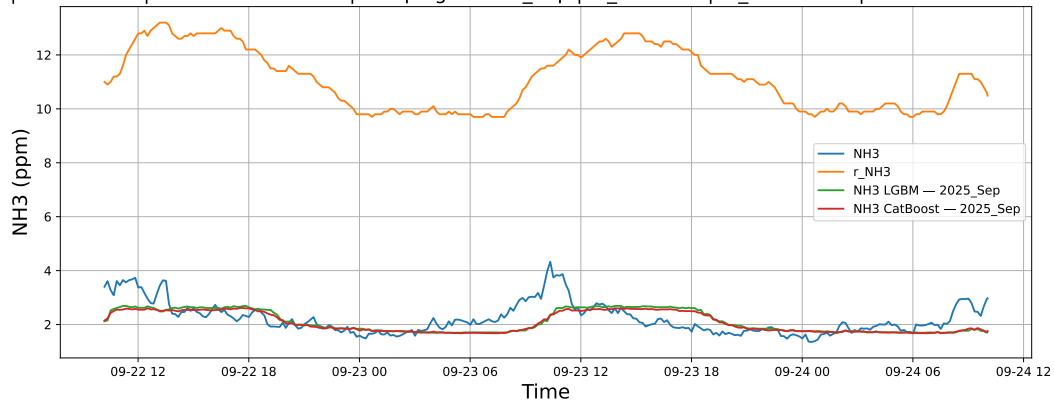
3.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



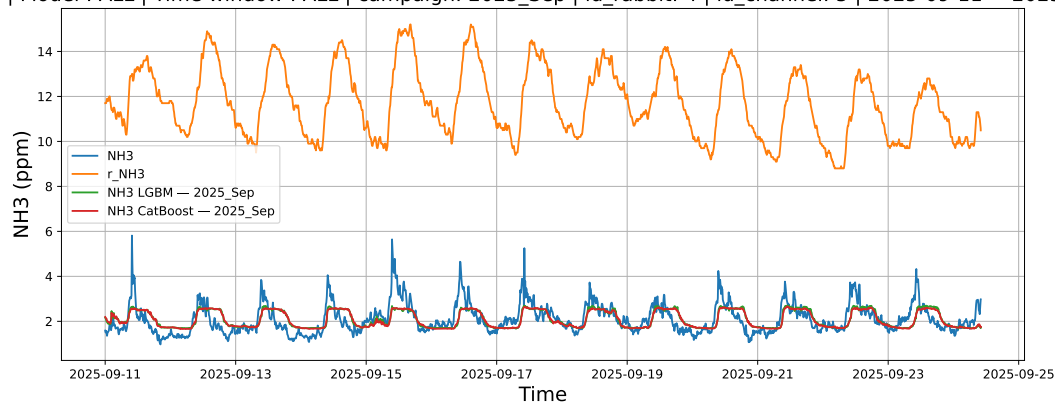
3.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



3.2.7.3 Time window : ALL

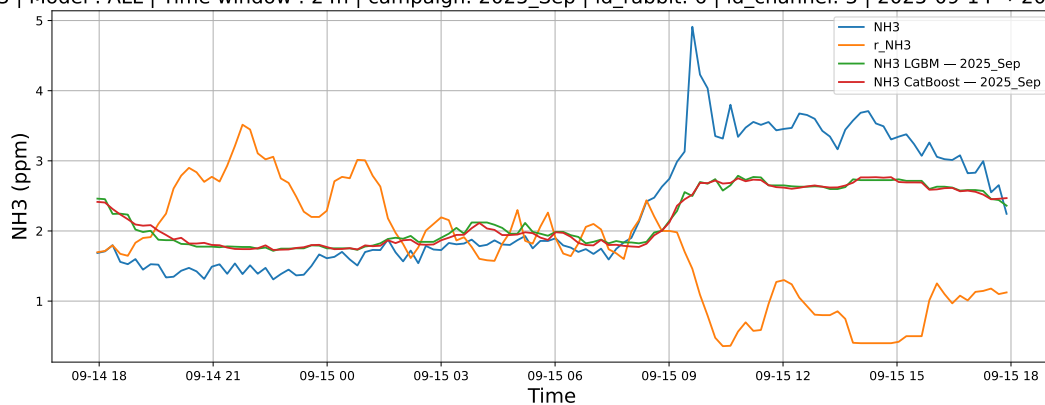
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



3.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

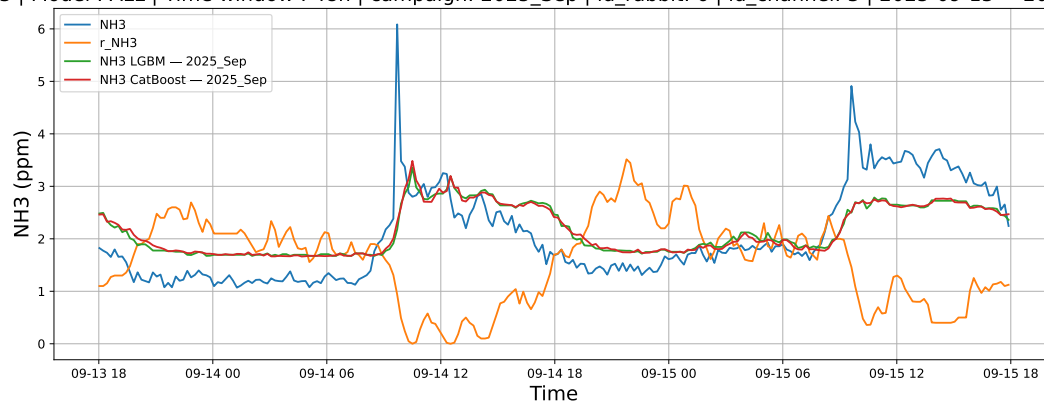
3.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



3.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



3.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

